Adverse selection without single crossing: monotone solutions*

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Abstract

The single crossing assumption simplifies the analysis of screening models as local incentive compatibility becomes sufficient for global incentive compatibility. If single crossing is violated, global incentive compatibility constraints have to be taken into account. This paper studies monotone solutions in a screening model that allows a one-time violation of single crossing.

It is shown that local and non-local incentive constraints distort the solution in opposite directions. Therefore, the optimal decision might involve distortions above as well as below the first best decision. Furthermore, the well known "no distortion at the top" property does not necessarily hold. It is shown that the decision can even be distorted above first best for all types. Sufficient conditions for existence, (strict) monotonicity and continuity of the solution are presented. A new necessary condition satisfied by such solutions is found. An algorithm based on this condition can calculate continuous and strictly monotone solutions.

Keywords: Spence-Mirrlees condition, global incentive compatibility, screening

JEL classification: C61, D82, D86

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1. Introduction

Screening models are among the most commonly used tools in microeconomics. In these models, a principal offers a menu of contracts from which an agent with private information about his "type" chooses his preferred option. Depending on the application, the type represents the production technology of a firm in models of regulation (Baron and Myerson, 1982; Laffont and Tirole, 1987), the productivity of a worker in an employment relationship (Guasch and Weiss, 1981) or the willingness to pay for a product in models of monopoly pricing (Mussa and Rosen, 1978).

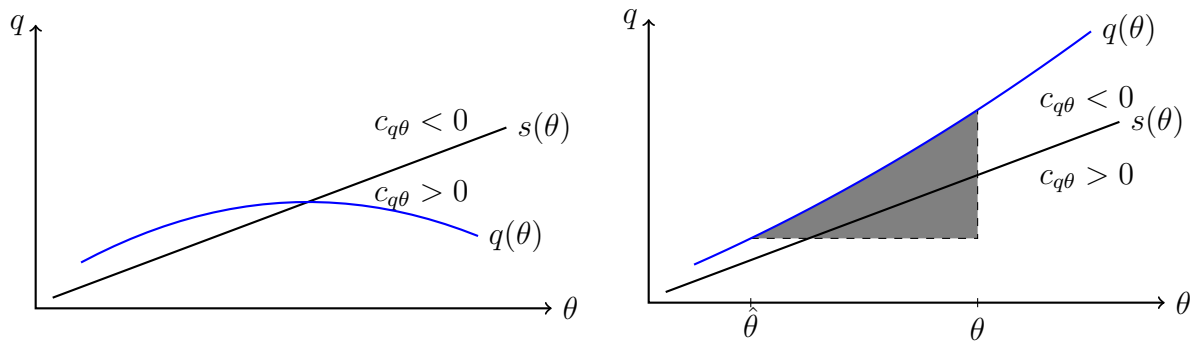
In all these applications, the authors assume single crossing (SC):[1] Types can be ordered according to their marginal rate of substitution between money and the decision, e.g. the quantity purchased in a monopoly pricing problem or the quantity produced in a regulation setup. With the commonly used quasilinear preferences, SC is equivalent to a type ordering according to marginal utilities/costs; e.g. a higher type has a higher marginal utility at every consumption level. The private information of an agent is then his eagerness to consume more.

While SC is a technically convenient assumption, we can think of many unobserved differences other than eagerness/efficiency. Some people are very eager at first but quickly saturated while others have more steady preferences. Compare, for example, single person households with families. At low quantities, single persons might have a higher marginal willingness to pay for standard groceries because of a higher income per household member. At high quantities, however, families will have the higher marginal willingness to pay as single persons are already saturated. This violates SC: Household types cannot be ordered according to their marginal willingness to pay.

As a second example, take firms with private information about their production technology. A capital intensive, fully automated production facility will – at normal output levels – have lower marginal costs than a labor intensive technology. But as soon as quantity approaches the capacity level of the capital intensive facility, the labor intensive production technology could have lower marginal costs.

This paper analyzes a screening model in which SC is violated. Agents have quasilinear preferences and a one-dimensional type. SC would require a constant sign of the cross derivative of the agent's cost function with respect to type and decision. By con-

[1] Other names for this assumption include "Spence-Mirrlees" or "constant sign" condition.



(a) inverse U-shape solution

(b) monotone solution

Figure 1: possible solution shapes

trast, my setting allows for a one time violation of SC: The type versus decision plane is separated by an increasing function s into two regions. The cross derivative is negative above s and positive below s, see figure 1.

The revelation principle implies that implementable menus of contracts can be viewed as mechanisms in which the agent is induced to truthfully reveal his type, i.e. the menu must be incentive compatible. With SC, the local first and second order conditions of the agent's maximization problem (maximizing utility over his type announcement) are necessary and sufficient for incentive compatibility. Without SC, these first and second order conditions are necessary but not sufficient for incentive compatibility.

Incentive compatibility requires the optimal decision function $q(\theta)$ to be increasing (decreasing) whenever $c_{q\theta}(q(\theta), \theta) < (>) 0$ and q is continuous at θ. A continuous optimal decision function (decision as function of type) will, therefore, be either inversely U-shaped or monotone. Araujo and Moreira (2010) analyze solutions in which the decision function is inversely U-shaped, see figure 1a. In this case, some contracts are chosen by two types and the incentive compatibility constraint is binding between any such two types. I analyze monotone solutions, that is, situations in which higher types are assigned a higher decision under the optimal mechanism, see figure 1b. In monotone solutions, incentive compatibility constraints can bind between types choosing different contracts. Sufficient conditions for the existence of a monotone and continuous solution are derived. I then present a new necessary condition satisfied by such solutions and introduce an algorithm to calculate strictly monotone and continuous solutions.

If SC holds, there is no distortion at the top and the distortion for all types goes in the same direction, e.g. all types consume/produce a quantity which is weakly below

their first best quantity. When SC is violated, neither result has to hold.

The reason is that binding non-local incentive constraints will counteract the normal distortion stemming from rent extraction motives. A rough intuition for this result is the following: Suppose the principal took only the first and second order conditions of the agent's maximization problem into account when designing the contract menu. The resulting decision function will be distorted below first best for almost all types because these distortions allow the principal to extract rents from the agent. If SC is violated, this resulting menu might, however, not be implementable: A type θ might benefit from choosing the contract of another type, say $\hat{\theta} < \theta$, instead of choosing the contract intended for him. To satisfy the incentive constraint of type θ, the principal must leave a higher rent to type θ (when choosing the contract intended for him). Reducing the normal distortion – or even distorting the decision in the opposite direction – will result in less rent extraction, leave a higher rent to type θ and, therefore, relax the incentive constraint.

The effect of non-local incentive constraints can be so strong that, in contrast to the previous literature, *all* types' decisions are distorted. This result could, for example, rationalize flat rate tariffs where a zero marginal price incentivizes overconsumption.

The following subsection gives two examples of preferences that violate single crossing. The related literature is reviewed in subsection 1.2 and the formal model is introduced in section 2. Section 3 derives results concerning the distortion in the optimal contract menu and provides a first order condition satisfied by monotone solutions. Section 4 gives sufficient conditions for the existence of a monotone solution and the continuity of the solution. Section 5 introduces a new necessary condition satisfied by strictly monotone and continuous solutions. This condition is based on a minimization program and leads – in combination with the first order condition of section 3 – to an algorithm to calculate such solutions. Section 6 discusses how the results depend on the assumptions as well as possible generalizations. Most proofs are relegated to the appendix. A webappendix includes solved numerical examples and those proofs that are straightforward extensions of other proofs in the paper or proofs in the literature.

1.1. Example setting where single crossing is violated

This section introduces a numerical example which are used for illustration later on. The webappendix contains more examples.

Consider a setting of monopoly regulation. The monopolist produces the good with two input factors which he uses in fixed proportions. The costs of the first input factor are proportional to output, e.g. unskilled labor can be hired in any quantity on a competitive labor market. The costs of the second input factor are convexly increasing in output, e.g. skilled labor is increasingly difficult to hire. The monopolist's private information is whether his production technology is more intense in the first or second factor. A cost function representing this is

$$c(q, \theta) = \theta q + \frac{q^2}{\theta} + \gamma(\theta)$$

where $\gamma(\theta)$ are fixed costs, q is quantity and θ is the type. The cross partial derivative $c_{q\theta}(q, \theta) = 1 - 2q/\theta^2$ can change sign and therefore SC is violated: Marginal costs can be in- or decreasing in type (depending on q). For low quantities, the linear part of the cost function dominates marginal costs and therefore high types have higher marginal costs. For high quantities, the convex part of the cost function is more relevant and therefore high types have lower marginal costs.

The cost function in this example is a simplified version of the flexible fixed cost quadratic cost function suggested by Baumol et al. (1982). Beard et al. (1991) estimate such a cost function for savings and loans associations. Interestingly, they allow for two unobservable types ("mixtures" in their language) of production technology in their estimation. It follows from their estimates in table 5 that mixture 1 has lower marginal costs at low output levels but higher marginal costs at high output levels. Hence, estimated cost functions in the savings and loan sector violate SC.

The example above illustrates a more general point: To avoid the difficulties of multidimensional screening, some papers collapse a multidimensional type into a one-dimensional type ; e.g. we could have used the cost function $\theta_1 q + \theta_2 q^2$ instead and assumed that θ_1 and θ_2 are negatively correlated. When there is a trade-off between being marginally efficient in either one or the other dimension, this collapsing will lead to a setting where SC is violated. Non-local incentive constraints will also bind in the cor-

responding multidimensional model. The multidimensional screening literature mainly sidesteps this complication, for example, by using linear utility/cost functions, see section 1.2.

1.2. Literature

Closest to this paper is Araujo and Moreira (2010). Their paper characterizes (inversely) U-shaped solutions in a setup with a one-time violation of SC. My paper complements their work by characterizing monotone solutions in the same model. Whether the optimal contract menu is monotone or (inversely) U-shaped depends on the utility functions of the agents. Roughly speaking, the shape of the solution is determined by the shape of the first best decision; see section 4.1 for details.

The main technical difference is that in my paper incentive constraints can bind between types choosing different contracts from the menu. Qualitatively, the solution in Araujo and Moreira (2010) features either a discontinuity or a bunching interval. Furthermore, there is a no distortion at the top result, i.e. the type with the highest first best decision[2] will be assigned this first best decision under the optimal mechanism. This paper shows that monotone solutions can be strictly monotone and continuous and therefore bunching and discontinuities are not a necessary implication of a violation of SC. Furthermore, I show that distortion at the top and even distortion of all types' decision is possible in monotone solutions. The (inverse) U-shape solution and its critical condition (see section 3.1 and the webappendix) has been applied in an insurance model (Araujo and Moreira, 2003), in non-linear monopoly pricing (Araujo et al., 2010) and signaling games (Araujo et al., 2007).

Several authors have analyzed perfect competition insurance models with discrete types where SC is violated. In their papers, private information has two dimensions and can take either a high or a low value in each dimension, i.e there are 2×2 types. In Smart (2000), the two dimensions are risk and risk aversion while in Wambach (2000) they are wealth and risk. Netzer and Scheuer (2010) model an additional labor supply decision and the two dimensions are productivity and risk. All three papers share a pooling result, i.e. if SC is violated two of the four types can be pooled. In contrast to

[2]In Araujo and Moreira (2010), the function $s(\theta)$ is downward sloping and the analyzed solution is actually U-shaped (not *inversely* U-shaped as it would be when applied to my setting). In this setting, the undistorted top type is then the type with the *lowest* decision.

my paper, these discrete type models have no distortion at the top and decisions of all other types are downward distorted.

Violations of SC are also related to the literature on multidimensional screening, see Armstrong (1996) and Rochet and Choné (1998) for seminal contributions, Rochet (2009) for a recent related paper and Rochet and Stole (2003) for a survey. As pointed out in the survey, "the problems arise not because of multiple dimensionality itself, but because of a commonly associated lack of exogenous type-ordering in multiple-dimensional environments." A violation of SC also conveys a lack of type-ordering. To make the relationship clear, think of a multidimensional, discrete type model. Clearly, one can reassign types to a one-dimensional parameter but with those reassigned types SC will normally be violated.

Generally speaking, multidimensionality leads to two new challenges concerning incentive compatibility: First, it is ex ante unclear which local incentive constraints are binding. Second, non-local incentive constraints can be binding. The literature has mainly focused on the first challenge[3] and provided sufficient conditions ensuring that non-local incentive constraints are not binding, see for example Carroll (2012). My paper analyzes a model where non-local incentive constraints are binding. Although the model is one-dimensional, this can be viewed as a natural starting point for solving more general setups.

The paper also relates to work relaxing the basic assumptions of the textbook screening model (Fudenberg and Tirole, 1991; Bolton and Dewatripont, 2005). Jullien (2000) allows for type dependent participation constraints while Hellwig (2010) analyzes the case of irregular type distributions, i.e. distributions with mass points and zero densities. Hellwig (2010) shows that this leads to discontinuities as well as bunching in the optimal decision schedule. Contrary to my paper, there is no distortion above first best and no distortion at the top. In Jullien (2000), distortion can be above as well as below first best. The reason is that incentive constraints can bind upward as well as downward. If a participation constraint binds in the interior, it is relaxed by increasing the decision of lower types: This increases the slope of the rent function and leads to higher rents for the interior type. I show that binding incentive constraints can also lead

[3]See Moore (1984) and Matthews and Moore (1987) for early exceptions to this in the discrete type case.

to distortion above first best although incentive constraints are only downward binding. The intuition is that if a type θ wants to misrepresent as a lower type $\hat{\theta}$, one can relax this incentive constraint by increasing the decisions of types between $\hat{\theta}$ and θ: This will increase the slope of the rent function and lead to higher rents for θ at his own contract. Hence, misrepresentation is less attractive. Furthermore, I show that all types can have distorted decisions which is – to the best of my knowledge – a new result in the screening literature.

Another motivation for studying models where SC is violated is that in some strands of the screening literature, e.g. the literature on common agency or on advantageous selection, violations of SC emerge naturally. Authors have – so far – restricted themselves to analyze models with specific functional forms, e.g. quadratic utility functions and uniformly distributed types, for which they could show that the agent's first order condition is sufficient for incentive compatibility, see Martimort and Stole (2009), Calzolari (2004), Jullien et al. (2007) or Hoffmann and Inderst (2011). In the signaling literature, SC fails in models where agents' signaling is motivated by social image concerns, see Bernheim (1994); Bagwell and Bernheim (1996) and Andreoni and Bernheim (2009). Consequently, violations of SC are also likely to matter in non-linear pricing problems with status goods.

2. Model

A principal and an agent contract on a decision $q \in \mathbb{R}_+$ and a monetary transfer $t \in \mathbb{R}$. The agent's utility is $\pi = t - c(q, \theta)$ where $\theta \in \Theta \equiv [\underline{\theta}, \bar{\theta}] \subset \mathbb{R}$ is the type of the agent which is his private information. The function $c(q, \theta)$ is assumed to be three times continuously differentiable with $c_\theta < 0$. The outside option is assumed to be zero for all types. The participation constraint can therefore only bind for the lowest type. Hence, any deviation from the standard solution will not be due to participation constraints binding in the interior, see Jullien (2000) for this, but to the violation of single crossing.

The principal's utility is $u(q, \theta) - t$ and is two times continuously differentiable. The principal has the prior distribution $F(\theta)$ with continuous density $f(\theta) > 0$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. To simplify the exposition, I assume full participation, i.e. the surplus from trade

is so high that it is not beneficial to exclude some types.[4]

The following assumptions are standard in the screening literature. They would ensure that the problem is well behaved if SC was satisfied. Section 6 discusses how the results of the paper extend when these assumptions are violated.

Assumption 1. *The principal's utility function satisfies $u_{q\theta} \geq 0$ and $u_{qq} \leq 0$. The agent's cost function satisfies $c_{qq} > 0$, $c_{qq\theta} < 0$ and $c_{q\theta\theta} > 0$. For all $\theta \in \Theta$, there exists an $s(\theta)$ defined by $c_{q\theta}(s(\theta), \theta) = 0$ which is strictly greater than 0. F satisfies the monotone hazard rate property: $f/(1-F)$ is non-decreasing.*

By making assumptions on the third derivatives of the cost function, I allow for a one time violation of SC; i.e. for a given type, $c_{q\theta}$ can change sign once when varying the decision q (see figure 1). Hence, the type decision plane is divided into two regions: One where $c_{q\theta} < 0$ and one where $c_{q\theta} > 0$. These two regions are separated by a strictly increasing function s defined by $c_{q\theta}(s(\theta), \theta) = 0$. It is assumed that $s > 0$, i.e. the violation of SC occurs for all types.

By the revelation principle, any contract menu can be mimicked by a direct revelation mechanism in which the agent truthfully reports his type (Myerson, 1979). A direct revelation mechanism consists of a pair of functions $(q, t) : \Theta \to \mathbb{R}_+ \times \mathbb{R}$ to which the principal commits.[5] The interpretation is that the agent announces a type $\hat{\theta}$ and this type announcement results in the allocation $(q(\hat{\theta}), t(\hat{\theta}))$. The agent truthfully reports his type if the *incentive compatibility constraint*

$$t(\theta) - c(q(\theta), \theta) \geq t(\hat{\theta}) - c(q(\hat{\theta}), \theta) \quad \text{for all } \theta, \hat{\theta} \in \Theta \tag{1}$$

is satisfied. In this case, the decision function $q : \Theta \to \mathbb{R}_+$ is said to be *implementable* through the transfer function $t : \Theta \to \mathbb{R}$. A contract (q, t) is *feasible* if it satisfies the agent's individual rationality constraint

$$t(\theta) - c(q(\theta), \theta) \geq 0 \quad \text{for all } \theta \in \Theta. \tag{IR}$$

[4]This is less restrictive than it might seem. By $c_\theta < 0$, only types at the low end could be excluded. If exclusion is optimal, the characterization in this paper applies to the set of not excluded types. Using the methods of this paper, one can calculate the solution for any given cutoff type and then maximize the principal's payoff over the cutoff type.

[5]I will concentrate on deterministic mechanisms. The webappendix gives a sufficient condition under which deterministic mechanisms are optimal.

A type θ agent will maximize $t(\hat{\theta}) - c(q(\hat{\theta}), \theta)$ over his type announcement $\hat{\theta}$. This leads to the necessary first and second order conditions of the incentive compatibility constraint.

Lemma 1. *Let q be a decision function implementable through transfers t. Then, the following consitions hold:*

- *Envelope Theorem/First order condition: The agent's rent function is given by*

$$\pi(\theta) \equiv t(\theta) - c(q(\theta), \theta) = \pi(\underline{\theta}) - \int_{\underline{\theta}}^{\theta} c_\theta(q(x), x)\, dx \quad \text{for all } \theta \in \Theta. \tag{2}$$

- *Monotonicity/Second order condition: Let q be right continuous with left-hand side limit at θ. If $q(\theta) > (<) s(\theta)$, then $q^-(\theta) \leq (\geq) q^+(\theta)$ and q is non-decreasing (non-increasing) on $(\theta, \theta + \varepsilon)$ for $\varepsilon > 0$ small enough.*[6]

- *q is bounded from above.*

One implication of the envelope condition that will be useful later on is that $\pi_\theta(\theta) = -c_\theta(q(\theta), \theta)$ for almost all types. The monotonicity condition in lemma 1 implies that the optimal decision function cannot have arbitrary shapes. Araujo and Moreira (2010) analyze inverse U-shape decisions, see figure 1a. In such a solution, one decision can be assigned to two types ("discrete pooling") and therefore these two types have the same contract. Clearly, the incentive compatibility constraint is binding between those two types. It turns out that these are the only binding non-local incentive constraints. I will analyze monotone solutions in this paper, see figure 1b.[7]

The violation of SC plays a role in monotone solutions although $c_{q\theta}(q(\theta), \theta) < 0$ for all types. To see this, rewrite (1) using (2) as

$$c(q(\hat{\theta}), \theta) - c(q(\hat{\theta}), \hat{\theta}) - \int_{\hat{\theta}}^{\theta} c_\theta(q(x), x)\, dx \geq 0.$$

This inequality can again be rewritten as

$$\Phi(\theta, \hat{\theta}) \equiv \int_{\hat{\theta}}^{\theta} \Big(c_\theta(q(\hat{\theta}), x) - c_\theta(q(x), x) \Big)\, dx = -\int_{\hat{\theta}}^{\theta} \int_{q(\hat{\theta})}^{q(x)} c_{q\theta}(y, x)\, dy\, dx \geq 0. \tag{NLIC}$$

[6] The superscript "–" ("+") indicates limits from below (above).

[7] I will focus on monotonically increasing solutions. It is easy to show that in solutions that are below $s(\theta)$ for all types (and therefore are decreasing) non-local incentive constraints do not bind. Hence, standard methods are sufficient to solve such problems.

It follows from (NLIC) that incentive compatibility between two types can be represented as an integral over the shaded area in figure 1b: If the integral of $c_{q\theta}$ over this shaded area is negative, incentive compatibility is satisfied for θ and $\hat{\theta}$. Hence, the region where $c_{q\theta} > 0$ is relevant for incentive compatibility although the solution does not pass through it.

(NLIC) also shows that the conditions in lemma 1 are sufficient for incentive compatibility if SC is satisfied. SC means that the integrand in (NLIC) has the same sign for any argument. Monotonicity of q implies then the inequality.

The principal's second best problem is to maximize

$$\int_{\underline{\theta}}^{\bar{\theta}} \left(u(q(\theta), \theta) - t(\theta)\right) f(\theta)\, d\theta$$

over $q(\cdot)$ and $t(\cdot)$ subject to incentive compatibility and individual rationality. Using (2) to eliminate t from the principal's objective and integrating by parts transforms the principal's objective into the usual virtual valuation and leads to the following second best problem:

$$\max_{q(\cdot),\pi(\underline{\theta})} \int_{\underline{\theta}}^{\bar{\theta}} \left(u(q(\theta), \theta) - c(q(\theta), \theta)\right) f(\theta) + (1 - F(\theta)) c_\theta(q(\theta), \theta)\, d\theta - \pi(\underline{\theta}) \qquad \text{(SB)}$$
$$s.t.\ \ (NLIC) \text{ and } (IR)$$

The space of decision functions is assumed to be the space of measurable functions. The individual rationality constraint can only bind for $\underline{\theta}$ because of the assumption $c_\theta < 0$. As leaving rents to the agent is costly to the principal, (IR) will bind for $\underline{\theta}$ and therefore $\pi(\underline{\theta}) = 0$ in the optimal contract menu. Note that (SB) is equivalent to the principal's original problem as every q that satisfies (NLIC) for all types θ, $\hat{\theta} \in \Theta$ is implementable through transfers given by (2).

I will refer to the constraint (NLIC) in problem (SB) as the *non-local incentive compatibility constraint*. If (NLIC) binds such that type θ is indifferent between his contract and the contract of $\hat{\theta}$, I will say that (NLIC) is *binding from θ to $\hat{\theta}$*. With *local incentive compatibility constraint* I will refer to the envelope and monotonicity condition in lemma 1.

Before turning to the analysis of the solution, two benchmarks are defined. The *first*

best decision denoted by $q^{fb}(\theta)$ is defined as $argmax_q u(q,\theta) - c(q,\theta)$ which is the decision maximizing joint surplus. Existence and uniqueness of $q^{fb}(\theta)$ follow from concavity; see assumption 1. As a second benchmark, the *relaxed solution* q^r is defined as the solution of the following *relaxed program* which neglects the constraint (NLIC):

$$\max_{q(\cdot)} \int_{\underline{\theta}}^{\bar{\theta}} \left(u(q(\theta),\theta) - c(q(\theta),\theta)\right) f(\theta) + (1 - F(\theta)) c_\theta(q(\theta),\theta)\, d\theta \tag{RP}$$

The inverse U-shape solutions in Araujo and Moreira (2010) occur when q^{fb} is inversely U-shaped and intersects s. In order to focus on monotone solutions, I require the first best decision to be above s. This will – together with assumption 1 – imply that q^{fb} and q^r are strictly monotone.

Assumption 2. *$s(\theta) < q^{fb}(\theta)$ for all $\theta \in \Theta$.*

Lemma 2. *The first best decision is strictly monotonically increasing. The relaxed program is strictly concave. The relaxed decision is strictly monotonically increasing and strictly above s.*

Lemma 2 implies that the relaxed decision q^r is implicitly defined by the pointwise first order condition

$$\left(u_q(q(\theta),\theta) - c_q(q(\theta),\theta)\right) f(\theta) + (1 - F(\theta)) c_{q\theta}(q(\theta),\theta) = 0. \tag{3}$$

If SC was satisfied, the relaxed decision would be the solution of the principal's problem. Without SC, non-local incentive constraints can be violated under the relaxed decision. The following section will show how the solution differs from q^r qualitatively as a result of binding non-local incentive constraints.

3. Optimal contract menu

3.1. Necessary conditions

This subsection presents necessary conditions which have to be met whenever a non-local incentive constraint is binding. Since these conditions are only a slight generalization of those presented in theorem 1A and 2 of Araujo and Moreira (2010), the presentation will be brief and more intuitive than formal.

Take an optimal decision function q and assume for now that q is continuous and strictly increasing. Let NLIC be satisfied with equality for types θ and $\hat{\theta}$; i.e. $\Phi(\theta, \hat{\theta}) = 0$. Φ has to be non-negative for all types by (NLIC). Therefore, $(\theta, \hat{\theta}) \in argmin_{(x,y)}\Phi(x, y)$ as $\Phi(\theta, \hat{\theta}) = 0$. If $\hat{\theta} \neq \bar{\theta}$ and $\theta \neq \underline{\theta}$, the following first order conditions for a minimum have to hold:[8]

$$\int_{q(\hat{\theta})}^{q(\theta)} -c_{q\theta}(y, \theta)\, dy \leq 0 \qquad \text{with "=" if } \theta < \bar{\theta} \tag{C1}$$

$$\int_{\hat{\theta}}^{\theta} c_{q\theta}(q(\hat{\theta}), x)\, dx \geq 0 \qquad \text{with "=" if } \hat{\theta} > \underline{\theta} \tag{C2}$$

Graphically, $\Phi(\theta, \hat{\theta}) = 0$ meant that the integral of $c_{q\theta}$ over the shaded area in figure 1b is zero. (C1) and (C2) imply that the integral of $c_{q\theta}$ over the right and the lower boundary of the shaded area in figure 1b is zero. If, for example, the integral in (C1) was positive and $\Phi(\theta, \hat{\theta}) = 0$, then incentive compatibility would be violated for $\theta - \varepsilon$ and $\hat{\theta}$ as $\Phi(\theta - \varepsilon, \hat{\theta}) \approx \Phi(\theta, \hat{\theta}) - \varepsilon \int_{q(\hat{\theta})}^{q(\theta)} -c_{q\theta}(y, \theta)\, dy$.

The graphical interpretation also allows to quickly generalize these conditions to decision functions that are neither continuous nor strictly increasing. This situation is depicted in figure 2. Assume $\Phi(\theta, \hat{\theta}_i) = 0$ for $i = 1, 2$. To satisfy (NLIC) for types close to θ, $\hat{\theta}_1$ and $\hat{\theta}_2$ the following conditions have to hold:

- $\int_{q(\hat{\theta}_i)}^{q^-(\theta)} c_{q\theta}(y, \theta)\, dy \geq 0$ as otherwise $\Phi(\theta - \varepsilon, \hat{\theta}_i) < 0$
- $\int_{q(\hat{\theta}_i)}^{q^+(\theta)} c_{q\theta}(y, \theta)\, dy \leq 0$ as otherwise $\Phi(\theta + \varepsilon, \hat{\theta}_i) < 0$
- $\int_{\hat{\theta}_1}^{\theta} c_{q\theta}(q(\hat{\theta}), x)\, dx \leq 0$ as otherwise $\Phi(\theta, \hat{\theta}_1 - \varepsilon) < 0$
- $\int_{\hat{\theta}_2}^{\theta} c_{q\theta}(q(\hat{\theta}), x)\, dx \geq 0$ as otherwise $\Phi(\theta, \hat{\theta}_2 + \varepsilon) < 0$

Given (C1) and (C2), Araujo and Moreira (2001) derive a third necessary condition for types at which (NLIC) binds using variational calculus. While (C1) and (C2) are purely driven by incentive compatibility, this third condition will be derived from the principal's optimization. The idea is to have a variation of the optimal decision around θ and $\hat{\theta}$ such that the two necessary conditions (C1) and (C2) are still satisfied. The derivation is sketched in the webappendix. The following variational condition results

[8] It will be shown later that the possibilities $\hat{\theta} = \bar{\theta}$ or $\theta = \underline{\theta}$ will not play a role in monotone decisions.

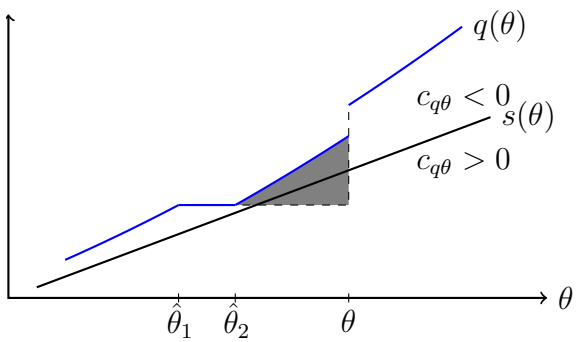


Figure 2: necessary conditions at discontinuity

for interior types θ, $\hat{\theta}$ such that (NLIC) binds between θ and $\hat{\theta}$ and q is strictly increasing in neighborhoods of θ and $\hat{\theta}$:

$$\frac{[u_q(q(\theta),\theta) - c_q(q(\theta),\theta)]f(\theta)}{c_{q\theta}(q(\theta),\theta)} + 1 - F(\theta) = \frac{[u_q(q(\hat{\theta}),\hat{\theta}) - c_q(q(\hat{\theta}),\hat{\theta})]f(\hat{\theta})}{c_{q\theta}(q(\hat{\theta}),\hat{\theta})} + 1 - F(\hat{\theta}) \quad \text{(C3)}$$

Section 3.2 will give a shadow value interpretation to the terms on both sides of (C3) and thereby provide some intuition for this condition.

3.2. Monotone solution

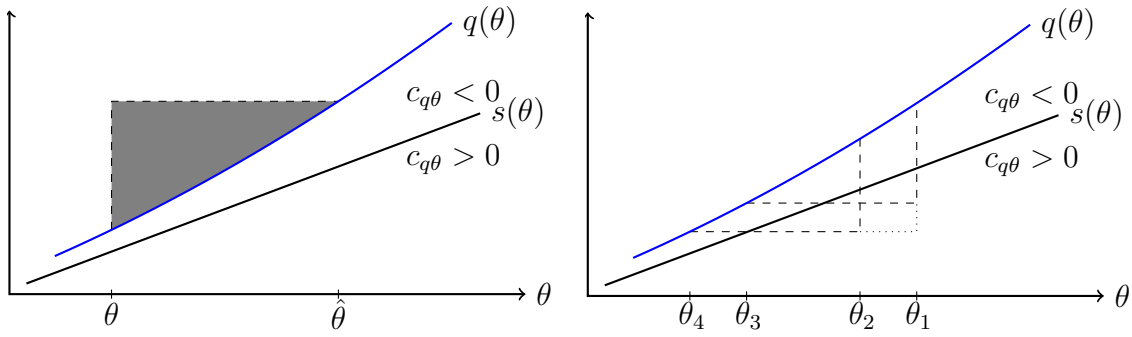
This section gives a characterization of monotonically increasing solutions to problem (SB). Lemmas 3–5 show that only a certain subset of non-local incentive constraints can be binding. This allows to identify implementable variations which are used to derive results on distortion and a first order condition (theorem 1). These results will be used later in the algorithm of section 5.[9]

This section assumes only monotonicity of the solution, i.e. non-differentiabilities and discontinuities are allowed. Section 4 will give sufficient conditions for the monotonicity of the solution. It should be pointed out that an increasing solution has to be above s as a result of the monotonicity condition in lemma 1.

Corollary 1. *Let q be an implementable and strictly increasing decision function. Then $q(\theta) \geq s(\theta)$ for all $\theta \in \Theta \setminus \{\underline{\theta}\}$.*

Lemma 3 implies that incentive constraints cannot be upward binding in monotone solutions. Put differently, no type will be indifferent between the contract designated

[9] A more direct approach to the problem would be to utilize the generalized Kuhn-Tucker theorem; see e.g. Luenberger (1968, ch. 9.4). However, the resulting first order condition appears to be intractable.



(a) no upwards binding (b) no overlap

Figure 3: non-binding constraints

for him and a contract that involves a higher decision. The intuition for lemma 3 is the same as in models with SC. Increasing the decision increases the costs for higher types less than for lower types. This is true as $c_{q\theta}(q', \theta) < 0$ for all $q' \geq q(\theta)$ in a monotone decision q, see figure 1b.

Lemma 3. *Let q be an increasing decision function such that $q(\theta) \geq s(\theta)$ for all $\theta \in \Theta \setminus \{\underline{\theta}\}$. Then, $\Phi(\theta', \hat{\theta}') > 0$ for any two types θ', $\hat{\theta}' \in \Theta$ such that $\theta' < \hat{\theta}'$ and $q^+(\theta') < q(\hat{\theta}')$.*

Proof. Let $\hat{\theta}' > \theta'$. Then, $\Phi(\theta', \hat{\theta}') = -\int_{\theta'}^{\hat{\theta}'} \int_{q(x)}^{q(\hat{\theta}')} c_{q\theta}(y, x)\, dy\, dx > 0$ where the inequality follows from the following two observations: First, monotonicity of q implies $q(x) \leq q(\hat{\theta}')$ for all $x \in (\theta', \hat{\theta}')$ and $q(x) < q(\hat{\theta}')$ for x close enough to θ' as $q^+(\theta') < q(\hat{\theta}')$. Second, $q(\theta) \geq s(\theta)$ for all $\theta \in (\theta', \hat{\theta}')$ implies that the integrand is strictly negative for almost all arguments (y, x). See figure 3a for a graphical representation. □

One of the results of this paper is that binding non-local incentive constraints can affect the solution without leading to irregularities, i.e. discontinuities or bunching. The following lemma shows that some irregularities can be ruled out on the grounds of incentive compatibility alone. It implies that the decision is continuous at a type to which (NLIC) is binding (assuming that the monotonicity constraint cannot bind for $\hat{\theta}$). Furthermore, the decision has to be strictly increasing in an open neighbohood of a type from which (NLIC) is binding (assuming that the decision is continuous at this type).

Lemma 4. *Let q be an implementable and increasing decision function. Suppose $\Phi(\theta, \hat{\theta}) = 0$ for θ, $\hat{\theta} \in \Theta$ such that $\theta > \hat{\theta}$ and $q(\theta) > q(\hat{\theta})$. If $\hat{\theta} > \underline{\theta}$ and there is no type $\tilde{\theta}$ such that either $q(\tilde{\theta}) = q^-(\hat{\theta})$ or $q(\tilde{\theta}) = q^+(\hat{\theta})$, then q is continuous at $\hat{\theta}$. If q is continuous at θ and $\theta < \bar{\theta}$, then there does not exist a type $\theta' \in \Theta \setminus \{\theta\}$ with $q(\theta') = q(\theta)$.*

The following lemma implies that binding non-local incentive constraints cannot overlap; i.e. if (NLIC) is satisfied with equality for θ and $\hat{\theta}$, then (NLIC) cannot hold with equality for a type in the interval $[\hat{\theta}, \theta]$ and a type outside this interval (abstracting from the trivial exception where an interval of types is assigned the same decision).

Lemma 5. *Let q be an implementable and increasing decision function. Suppose $\Phi(\theta, \hat{\theta}) = 0$ for θ, $\hat{\theta} \in \Theta$ such that $\theta > \hat{\theta}$. Then, $\Phi(\theta', \hat{\theta}') > 0$ for any pair of types θ', $\hat{\theta}'$ such that $\theta' \in [\hat{\theta}, \theta)$, $\hat{\theta}' < \hat{\theta}$ and $q(\hat{\theta}) > q(\hat{\theta}')$. Also, $\Phi(\theta', \hat{\theta}') > 0$ for any pair of types $\theta', \hat{\theta}'$ such that $\hat{\theta}' \in (\hat{\theta}, \theta]$, $\theta' > \theta$ and $q(\hat{\theta}') > q(\hat{\theta})$.*

Proof. The proof is by contradiction. Suppose, contrary to the lemma, there are types $\theta_1 > \theta_2 \geq \theta_3 > \theta_4$ such that $\Phi(\theta_1, \theta_3) = \Phi(\theta_2, \theta_4) = 0$ and $q(\theta_3) > q(\theta_4)$. Then, the incentive constraint between θ_1 and θ_4 will be violated, i.e. $\Phi(\theta_1, \theta_4) < 0$:

$$\begin{aligned}
\Phi(\theta_1, \theta_4) &= -\int_{\theta_4}^{\theta_1} \int_{q(\theta_4)}^{q(x)} c_{q\theta}(y, x)\, dy\, dx \\
&= -\int_{\theta_4}^{\theta_2} \int_{q(\theta_4)}^{q(x)} c_{q\theta}(y, x)\, dy\, dx - \int_{\theta_2}^{\theta_1} \int_{q(\theta_4)}^{q(\theta_3)} c_{q\theta}(y, x)\, dy\, dx - \int_{\theta_2}^{\theta_1} \int_{q(\theta_3)}^{q(x)} c_{q\theta}(y, x)\, dy\, dx \\
&= -\int_{\theta_4}^{\theta_2} \int_{q(\theta_4)}^{q(x)} c_{q\theta}(y, x)\, dy\, dx - \int_{\theta_2}^{\theta_1} \int_{q(\theta_4)}^{q(\theta_3)} c_{q\theta}(y, x)\, dy\, dx \\
&\quad + \int_{\theta_3}^{\theta_2} \int_{q(\theta_3)}^{q(x)} c_{q\theta}(y, x)\, dy\, dx - \int_{\theta_3}^{\theta_1} \int_{q(\theta_3)}^{q(x)} c_{q\theta}(y, x)\, dy\, dx \\
&= -\Phi(\theta_2, \theta_3) - \int_{\theta_2}^{\theta_1} \int_{q(\theta_4)}^{q(\theta_3)} c_{q\theta}(y, x)\, dy\, dx < 0
\end{aligned}$$

The second and third equality are simple splitting up the integral steps (and can be visualized in figure 3b), the fourth uses the fact that $\Phi(\theta_1, \theta_3) = \Phi(\theta_2, \theta_4) = 0$ and the last inequality follows from the incentive compatibility between θ_2 and θ_3 as well as the following idea: $\Phi(\theta_2, \theta_4) = 0$ and the fact that θ_2 is interior imply $\int_{q(\theta_4)}^{q^-(\theta_2)} c_{q\theta}(y, \theta_2)\, dy \geq 0$ by (C1) (with equality if q is continuous at θ_2). By the monotonicity of q, $q(\theta_3) \leq q^-(\theta_2)$ and therefore $\int_{q(\theta_4)}^{q(\theta_3)} c_{q\theta}(y, \theta_2)\, dy \geq 0$ (see figure 3b).[10] The inequality above follows then from $c_{q\theta\theta} > 0$. □

As a special case, i.e. with $\hat{\theta}' = \theta$, the preceding lemma includes the following: If θ is indifferent between his and $\hat{\theta}$'s contract, then no other type θ' is indifferent between his contract and θ's contract, i.e. $\Phi(\theta', \theta) > 0$ for all $\theta' \in \Theta \setminus \{\theta\}$. The top panel of figure

[10] Note that q has to be continuous at θ_2 if $\theta_2 = \theta_3$ by lemma 4.

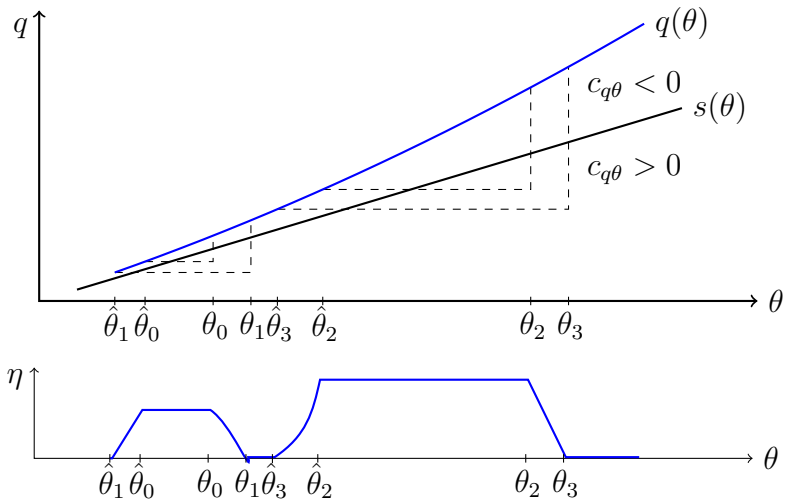


Figure 4: how incentive constraints can bind

4 summarizes the previous lemmas by showing how non-local incentive compatibility constraints can bind.

The previous three lemmas applied to any increasing (and implementable) decision function. The following results focus on the optimal decision function. Before stating the main result of this section, I want to clarify the terms used in the theorem. The theorem restricts itself to càdlàg functions.[11] For monotone solutions, this is, however, without loss of generality: A monotone function q is continuous almost everywhere and both one-sided limits exist everywhere. Hence, assigning the right limit decision to types where q is discontinuous will not affect the principal's payoff. Incentive compatibility of this right limit contract is implied by incentive compatibility of the contracts of slightly higher types.

Theorem 1. *Let $q \geq s$ be an increasing solution to problem (SB). Assume that q is a càdlàg function. Then $q \geq q^r$ and*

$$[u_q(q(\theta), \theta) - c_q(q(\theta), \theta)]f(\theta) + (1 - F(\theta))c_{q\theta}(q(\theta), \theta) = \eta(\theta)c_{q\theta}(q(\theta), \theta) \tag{4}$$

holds where $\eta(\theta)$ is a non-negative, almost everywhere differentiable càdlàg function with the following properties:

- *Let (θ_1, θ_2) be an interval such that q is strictly increasing on (θ_1, θ_2) and $\Phi(\theta', \hat{\theta}') > 0$ for every $\hat{\theta}' \in (\theta_1, \theta_2)$ and every $\theta' \in \Theta \setminus \{\hat{\theta}'\}$. Then, η is non-increasing on*

[11] A càdlàg function is defined as a function that is right continuous with left hand side limit at each point of the domain.

(θ_1, θ_2).

- *Let (θ_1, θ_2) be an interval such that q is strictly increasing on (θ_1, θ_2) and $\Phi(\theta', \hat{\theta}') > 0$ for every $\theta' \in (\theta_1, \theta_2)$ and every $\hat{\theta}' \in \Theta \setminus \{\theta'\}$. Then, η is non-decreasing on (θ_1, θ_2).*

- *Let $\Phi(\theta', \hat{\theta}') = 0$ and let q be strictly increasing in open neighborhoods around θ' and $\hat{\theta}'$. Then, η is non-increasing in an open neighborhood of θ'. Furthermore, η is non-decreasing in an open neighborhood of $\hat{\theta}'$.*

- *$\eta^-(\bar{\theta})$ is zero if $\Phi(\bar{\theta}, \hat{\theta}) > 0$ for all $\hat{\theta} < \bar{\theta}$.*

- *$\eta^+(\underline{\theta})$ is zero if $\Phi(\theta, \underline{\theta}) > 0$ for all $\theta > \underline{\theta}$.*

- *Let (θ_1, θ_2) be an interval such that q is constant on (θ_1, θ_2) but q is neither constant on $(\theta_1 - \varepsilon, \theta_2)$ nor on $(\theta_1, \theta_2 + \varepsilon)$ for any $\varepsilon > 0$. Then, $\eta^+(\theta_1) \leq \eta^-(\theta_2)$.*

- *Let θ' be a type such that $q^-(\theta') < q^+(\theta')$. Then, $\eta^-(\theta') \geq \eta^+(\theta')$.*

The theorem states that $q \geq q^r$. Put differently, distortion from binding non-local incentive constraints counteracts the usual downward distortion in (RP). The intuition for this can be illustrated with figure 1b. (NLIC) is violated if the integral of $c_{q\theta}$ over shaded area is positive. To satisfy incentive compatibility the decision can be raised for all types between $\hat{\theta}$ and θ. The additional shaded area features $c_{q\theta} < 0$ and therefore the incentive problem is mitigated.[12]

One noteworthy point is that the incentive constraint is mainly relaxed by increasing the decision of types at which the incentive constraint is non-binding; i.e. if (NLIC) is binding from θ' to $\hat{\theta}'$, it is less $q(\theta')$ and $q(\hat{\theta}')$ that have to be increased but q for the types between $\hat{\theta}'$ and θ'. To see the intuition, recall that $\pi_\theta(\theta) = -c_\theta(q(\theta), \theta)$ for almost all types and that $c_{q\theta}(q(\theta), \theta) < 0$. Increasing q will raise the slope of the rent function π as $c_{qq\theta} < 0$. An increased decision for types in $(\hat{\theta}', \theta')$ will therefore increase the rent of θ' at his own contract which relaxes the non-local incentive constraint.

Before giving an intuitive interpretation of the properties of η, let me briefly sketch the idea behind the theorem. Given the solution q, one can simply define η by (4).

[12] In models with single crossing, the decision can be below q^r for some types if the monotonicity constraint is violated under q^r. This does, however, not contradict the theorem because the assumptions of this paper imply that q^r is strictly increasing (lemma 2).

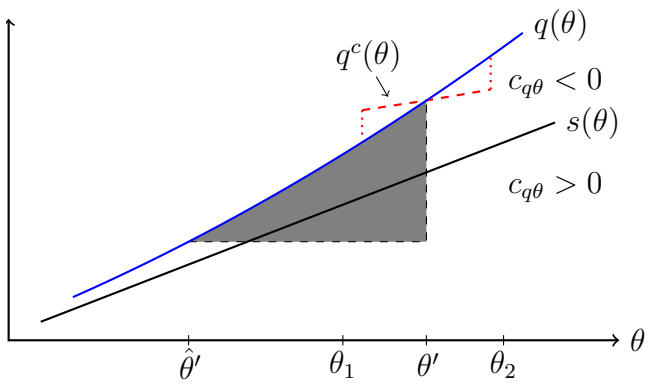


Figure 5: implementable changes

The properties of η are derived by showing that q could be changed in a way that (i) increases the principal's payoff if these properties were not satisfied and (ii) is incentive compatible.

To be more specific, take the first bullet point and consider the variation depicted in figure 5, i.e. take a type $\theta' \in (\theta_1, \theta_2)$ and increase the decision for types slightly below θ' and decrease the decision for types slightly above θ'. Choose the variation such that $\int_{\theta_1}^{\theta_2} \int_{q(x)}^{q^c(x)} c_{q\theta}(y, x)\, dy\, dx = 0$. The variation would increase the payoff of the principal if η was increasing on (θ_1, θ_2): (4) implies that the derivative of the virtual valuation with respect to $q(\theta)$ is $\eta(\theta) c_{q\theta}(q(\theta), \theta)$. Hence, the change in the virtual valuation of θ is approximately $\int_{q(\theta)}^{q^c(\theta)} \eta(\theta) c_{q\theta}(y, \theta)\, dy$ if $q^c(\theta)$ is close to $q(\theta)$. The overall change in the principal's payoff caused by the change in the decision can therefore be approximated by $\int_{\theta_1}^{\theta_2} \int_{q(x)}^{q^c(x)} \eta(x) c_{q\theta}(y, x)\, dy\, dx$. If η was strictly increasing on (θ_1, θ_2), this would be positive because $\int_{\theta_1}^{\theta_2} \int_{q(x)}^{q^c(x)} c_{q\theta}(y, x)\, dy\, dx = 0$ and $\int_{q(x)}^{q^c(x)} c_{q\theta}(y, x)\, dy$ is positive (negative) for high (low) x; see figure 5. The changed decision is implementable (if q^c is close to q): As q^c is increasing, upward incentive constraints are satisfied (lemma 3). As – by assumption – no type was indifferent between his contract and the contract of $\hat{\theta} \in (\theta_1, \theta_2)$, the same will be true for the changed decision if the variation is chosen small enough. Finally, if in the original decision types in $\theta' \in (\theta_1, \theta_2)$ were indifferent between their contract and the contract of some $\hat{\theta}' < \theta'$, then the variation will relax this incentive constraint. The argument for the other points is similar although the implementable variation differs.

The properties of η have an intuitive interpretation. The left hand side of (4) is well known from models with SC: Increasing $q(\theta)$ affects the surplus from type θ but also the rents of all types above θ. Marginally increasing $q(\theta)$ will also relax all non-local

incentive constraints from types $\theta' > \theta$ to types $\hat{\theta}' < \theta$, see figure 1b. As these incentive constraints can be expressed as integrals over $c_{q\theta}$ (see equation (NLIC)), the "amount" by which those non-local incentive constraints are relaxed is given by $c_{q\theta}(q(\theta), \theta)$ which can be found on the right hand side of (4). Consequently, $\eta(\theta)$ could be interpreted as the shadow value of all non-local incentive constraints from types $\theta' > \theta$ to types $\hat{\theta}' < \theta$.

Consider a type θ such that (NLIC) binds from θ to $\hat{\theta}$. Take a type slightly above θ, say $\theta + \varepsilon$ and a type slightly below θ, say $\theta - \varepsilon$. Because of lemmas 3 and 5, all incentive constraints relaxed by increasing $q(\theta + \varepsilon)$ are also relaxed by increasing $q(\theta - \varepsilon)$. The incentive constraint binding from θ is relaxed by increasing $q(\theta - \varepsilon)$ but not by increasing the decision of type $\theta + \varepsilon$. Hence, the shadow value of non-local incentive constraints has to be higher at $\theta - \varepsilon$ than at $\theta + \varepsilon$. This is in line with the third property of η given in theorem 1: η is non-increasing in a neighborhood of θ.

From this intuition, it seems that η should be strictly lower for types in $(\theta, \theta + \varepsilon)$ than for types in $(\theta - \varepsilon, \theta)$ if (NLIC) binds from θ (in contrast to η being non-increasing in an open neighbohood of θ). Following the usual interpretation of shadow values, one can distinguish between types θ where merely $\Phi(\theta, \hat{\theta}) = 0$ and types from which a non-local incentive constraint binds in the sense that it affects the solution: I will say that (NLIC) binds from θ if $\Phi(\theta, \hat{\theta}) = 0$ for some $\hat{\theta} \neq \theta$ and either η is strictly higher for types in $(\theta - \varepsilon, \theta)$ than for types in $(\theta, \theta + \varepsilon)$ for some $\varepsilon > 0$, or $\theta = \bar{\theta}$. If η is constant (but not necessarily 0!) on an open interval of types, I will say that (NLIC) is slack for the types in this interval.

The shadow value interpretation also provides some intuition for the necessary condition (C3) which states that $\eta(\theta) = \eta(\hat{\theta})$ when (NLIC) binds from θ to $\hat{\theta}$. This makes sense in light of lemma 5. Because there is no overlap in binding incentive constraints, the non-local incentive constraints relaxed by increasing $q(\theta)$ are the same as the ones relaxed by increasing $q(\hat{\theta})$. Consequently, the shadow value of relaxing those constraints is the same for the two types.

Also the two properties for the boundary types are straightforward: Increasing the boundary types' decision does not affect non-local incentive constraints of other types.

Some of the properties of η in theorem 1 hold only at types where the decision is strictly increasing. The reason is that, the way (4) is written, η captures not only the effect of non-local incentive constraints but also the effect of the (local) monotonicity con-

straint. If one wants to avoid this cluttering of effects, it is straightforward to explicitly introduce the monotonicity constraint. This is done in section B.3 of the webappendix.

3.3. Distortion at the top

The idea underlying distortion in models with SC is that downward distortion at a certain type allows to reduce the rent of all higher types. As one moves across types towards $\bar{\theta}$, there are less and less higher types and therefore the distortion vanishes. Eventually, there is no distortion for $\bar{\theta}$. This is the famous "no distortion at the top" result.

Now consider the case where the non-local incentive constraint binds from $\bar{\theta}$ to a mass of types $\hat{\theta}$ (or to $\underline{\theta}$). Then the shadow value $\eta(\theta)$ will be strictly positive and bounded away from 0 for types slightly below $\bar{\theta}$. Hence, these types have a decision $q(\theta)$ which is at least ε above their relaxed decision $q^r(\theta)$ for some $\varepsilon > 0$. The same has then to apply for $\bar{\theta}$ because of the monotonicity constraint. Put differently, $\eta(\bar{\theta}) > 0$ and therefore $q(\bar{\theta}) > q^{fb}(\bar{\theta})$: There is distortion at the top.

The main difference to the "no distortion at the top" intuition is the following: The effect of distortion on the binding non-local incentive constraint does not necessarily vanish as we move across types towards $\bar{\theta}$. Increasing the decision of any type below $\bar{\theta}$ will – through the slope of the rent function – increase the rent of $\bar{\theta}$ and therefore relax the binding non-local incentive constraints. Note that the distortion created by (NLIC) is upwards, i.e. there is no downward distortion at the top type as $q \geq q^r$ (theorem 1).

A simple sufficient condition for "no distortion at the top" can be formulated using (C1).

$$\int_{q^r(\underline{\theta})}^{q^{fb}(\bar{\theta})} c_{q\theta}(q, \bar{\theta})\, dq \leq 0 \tag{5}$$

is sufficient for $q(\bar{\theta}) = q^{fb}(\bar{\theta})$: The reason is that (C1) cannot hold with inequality in this case as $q(\bar{\theta}) \geq q^r(\bar{\theta}) = q^{fb}(\bar{\theta})$ and $q(\hat{\theta}) \geq q^r(\hat{\theta}) \geq q^r(\underline{\theta})$ by theorem 1.

4. Sufficient conditions for monotone contract menus

The previous section focused on distortions in monotone solutions and derived necessary conditions. This section gives sufficient conditions under which the solution to problem (SB) is monotone and continuous. These conditions have to be checked before using the methods of section 5 to calculate a solution.

4.1. Existence and monotonicity

As already indicated, solutions can be monotone or inversely U-shaped or even jumping over $s(\theta)$ discontinuously. It is therefore useful to have a sufficient condition under which the solution is monotone. The assumptions in section 2 alone are not sufficient because they cannot rule out that q discontinuously jumps across s. One technical condition has to be added to rule this case out. Given this, assumption 2 ensures that the solution is not inversely U-shaped but monotone.

To state this technical condition two associated decisions have to be defined for each decision $y \in [0, s(\theta)]$: First, $q^s(y,\theta) \geq s(\theta)$ is defined by $c_\theta(y,\theta) = c_\theta(q^s(y,\theta),\theta)$; i.e. $q^s(y,\theta)$ is the decision that would lead to the same slope of the rent function $\pi(\theta)$ as y at type θ. Second, $q^v(y,\theta) \geq s(\theta)$ is the decision which gives the same virtual valuation as y, i.e. $[u(y,\theta) - c(y,\theta)]f(\theta) + (1 - F(\theta))c_\theta(y,\theta) = [u(q^v(y,\theta),\theta) - c(q^v(y,\theta),\theta)]f(\theta) + (1 - F(\theta))c_\theta(q^v(y,\theta),\theta)$. Since $c_\theta(q,\theta)$ and (RP) are strictly concave, the two associated decisions q^s and q^v are well defined. By the strict concavity of c_θ and (RP), the condition below ensures that the principal's objective is higher if he assigns $q^s(y,\theta) > s(\theta)$ instead of $y < s(\theta)$ to type θ.

Proposition 1. *Assume $q^v(y,\theta) \geq q^s(y,\theta)$ for all $y \in [0, s(\theta)]$ and all $\theta \in \Theta$. Then a solution $q \geq s$ to problem (SB) exists and q is monotonically increasing.*

Note that the imposed condition is automatically satisfied for y close to $s(\theta)$ as $q^r(\theta) > s(\theta)$ by lemma 2. To illustrate, take the example from section 1.1 and assume that $u(q,\theta) = Sq$ with $S > 0$. Straightforward calculation shows that in this case $q^v(y,\theta) = 2q^r(\theta) - y$ and $q^s(y,\theta) = \theta^2 - y = 2s(\theta) - y$. Hence, $q^v(y,\theta) > q^s(y,\theta)$ by lemma 2 and – regardless of the type distribution – the conditions of the proposition are satisfied.

4.2. Continuity

This subsection gives two sufficient conditions for continuity of the solution to problem (SB). The first condition is loosely based on the idea of having a one-to-one relationship between η and q for a given type θ; i.e. the idea that for a given type θ and $\eta(\theta) \geq 0$, equation (4) yields a unique solution for $q(\theta)$. The condition in the proposition ensures this and also ascertains that this relationship is monotonic, i.e. a higher $\eta(\theta)$ is associated with a higher $q(\theta)$.

Proposition 2. *A monotone solution q to problem (SB) is continuous if*

$$\frac{u_{qq}(x,\theta) - c_{qq}(x,\theta)}{c_{qq\theta}(x,\theta)} > \frac{u_q(x,\theta) - c_q(x,\theta)}{c_{q\theta}(x,\theta)} \tag{CVR}$$

holds for all types and all $x \geq q^{fb}(\theta)$.[13]

To illustrate, take the cost function in the example in section 1.1 and assume that $u(q,\theta) = Sq$. It turns out that (CVR) is equivalent to the condition for $q^{fb}(\theta) > s(\theta)$, i.e. $S > 2\bar{\theta}$.[14]

The following proposition gives an alternative condition under which the optimal solution is below the first best decision. Having a solution below first best turns out to be sufficient for continuity and strict monotonicity of the solution. This is in itself remarkable. As the relaxed solution is below first best, one should expect the solution to be below first best whenever non-local incentive constraints are not violated "too much" by the relaxed solution. Hence, there is a broad class of problems in which the solution will be strictly monotone and continuous. Furthermore, the proof of the following proposition shows that the property holds also locally. That is, if the decision is below q^{fb} on some interval (θ_1, θ_2), then the decision will be strictly increasing and continuous on (θ_1, θ_2).

Before stating the proposition some additional notation is needed. Define $q^m(\theta) < s(\theta)$ such that $c_\theta(q^{fb}(\theta),\theta) = c_\theta(q^m(\theta),\theta)$, i.e. $q^{fb}(\theta) = q^s(q^m(\theta),\theta)$. Hence, q^m is the decision function below the sign switch decision s which would lead to the same slope of the rent function as q^{fb}. If $c_\theta(q^{fb}(\theta),\theta) < c_\theta(q,\theta)$ for all $q \in \mathbb{R}_+$, let $q^m(\theta)$ be 0. By $c_{qq\theta} < 0$, q^m is uniquely defined.

Proposition 3. *Let q be an increasing solution to problem (SB) and let θ be a type such that $q^+(\theta) \leq q^{fb}(\theta)$. Then q is continuous at θ. q is strictly increasing on every interval (θ_1, θ_2) such that $q(\theta) \leq q^{fb}(\theta)$ for all $\theta \in (\theta_1, \theta_2)$. Assume that q^m is non-decreasing and that there is no distortion at the top.*[15] *Then, the optimal solution is below first best and therefore continuous and strictly increasing.*

[13] The condition has actually only to hold for $x \in [q^{fb}(\theta), \bar{q}]$ where $\bar{q}$ is defined as in the webappendix (section B.1). Also note that (CVR) is automatically satisfied for $x \in (s(\theta), q^{fb}(\theta))$ because the left hand side of (CVR) is positive and the right hand side is negative for these x.

[14] In fact, this also holds true if q^2 in the cost function is replaced by any increasing and convex function.

[15] Recall that (5) is a sufficient condition for the latter.

One example for a class of function where q^m is increasing are cost functions of the form $c(q,\theta) = \theta q + \phi(q - \alpha\theta) + \gamma(\theta)$ where $\phi(\cdot)$ is a function of which the first three derivatives are positive. The interpretation of this cost function is that there is a "normal scale" of $\alpha\theta$. Producing above this normal scale is increasingly costly. Type reflects a trade-off between the size of the normal scale and marginal cost when producing within the normal scale. With such a cost function, any increasing and concave benefit function u with $u_{q\theta} = 0$ and $q^{fb} > s$ yields an increasing q^m.

5. Continuous and strictly monotone solutions

This section introduces an additional necessary condition that a continuous solution satisfies. It is then illustrated that this condition can be used to calculate the solution of problem (SB). Put differently, the necessary conditions derived in this paper are sufficient under certain assumptions.

Throughout this section it is assumed that either (CVR) or the conditions of proposition 3, i.e. q^m increasing and no distortion at the top, hold. These conditions imply that, for a given type, (4) defines a one-to-one relationship between η and q. This property is repeatedly used when proving the following results. It is also assumed that problem (SB) has a strictly increasing solution.[16] The problem has an increasing solution if the conditions in propositions 1 are satisfied. *Strict* monotonicity does not pose an additional restriction if the assumptions of proposition 3 are satisfied. In case (CVR) is satisfied, I will assume additionally that $q(\theta,\eta)$ is strictly increasing in type for $\eta \geq 0$ where $q(\theta,\eta)$ is with a slight abuse of notation defined as the q solving[17]

$$\{u_q(q,\theta) - c_q(q,\theta)\}f(\theta) + (1 - F(\theta) - \eta)c_{q\theta}(q,\theta) = 0.$$

This can be shown to be sufficient for strict monotonicity of q in case (CVR) holds; see lemma WA5 in the webappendix. These sufficient conditions should be checked before using the algorithm below to determine the solution of a specific problem.

[16] The methods introduced here can sometimes also be used in cases where the monotonicity constraint is binding; see the webappendix for an example.

[17] By lemma WA2, the optimal decision is bounded by $\bar{q}$. Under (CVR) this implies that η is bounded by some η^{max} and only $\eta \leq \eta^{max}$ will be relevant in the following. Under (CVR) $q(\theta,\eta)$ is uniquely defined for all $\eta \leq \eta^{max}$. Under the assumptions of proposition 3 only $\eta \leq 1 - F(\theta)$ will be relevant for θ. For these η (and given θ), $q(\theta,\eta) \in (s(\theta), q^{fb}(\theta)]$ is again uniquely defined.

The rough idea behind the theorem is the following. Take the optimal contract q as given and say (NLIC) binds from θ' to $\hat{\theta}'$. Then $\Phi(\theta', \hat{\theta}') = 0$ and $(\theta', \hat{\theta}')$ minimize Φ which implies $\Phi_\theta(\theta, \hat{\theta}') \leq 0$ for $\theta \in (\theta' - \varepsilon, \theta')$ and $\Phi_\theta(\theta, \hat{\theta}') \geq 0$ for $\theta \in (\theta', \theta' + \varepsilon)$ for some $\varepsilon > 0$. Let both θ' and $\hat{\theta}'$ have the same η, call it η', which is implied by (C3) if both types are interior. I will now argue that $(\theta', \hat{\theta}')$ still minimizes Φ if the optimal decision q is manipulated in the following way: According to theorem 1, η is non-increasing in an open neighborhood of θ'. Now manipulate q such that η defined by (4) is constant (and equal to η') in some neighborhood of θ'. This implies that the decision of types slightly below θ' is decreased and the decision of types slightly above θ' is increased. This manipulation decreases $\Phi_\theta(\theta, \hat{\theta}')$ for $\theta \in (\theta' - \varepsilon, \theta')$, increases $\Phi_\theta(\theta, \hat{\theta}')$ for types $\theta \in (\theta', \theta' + \varepsilon)$ and leaves $\Phi_\theta(\theta, \hat{\theta}')$ zero for $\theta = \theta'$. Consequently, $(\theta', \hat{\theta}')$ is a local minimizer of this manipulated Φ. The following theorem shows that this holds still true if we manipulate q not only in a neighborhood of θ but for all types, namely if all types are assigned η'. The theorem, gives, for this case, even some non-local minimizer properties for $(\theta', \hat{\theta}')$. I use the following notation in the remainder of this section: $(\theta(\eta), \hat{\theta}(\eta))$ denotes the minimizer of $\Phi^\eta(\theta, \hat{\theta})$ which is defined as $\Phi(\theta, \hat{\theta})$ under $q(\theta, \eta)$:

$$\Phi^\eta(\theta, \hat{\theta}) = -\int_{\hat{\theta}}^{\theta} \int_{q(\hat{\theta},\eta)}^{q(x,\eta)} c_{q\theta}(y, x)\, dy\, dx.$$

Theorem 2. *Let q solve problem (SB). Assume that (NLIC) binds from θ' to $\hat{\theta}'$. Then, $(\theta', \hat{\theta}')$ minimize $\Phi^{\eta'}(\theta, \hat{\theta})$ on $[\hat{\theta}' - \varepsilon, \theta' + \varepsilon]$ for some $\varepsilon > 0$ where $\eta' = min(\eta(\theta'), \eta(\hat{\theta}'))$.*[18] *Furthermore, $\Phi^{\eta'}(\theta', \hat{\theta}') \leq 0$.*

The theorem says that types for which (NLIC) binds under the optimal decision are minimizers of their respective Φ^η. The inequality condition is useful as it can help to determine the maximum of $\eta(\theta)$, which I will call $\bar{\eta}$, in the solution: Because of lemma 5 and theorem 1, (NLIC) will not bind to or from types in $(\hat{\theta}(\bar{\eta}), \theta(\bar{\eta}))$ and η is therefore constant on $(\hat{\theta}(\bar{\eta}), \theta(\bar{\eta}))$. Therefore, $\Phi(\theta(\bar{\eta}), \hat{\theta}(\bar{\eta})) = \Phi^{\bar{\eta}}(\theta(\bar{\eta}), \hat{\theta}(\bar{\eta}))$ and incentive compatibility requires then that the inequality in the theorem holds with equality for $\eta' = \bar{\eta}$.

For a given η, Φ^η and its minimizers can be determined without knowing the optimal decision. Suppose there is – for each $\eta \geq 0$ – only one local minimizer of Φ^η and that

[18] If θ' and $\hat{\theta}'$ are interior, (C3) implies $\eta' = \eta(\theta') = \eta(\hat{\theta}')$.

the condition above determines $\bar{\eta}$. Then the solution can be constructed: For $\eta \in [0, \bar{\eta}]$, the optimal decision for the Φ^η minimizing types is their respective $q(\theta, \eta)$. The optimal decision for those types at which (NLIC) is slack is then determined by (4) as theorem 1 implies that η is constant on intervals of types for which (NLIC) is slack (and therefore determined by the closest type from/to which (NLIC) binds).

For the remainder of this section, I will make the assumption that Φ^η has only one local minimizer $(\theta(\eta), \hat{\theta}(\eta))$ for all $\eta \geq 0$. The case of several minimizers is briefly discussed at the end of this section and more extensively in the webappendix. The idea above leads then to the following algorithm:

1. Solve (4) for $q(\theta, \eta)$ (assuming $\eta \geq 0$) and calculate $\Phi^\eta(\theta, \hat{\theta})$.

2. Minimize $\Phi^\eta(\theta, \hat{\theta})$ over θ and $\hat{\theta}$. The domain of the minimization is $[\underline{\theta}, \bar{\theta}]$ if (CVR) is satisfied. If the conditions of proposition 3 are satisfied, the domain is $[\underline{\theta}, \theta'(\eta)]$ where $\theta'(\eta)$ is defined by $1 - F(\theta') = \eta$.

3. Define $\bar{\eta}$ as the η solving $\Phi^\eta(\theta(\eta), \hat{\theta}(\eta)) = 0$.

4. The solution is then the following:

   (a) types from/to which non-local incentive constraints are binding: For all $\eta \in [0, \bar{\eta}]$, $q(\theta(\eta)) = q(\theta(\eta), \eta)$ and $q(\hat{\theta}(\eta)) = q(\hat{\theta}(\eta), \eta)$ (and therefore $\eta(\theta(\eta)) = \eta$ and $\eta(\hat{\theta}(\eta)) = \eta$).[19]

   (b) types with $\eta = 0$: For types above $\theta(0)$ and below $\hat{\theta}(0)$, the optimal decision is the relaxed decision.[20]

   (c) types where non-local incentive constraints are slack but $\eta > 0$: Let $\tilde{\Theta}$ be the set of all types that were assigned a decision in (a) or (b). For $\theta \notin \tilde{\Theta}$, define $\tilde{\theta}(\theta) = \min \tilde{\Theta}$ s.t.: $\tilde{\theta}(\theta) \geq \theta$. Then $\eta(\theta) = \eta(\tilde{\theta}(\theta))$ for $\theta \notin \tilde{\Theta}$. The optimal decision for these types is $q(\theta, \eta(\theta))$.

Proposition 4. *The outcome of the algorithm solves problem (SB).*

[19] The boundary types $\bar{\theta}$ and $\underline{\theta}$ can minimize Φ^η for several values of η. In this case, $q(\bar{\theta}) = q(\bar{\theta}, \eta')$ (and $q(\underline{\theta}) = q(\underline{\theta}, \eta')$ respectively) where η' is the highest η for which $\bar{\theta} = \theta(\eta)$ (and $\underline{\theta} = \hat{\theta}(\eta)$ respectively).

[20] This is not in conflict with (a) as $\theta(\eta)$ is increasing and $\hat{\theta}(\eta)$ is decreasing in η; see the proof of proposition 4.

I want to illustrate the algorithm with a numerical example. Amend the example of section 1.1 by assuming $c(q,\theta) = q\theta + \frac{1}{2}\frac{q^2}{\theta} - \frac{\theta}{10}$, $u(q,\theta) = \frac{21}{10}q\theta$ and let types be distributed uniformly between 1/4 and 1/2.

Hence, $s(\theta) = \theta^2$ and the first best decision is $q^{fb}(\theta) = \frac{11}{10}\theta^2$. The relaxed solution is $q^r(\theta) = \theta^2 + \frac{1}{5}\theta^3$. The example is illustrated in figure 6. As the first best decision is close to s for all types, one would expect distortion above first best and (NLIC) to bind from the top to the bottom type. Indeed this is shown below. According to proposition 1, the solution is monotone because $q^v(q,\theta) = -q + 2\theta^2 + \frac{2}{5}\theta^3 > -q + 2\theta^2 = q^s(q,\theta)$.

Also (CVR) is satisfied as it boils down to

$$\frac{\frac{-1}{\theta}}{\frac{-1}{\theta^2}} > \frac{\frac{11}{10}\theta - \frac{q}{\theta}}{1 - \frac{q}{\theta^2}} \qquad \Leftrightarrow \qquad 0 > \frac{\frac{1}{10}\theta^2}{\theta^2 - q}$$

which holds for all $q > s(\theta) = \theta^2$. Hence, the solution is continuous. Next use the algorithm to compute a solution:

1. Solving (4) for q yields

$$q(\theta,\eta) = \theta^2 + \frac{2\theta^3}{5(2-\eta)}$$

which is strictly increasing for $\eta < 2$. Computing Φ^η is slightly cumbersome but possible:

$$\begin{aligned}\Phi^\eta(\theta,\hat\theta) &= \int_{\hat\theta}^{\theta} \Big(c_\theta(q(\hat\theta,\eta),x) - c_\theta(q(x,\eta),x)\Big)\, dx \\ &= \hat\theta^2\left(\theta - \frac{4}{3}\hat\theta\right) + \frac{2\hat\theta^3(\theta - 2\hat\theta)}{5(2-\eta)} + \frac{\hat\theta^4 + \frac{4\hat\theta^5}{5(2-\eta)} + \frac{4\hat\theta^6}{25(2-\eta)^2}}{2\theta} - \frac{\theta^3}{6} + \frac{2\theta^5 - 12\hat\theta^5}{125(2-\eta)^2}\end{aligned}$$

2. The first order conditions for minimization, i.e. the equivalents of (C1) and (C2), are

$$\hat\theta^2 + \frac{2\hat\theta^3}{5(2-\eta)} - \frac{1}{2\theta^2}\left(\hat\theta^4 + \frac{4\hat\theta^5}{5(2-\eta)} + \frac{4\hat\theta^6}{25(2-\eta)^2}\right) \le \frac{\theta^2}{2} - \frac{2\theta^4}{25(2-\eta)^2} \quad \text{with "=" if } \theta < \bar\theta \tag{6}$$

$$\theta + \frac{1}{\theta}\left(\hat\theta^2 + \frac{2\hat\theta^3}{5(2-\eta)}\right) \ge 2\hat\theta + \frac{2\hat\theta^2}{5(2-\eta)} \quad \text{with "=" if } \hat\theta > \underline{\theta}. \tag{7}$$

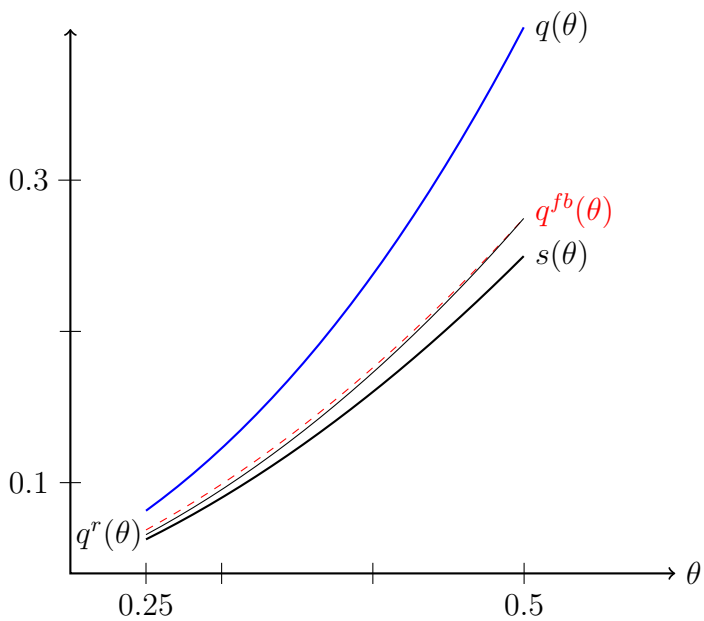


Figure 6: numerical example: all types are distorted

It is easy to verify that the first order conditions are satisfied for the boundary types $\bar{\theta} = 1/2$ and $\underline{\theta} = 1/4$ for all $\eta \leq 1.7$. It is a bit more tedious to verify that $(\bar{\theta}, \underline{\theta})$ is the unique local minimizer (for $\eta \leq 1.7$), i.e. that there are no other local minimizers. This is done in the webappendix. For $\eta > 1.7$, minimizers are interior. As those minimizers are not needed in the following, I do not derive them here.

3. For $\bar{\eta} \approx 1.67$, $\Phi^{\bar{\eta}}(\bar{\theta}, \underline{\theta}) = 0$. Hence, (NLIC) will bind only from $\bar{\theta}$ to $\underline{\theta}$.

4. For all types, $\eta(\theta) = 1.67$. Hence,

$$q(\theta) = \theta^2 + \frac{2\theta^3}{5(2 - 1.67)}.$$

Since $\eta(\theta) > 1 \geq 1 - F(\theta)$ for all θ, all types have decisions above first best in the example. This result differs from results in the literature on type dependent participation constraints, see for example Jullien (2000). In this literature, the decision can also be distorted above as well as below first best but there is always at least one type with an undistorted decision. Also in the inverse U-shape solutions of Araujo and Moreira (2010) there is always one type with an undistorted decision.

Theorem 3. *The solution q of problem (SB) can be such that $q(\theta) > q^{fb}(\theta)$ for all $\theta \in \Theta$.*

It was assumed that for all η there was a unique local minimizer of Φ^η. If instead there are several Φ^η minimizers $(\theta_i(\eta), \hat{\theta}_i(\eta))$ for a given η, theorem 2 alone does not necessarily determine for which of these minimizers (NLIC) binds. A variation of the algorithm can still be used if it can be determined for which minimizers (NLIC) binds and for which (NLIC) does not bind. While this is a hard problem in general, similar techniques as in the proof of theorem 2, i.e. utilizing the one-to-one relationship between q and η and incentive compatibility, can be used in special cases. The supplementary material to this paper contains several such results for specific configurations of Φ^η minimizers, e.g. nested minimizers and overlapping minimizers.

6. Conclusion

The paper characterizes monotone solutions in a screening environment where SC is violated. This implies that non-local incentive constraints can be binding. The distortion caused by non-locally binding incentive constraints can counteract the normal rent extraction distortion. The solution can therefore be partly above as well as below the first best decision. If a non-local incentive constraint is binding from the highest type to the lowest type, the decision of all types can be upward distorted.

To conclude, I discuss relaxing the assumptions, possible generalizations and applications. First, note that for the results in section 3 the assumptions on $u_{q\theta}$, u_{qq}, c_{qq} and the monotone hazard rate assumption in assumption 1 can be relaxed as long as the relaxed program remains concave and the relaxed decision remains strictly increasing, i.e. as long as the properties in lemma 2 hold. These properties of the relaxed program are used for showing theorem 1. The assumption that $u_{qq} - c_{qq} \leq 0$ and the monotone hazard rate assumption are only used in the proof of proposition 3. The monotonicity properties of η in theorem 1 hold more generally. The only requirement is that the changed decisions in the proof (as for example indicated in figure 5) are implementable. Implementability follows from lemmas 3– 5 which only require that the third derivatives $c_{qq\theta}$ and $c_{q\theta\theta}$ do not change sign.

This leads to the question how important the assumptions on the third derivatives of c are. It is not important which signs these third derivatives have as long as they do not switch sign. This is illustrated with an example in the webappendix. However, the

assumption that they do not switch sign is necessary to make the analysis tractable. For illustration, suppose that the third derivatives of c can switch sign in the relevant range of q and θ. In this case, there could be a second function $s_2(\theta)$ where $c_{q\theta}$ changes sign. Consider figure 3a and imagine such an s_2 function somewhere above q and going through the shaded area. It is clear that in this case upwards binding incentive constraints can no longer be ruled out, i.e. lemma 3 no longer holds (also lemma 5 is no longer valid). Nevertheless, theorem 1 will not become irrelevant in such a framework: As long as the changed decisions (as in figure 5) are implementable, theorem 1 holds. In particular, η has to be constant on an interval of types for which all non-local incentive constraints are slack. Similarly, η has to be non-increasing in a neighborhood of a type θ satisfying (i) $\Phi(\theta, \hat{\theta}) = 0$ for some $\hat{\theta} < \theta$ and (ii) $\Phi(\theta', \theta) > 0$ for all $\theta' \neq \theta$.

Possible applications can be found in various fields of economics. While the paper uses the notation of a regulation or procurement setting, the same model is applicable, for example, in models of labor, insurance or monopoly pricing. It is hard to judge how common, for example, distortions above first best are in these applications. However, there is some evidence from non-linear pricing. Telecommunication could be an example for this: Consumers often buy packages where an additional unit of calling (or internet use) is for free. If the marginal costs of the provider are only ε above zero, such a price scheme will lead to consumption above the socially optimal consumption.

The paper derives a new necessary condition that is based on minimizing the incentive compatibility constraint for a given shadow value. This condition is sufficient to calculate solutions that are continuous and strictly monotone. Future research will determine to what extent this condition can be generalized. It might be particularly interesting to see whether a similar procedure can be used in multidimensional screening models. More straightforward applications include areas in which violations of single crossing and binding non-local incentive constraints emerge naturally. This has been documented in common-agency games (Martimort and Stole, 2009), provision of pre-sale information (Hoffmann and Inderst, 2011) and dynamic mechanism design problems with serially correlated types (Battaglini and Lamba, 2012).

Appendix: Proofs

Proof of lemma 1 and 2: see webappendix

Proof of corollary 1: Suppose there was a type $\theta' > \underline{\theta}$ such that $q(\theta') < s(\theta')$. Monotonicity of q and continuity of s then imply $q(\theta) < s(\theta)$ for all $\theta \in (\theta' - \varepsilon, \theta')$ for some $\varepsilon > 0$. As q is increasing, q is continuous almost everywhere. Hence, there exists a $\theta'' \in (\theta' - \varepsilon, \theta')$ such that q is continuous at θ''. The monotonicity condition of lemma 1 is violated at θ'' contradicting that q is implementable. □

Proof of lemma 4: Let $\Phi(\theta, \hat{\theta}) = 0$. First, it is shown that there cannot be a discontinuity at $\hat{\theta}$ under the conditions of the lemma. By way of contradiction, suppose that q is discontinuous at $\hat{\theta}$, i.e. $q^-(\hat{\theta}) < q^+(\hat{\theta})$. As q is increasing, either (i) $q(\hat{\theta}) = q^-(\hat{\theta})$ or (ii) $q(\hat{\theta}) = q^+(\hat{\theta})$ or (iii) $q(\hat{\theta}) \in (q^-(\hat{\theta}), q^+(\hat{\theta}))$.

In case (i), it must hold that $\int_{\hat{\theta}}^{\theta} c_{q\theta}(q^-(\hat{\theta}), x)\, dx \leq 0$ which is just (C2) adapted to apply for a right hand side discontinuity; i.e. if this did not hold, $\Phi(\theta, \hat{\theta} - \varepsilon) < 0$ for $\varepsilon > 0$ small enough which violates (NLIC). But then $\int_{\hat{\theta}}^{\theta} \int_{q^-(\hat{\theta})}^{q^+(\hat{\theta})} c_{q\theta}(y, x)\, dy\, dx < 0$ from $c_{qq\theta} < 0$. Hence, $\Phi(\theta, \hat{\theta}) + \int_{\hat{\theta}}^{\theta} \int_{q^-(\hat{\theta})}^{q^+(\hat{\theta})} c_{q\theta}(y, x)\, dy\, dx < 0$ as $\Phi(\theta, \hat{\theta}) = 0$ by assumption. This implies $\Phi(\theta, \hat{\theta} + \varepsilon) < 0$ for $\varepsilon > 0$ small enough as $\lim_{\varepsilon \to 0^+} \Phi(\theta, \hat{\theta} + \varepsilon) = \Phi(\theta, \hat{\theta}) + \int_{\hat{\theta}}^{\theta} \int_{q^-(\hat{\theta})}^{q^+(\hat{\theta})} c_{q\theta}(y, x)\, dy\, dx$, i.e. incentive compatibility is violated from θ to types slightly above $\hat{\theta}$. This is the desired contradiction.

In case (ii), it must hold that $\int_{\hat{\theta}}^{\theta} c_{q\theta}(q^+(\hat{\theta}), x)\, dx \geq 0$. But then $\int_{\hat{\theta}}^{\theta} \int_{q^-(\hat{\theta})}^{q^+(\hat{\theta})} c_{q\theta}(y, x)\, dy\, dx > 0$ from $c_{qq\theta} < 0$. Consequently, $\lim_{\varepsilon \to 0^+} \Phi(\theta, \hat{\theta} - \varepsilon) = \Phi(\theta, \hat{\theta}) - \int_{\hat{\theta}}^{\theta} \int_{q^-(\hat{\theta})}^{q^+(\hat{\theta})} c_{q\theta}(y, x)\, dy\, dx < 0$ and therefore incentive compatibility is violated from θ to types slightly below $\hat{\theta}$.

In case (iii), the same arguments as in case (i) apply if $\int_{\hat{\theta}}^{\theta} c_{q\theta}(q(\hat{\theta}), x)\, dx \leq 0$ while the same arguments as in case (ii) apply if $\int_{\hat{\theta}}^{\theta} c_{q\theta}(q(\hat{\theta}), x)\, dx > 0$.

Second, it is shown that $\theta < \bar{\theta}$ cannot be assigned the same decision as a type θ' if q is continuous at θ. Suppose, by way of contradiction, $q(\theta) = q(\theta')$ for some type $\theta' \neq \theta$. Note that, since q is increasing, $q(x) = q(\theta) = q(\theta')$ for all $x \in [\min\{\theta, \theta'\}, \max\{\theta, \theta'\}]$. First, let $\theta < \theta'$. But then $\Phi(\theta', \hat{\theta}) < 0$ and (NLIC) is violated as

$$\begin{aligned}
\Phi(\theta', \hat{\theta}) &= -\int_{\hat{\theta}}^{\theta'} \int_{q(\hat{\theta})}^{q(x)} c_{q\theta}(y, x)\, dy\, dx \\
&= \Phi(\theta, \hat{\theta}) - \int_{\theta}^{\theta'} \int_{q(\hat{\theta})}^{q(\theta)} c_{q\theta}(y, x)\, dy\, dx < 0
\end{aligned}$$

where the last inequality follows from (C1) and $c_{q\theta\theta} > 0$.

Next, let $\theta > \theta'$. From condition (C1) and $c_{q\theta\theta} > 0$, it follows that $\int_{q(\hat{\theta})}^{q(\theta)} c_{q\theta}(y,x)\,dy < 0$ for any $x \in (\theta', \theta)$. Hence, $\Phi(\theta', \hat{\theta}) = \Phi(\theta, \hat{\theta}) + \int_{\theta'}^{\theta} \int_{q(\hat{\theta})}^{q(\theta)} c_{q\theta}(y,x)\,dy\,dx < 0$ and (NLIC) is violated. □

Proof of theorem 1: To show that $q \geq q^r$ suppose, by way of contradiction, that $q(\theta') < q^r(\theta')$ for some $\theta' \in \Theta$. The set $\{\theta : q(\theta) < q^r(\theta)\}$ must then have positive Lebesgue measure because q is increasing and cadlag and q^r is continuous. Consider the decision function $q^*(\theta)$ where $q^*(\theta) = max\{q(\theta), q^r(\theta)\}$.

q^* achieves a strictly higher objective value than q in problem (SB) because $q^r(\theta)$ is defined as the pointwise maximizer of the objective. The improvement is strict as q and q^* differ on a set of positive measure and the objective is strictly concave in q (see lemma 2). This will contradict the optimality of q if q^* also satisfies (NLIC).

Since q and q^r are both increasing, q^* is also increasing. By lemma 3, q^* is incentive compatible if

$$\Phi^*(\theta, \hat{\theta}) = -\int_{\hat{\theta}}^{\theta} \int_{q^*(\hat{\theta})}^{q^*(x)} c_{q\theta}(y,x)\,dy\,dx \geq 0$$

for arbitrary types θ and $\hat{\theta} < \theta$. If $q^*(\hat{\theta}) = q(\hat{\theta})$, $q^*(x) \geq q(x)$ implies $\Phi^*(\theta, \hat{\theta}) \geq \Phi(\theta, \hat{\theta}) \geq 0$: As $q(x) \geq s(x)$ for all $x \in (\hat{\theta}, \theta)$, $\Phi^*(\theta, \hat{\theta}) - \Phi(\theta, \hat{\theta}) = -\int_{\hat{\theta}}^{\theta} \int_{q(x)}^{q^*(x)} c_{q\theta}(y,t)\,dy\,dx \geq 0$.

If $q^*(\hat{\theta}) > q(\hat{\theta})$ (and therefore $q^*(\hat{\theta}) = q^r(\hat{\theta})$), there are three possibilities: (i) There exists a type $\theta' \in (\hat{\theta}, \theta)$ with $q(\theta') = q^*(\hat{\theta})$, (ii) all types $\theta' \in (\hat{\theta}, \theta)$ have $q(\theta') < q^*(\hat{\theta})$ and (iii) there are types $\theta' \in (\hat{\theta}, \theta)$ with $q(\theta') > q^*(\hat{\theta})$ but no type θ' with $q(\theta') = q^*(\hat{\theta})$, hence q is discontinuous on $[\hat{\theta}, \theta)$.

If (i), then $\Phi(\theta, \theta') \geq 0$ implies incentive compatibility as $\Phi^*(\theta, \hat{\theta}) - \Phi(\theta, \theta') = \int_{\hat{\theta}}^{\theta'} \int_{q(\theta')}^{q^*(x)} -c_{q\theta}(y,t)\;dy\,dx + \int_{\theta'}^{\theta} \int_{q(x)}^{q^*(x)} -c_{q\theta}(y,t)\;dy\,dx \geq 0$ where the first double integral is positive because of $s(x) < s(\theta') \leq q(\theta') = q^*(\hat{\theta}) \leq q^*(x)$ for all $x \in (\hat{\theta}, \theta')$ and the second one is positive by $q^*(x) \geq q(x) \geq s(x)$ for all $x \in (\theta', \theta)$.

In case (ii), $q^*(\theta') \geq q^*(\hat{\theta})$ for all $\theta' \in (\hat{\theta}, \theta)$ since q^* is increasing. The inequalities $q^*(\theta') \geq q^*(\hat{\theta}) > q(\theta') \geq s(\theta')$ imply then incentive compatibility as the integrand in Φ^* is non-positive for all arguments.

In case (iii), define $\theta' = sup\{x \in [\hat{\theta}, \theta) : \; q(x) < q^*(\hat{\theta})\}$. Note that q is discontinuous at θ'. Incentive compatibility between θ and θ' under q implies $\int_{\theta'}^{\theta} \int_{q^-(\theta')}^{q(x)} c_{q\theta}(y,x)\;dy\,dx \leq 0$ as well as $\int_{\theta'}^{\theta} \int_{q^+(\theta')}^{q(x)} c_{q\theta}(y,x)\;dy\,dx \leq 0$. From $c_{qq\theta} < 0$ and $q^-(\theta') \leq q^*(\hat{\theta}) \leq q^+(\theta')$,

it follows that $\int_{\theta'}^{\theta}\int_{q^*(\hat{\theta})}^{q(x)} c_{q\theta}(y,x)\ dy\, dx \leq 0$. Incentive compatibility is satisfied because $q^*(x) \geq q^*(\hat{\theta})$ for all $x \in (\hat{\theta}, \theta')$ implies $\Phi^*(\theta, \hat{\theta}) \geq -\int_{\theta'}^{\theta}\int_{q^*(\hat{\theta})}^{q(x)} c_{q\theta}(y,x)\ dy\, dx \geq 0$. Hence, $q \geq q^r$.

Next we consider the properties of η. Define η by equation (4). As q is monotonic, it is differentiable almost everywhere and continuous almost everywhere. These two properties as well as the càdlàg property are then also inherited by η. It is shown in the supplementary material that q is bounded which is then also true for η. As the left hand side of (4) is decreasing in q (lemma 2) and $c_{q\theta}(q(\theta), \theta) < 0$ by $q \geq q^r$, $\eta(\theta) \geq 0$ follows from $q \geq q^r$.

To turn to the first bullet point, let q be strictly increasing on an interval (θ_1, θ_2) and assume that $\Phi(\theta', \hat{\theta}') > 0$ for every $\hat{\theta}' \in (\theta_1, \theta_2)$ and every $\theta' \in \Theta \setminus \{\hat{\theta}'\}$. Suppose, contrary to the theorem, that there exist types θ' and θ'' both in (θ_1, θ_2) with $\theta'' > \theta'$ and $\eta(\theta'') > \eta(\theta')$. Then, by the right continuity of η, there exists an $\varepsilon > 0$ and an $\tilde{\eta}$ such that $\eta(\theta) > \tilde{\eta}$ for all $\theta \in [\theta'', \theta'' + \varepsilon]$ and $\eta(\theta) < \tilde{\eta}$ for all $\theta \in [\theta', \theta' + \varepsilon]$ as well as $\theta'' + \varepsilon < \theta_2$. Consider changing the decision to

$$q^c(\theta, \varepsilon') = \begin{cases} \varepsilon' q(\theta' + \varepsilon) + (1 - \varepsilon') q(\theta) & \text{for } \theta \in [\theta', \theta' + \varepsilon] \\ \delta(\varepsilon') q(\theta'') + (1 - \delta(\varepsilon')) q(\theta) & \text{for } \theta \in [\theta'', \theta'' + \varepsilon) \\ q(\theta) & \text{else} \end{cases}$$

where $\delta(\varepsilon') : \mathbb{R}_+ \to \mathbb{R}_+$ is chosen such that

$$\int_{\theta'}^{\theta'+\varepsilon}\int_{q(x)}^{q^c(x,\varepsilon')} c_{q\theta}(y,x)\ dy\ dx = \int_{\theta''}^{\theta''+\varepsilon}\int_{q^c(x,\varepsilon')}^{q(x)} c_{q\theta}(y,x)\ dy\ dx. \tag{8}$$

For $\varepsilon' \geq 0$ small enough, this equation defines δ uniquely as $q(\theta) \geq q^r(\theta) > s(\theta)$ by lemma 2 which implies that both integrands are negative for all arguments. Therefore, the left hand side is zero for $\varepsilon' = 0$ and strictly decreasing in ε' by the strict monotonicity of q on (θ_1, θ_2). The right hand side is – through q^c – strictly decreasing in δ and zero for $\delta = 0$. Note that $\delta(\varepsilon')$ is, by its definition (8), differentiable and its derivative is

$$\delta'(\varepsilon') = \frac{\int_{\theta'}^{\theta'+\varepsilon} \left[c_{q\theta}(q^c(x,\varepsilon'), x)(q(\theta' + \varepsilon) - q(x))\right]\ dx}{\int_{\theta''}^{\theta''+\varepsilon} \left[c_{q\theta}(q^c(x,\varepsilon'), x)(q(x) - q(\theta''))\right]\ dx} > 0.$$

The change in the principal's payoff caused by the change in the decision function is

$$\Delta(\varepsilon') = \int_{\underline{\theta}}^{\bar{\theta}} \int_{q(x)}^{q^c(x,\varepsilon')} ((u_q(y,x) - c_q(y,x))f(x) + (1 - F(x))c_{q\theta}(y,x))\, dy\, dx.$$

Note that the integrand in the last expression equals $\eta(x)c_{q\theta}(q(x),x)$ if $y = q(x)$ by the definition of η in (4). Now consider the derivative of Δ at $\varepsilon' = 0$:

$$\begin{aligned}\Delta'(0) =& \int_{\theta'}^{\theta'+\varepsilon} (\eta(x)c_{q\theta}(q(x),x)(q(\theta'+\varepsilon) - q(x)))\, dx + \int_{\theta''}^{\theta''+\varepsilon} (\eta(x)c_{q\theta}(q(x),x)(q(\theta'') - q(x))\delta'(0))\, dx \\ >& \tilde{\eta}\left(\int_{\theta'}^{\theta'+\varepsilon} (c_{q\theta}(q(x),x)(q(\theta'+\varepsilon) - q(x)))\, dx + \delta'(0)\int_{\theta''}^{\theta''+\varepsilon} (c_{q\theta}(q(x),x)(q(\theta'') - q(x)))\, dx\right) \\ =& \tilde{\eta}\left(\int_{\theta'}^{\theta'+\varepsilon} (c_{q\theta}(q(x),x)(q(\theta'+\varepsilon) - q(x)))\, dx - \int_{\theta'}^{\theta'+\varepsilon} (c_{q\theta}(q(x),x)(q(\theta'+\varepsilon) - q(x)))\, dx\right) = 0\end{aligned}$$

where the inequality holds because $\tilde{\eta} > \eta(x)$ for $x \in [\theta', \theta' + \varepsilon]$ and $\tilde{\eta} < \eta(x)$ for $x \in [\theta'', \theta'' + \varepsilon]$ and because the integrand of the first (second) integral is negative (positive) for all x. As $\Delta(0) = 0$, the principal's payoff under $q^c(\cdot, \varepsilon')$ is higher than his payoff under q for $\varepsilon' > 0$ sufficiently small.

The changed decision is implementable: By lemma 3, upwards incentive constraints cannot be violated since the changed decision inherits monotonicity from the original decision. Incentive constraints from types $\theta \notin (\theta', \theta'' + \varepsilon)$ to types $\hat{\theta} \notin (\theta', \theta'' + \varepsilon)$ are – by the definition of $\delta(\varepsilon')$ – not affected by the change. Incentive constraints from types $\theta \in (\theta', \theta'' + \varepsilon)$ to types $\hat{\theta} < \theta$ are relaxed as $\Phi^c(\theta, \hat{\theta}) = \Phi(\theta, \hat{\theta}) - \int_{\theta'}^{\theta} \int_{q(x)}^{q^c(x)} c_{q\theta}(y,x)\, dy\, dx > \Phi(\theta, \hat{\theta})$ where Φ^c is Φ under the decision q^c; see figure 5.

By assumption, $\Phi(\theta, \hat{\theta}) > 0$ for all $\hat{\theta} \in (\theta', \theta'' + \varepsilon)$ and $\theta \neq \hat{\theta}$ under the original decision and this also holds true for the changed decision if $\varepsilon' > 0$ is sufficiently small: For ε' small enough, $q^c > s$ which implies that there exists an $\varepsilon'' > 0$ such that $\Phi^c(\theta, \hat{\theta}) > 0$ for all $\theta \in \Theta$ and $\hat{\theta} \in (\theta - \varepsilon'', \theta)$. The assumption $\Phi(\theta, \hat{\theta}) > 0$ for all $\hat{\theta} \in (\theta', \theta'' + \varepsilon)$ and the continuity of Φ in its first argument imply that $\min_{\theta \in [\hat{\theta} + \varepsilon'', \bar{\theta}], \hat{\theta} \in [\theta', \theta'' + \varepsilon]} \Phi(\theta, \hat{\theta}) > 0$. As Φ^c is continuous in ε' and $\Phi^c = \Phi$ for $\varepsilon' = 0$, (NLIC) is slack for $\hat{\theta} \in [\theta', \theta'' + \varepsilon]$ and $\theta > \hat{\theta}$ for $\varepsilon' > 0$ small enough. Therefore, q^c is implementable and increases the principal's payoff for $\varepsilon' > 0$ small enough which contradicts the optimality of q. Hence, η has to be non-increasing on (θ_1, θ_2).

The second statement is proven similarly. Let q be strictly increasing on (θ_1, θ_2) and

let $\Phi(\theta', \hat{\theta}') > 0$ for every $\theta' \in (\theta_1, \theta_2)$ and every $\hat{\theta}' \in \Theta \setminus \{\theta'\}$. Suppose, contrary to the theorem, that there were $\theta_1 < \theta' < \theta'' < \theta_2$ such that $\eta(\theta') > \eta(\theta'')$. Then there exists an $\varepsilon > 0$ and $\tilde{\eta}$ such that $\eta(\theta) > \tilde{\eta}$ for $\theta \in [\theta', \theta' + \varepsilon]$ and $\eta(\theta) < \tilde{\eta}$ for $\theta \in [\theta'', \theta'' + \varepsilon]$. Consider the changed decision

$$q^c(\theta, \varepsilon') = \begin{cases} \varepsilon' q(\theta') + (1 - \varepsilon') q(\theta) & \text{for } \theta \in [\theta', \theta' + \varepsilon) \\ \delta(\varepsilon') q(\theta'' + \varepsilon) + (1 - \delta(\varepsilon')) q(\theta) & \text{for } \theta \in [\theta'', \theta'' + \varepsilon) \\ q(\theta) & \text{else} \end{cases}$$

where $\delta(\varepsilon') : \mathbb{R}_+ \to \mathbb{R}_+$ is chosen such that (8) holds. This yields

$$\delta'(\varepsilon') = \frac{\int_{\theta'}^{\theta'+\varepsilon} [c_{q\theta}(q^c(x, \varepsilon'), x)(q(\theta') - q(x))] \; dx}{\int_{\theta''}^{\theta''+\varepsilon} [c_{q\theta}(q^c(x, \varepsilon'), x)(q(x) - q(\theta'' + \varepsilon))] \; dx} > 0.$$

Defining $\Delta(\varepsilon')$ as above, its derivative at $\varepsilon' = 0$ is then

$$\begin{aligned} \Delta'(0) = & \int_{\theta'}^{\theta'+\varepsilon} (\eta(x) c_{q\theta}(q(x), x)(q(\theta') - q(x))) \; dx + \int_{\theta''}^{\theta''+\varepsilon} (\eta(x) c_{q\theta}(q(x), x)(q(\theta'' + \varepsilon) - q(x)) \delta'(0)) \; dx \\ > & \tilde{\eta} \left(\int_{\theta'}^{\theta'+\varepsilon} (c_{q\theta}(q(x), x)(q(\theta') - q(x))) \; dx + \delta'(0) \int_{\theta''}^{\theta''+\varepsilon} (c_{q\theta}(q(x), x)(q(\theta'' + \varepsilon) - q(x))) \; dx \right) \\ = & 0. \end{aligned}$$

The changed decision satisfies (NLIC): Incentive constraints from types in (θ_1, θ_2) cannot be violated by assumption for $\varepsilon' > 0$ small enough. Note that for every $\hat{\theta} \in [\theta', \theta'' + \varepsilon]$ there exists a type $\tilde{\theta}(\hat{\theta}, \varepsilon') \in [\theta', \theta'' + \varepsilon]$ such that either $q^c(\hat{\theta}, \varepsilon') = q(\tilde{\theta}(\hat{\theta}))$ or $q^-(\tilde{\theta}(\hat{\theta})) \leq q^c(\hat{\theta}, \varepsilon') < q^+(\tilde{\theta}(\hat{\theta}))$. In the latter case, lemma 4 implies that $\Phi(\theta, \tilde{\theta}(\hat{\theta}, \varepsilon')) > 0$ and $\int_{\tilde{\theta}(\hat{\theta}, \varepsilon')}^{\theta} \int_{q^-(\tilde{\theta}(\hat{\theta}, \varepsilon'))}^{q(x)} -c_{q\theta}(y, x) \, dy \, dx > 0$ for all $\theta > \tilde{\theta}(\hat{\theta}, \varepsilon')$. By $c_{qq\theta} < 0$, these two inequalities imply $\int_{\tilde{\theta}(\hat{\theta}, \varepsilon')}^{\theta} \int_{z}^{q(x)} -c_{q\theta}(y, x) \, dy \, dx > 0$ for all $z \in [q^-(\tilde{\theta}(\hat{\theta}, \varepsilon')), q^+(\tilde{\theta}(\hat{\theta}, \varepsilon'))]$.

Incentive constraints from $\theta \geq \theta'' + \varepsilon$ to $\hat{\theta} \in [\theta' + \varepsilon, \theta'' + \varepsilon]$ are not violated because $\Phi^c(\theta, \hat{\theta}) = \int_{\tilde{\theta}(\hat{\theta}, \varepsilon')}^{\theta} \int_{q^c(\hat{\theta}, \varepsilon')}^{q(x)} -c_{q\theta}(y, x) \, dy \, dx - \int_{\hat{\theta}}^{\tilde{\theta}(\hat{\theta}, \varepsilon')} \int_{q^c(\hat{\theta}, \varepsilon')}^{q^c(x, \varepsilon')} c_{q\theta}(y, x) \, dy \, dx - \int_{\tilde{\theta}(\hat{\theta}, \varepsilon')}^{\theta''+\varepsilon} \int_{q(x)}^{q^c(x, \varepsilon')} c_{q\theta}(y, x) \, dy \, dx$ $\geq \int_{\tilde{\theta}(\hat{\theta}, \varepsilon')}^{\theta} \int_{q^c(\hat{\theta}, \varepsilon')}^{q(x)} -c_{q\theta}(y, x) \, dy \, dx \geq 0$ where the last inequality follows from incentive compatibility of q and the previous paragraph. Incentive constraints from $\theta \geq \theta'' + \varepsilon$ to $\hat{\theta} \in [\theta', \theta' + \varepsilon)$ are not violated because $\Phi^c(\theta, \hat{\theta}) = \int_{\tilde{\theta}(\hat{\theta}, \varepsilon')}^{\theta} \int_{q^c(\hat{\theta}, \varepsilon')}^{q(x)} -c_{q\theta}(y, x) \, dy \, dx +$ $\int_{\tilde{\theta}(\hat{\theta}, \varepsilon')}^{\theta'+\varepsilon} \int_{max(q^c(\hat{\theta}, \varepsilon'), q^c(x, \varepsilon'))}^{q(x)} c_{q\theta}(y, x) \, dy \, dx - \int_{\theta''}^{\theta''+\varepsilon} \int_{q(x)}^{q^c(x, \varepsilon')} c_{q\theta}(y, x) \, dy \, dx \geq \int_{\tilde{\theta}(\hat{\theta}, \varepsilon')}^{\theta} \int_{q^c(\hat{\theta}, \varepsilon')}^{q(x)} -c_{q\theta}(y, x) \, dy \, dx \geq$

0 where the first inequality follows from the definition of δ and the second by incentive compatibility of q. Note that θ' and θ'' can be chosen such that $c_{q\theta}(q(\theta'), \theta'' + \varepsilon) \leq 0$ which implies that $\Phi(\theta, \hat{\theta}) > 0$ for $\theta \in [\theta', \theta'' + \varepsilon]$ and $\hat{\theta} \in [\theta', \theta)$. As q^c is monotone, upward incentive constraints are slack (lemma 3) and for all other types Φ did not change when changing q to q^c.

Turning to the next statement in the theorem, let $\Phi(\theta', \hat{\theta}') = 0$ and let q be strictly increasing in neighborhoods of θ' and $\hat{\theta}'$. Note that lemma 5 implies that there has to be an open neighborhood of θ' such that $\Phi(\theta, \hat{\theta}) > 0$ for all $\hat{\theta}$ in this neighborhood and $\theta \neq \hat{\theta}$: Otherwise, there would be a sequence $((\theta_n, \hat{\theta}_n))_{n=1}^{\infty}$ such that $\Phi(\theta_n, \hat{\theta}_n) = 0$ and $\lim_{n\to\infty} \hat{\theta}_n = \theta'$. But then – by the continuity[21] of Φ – $\Phi(\tilde{\theta}, \theta') = 0$ violating lemma 5 where $\tilde{\theta}$ is the limit of a converging subsequence of $(\theta_n)_{n=1}^{\infty}$. Such a convergent subsequence exists by the Bolzano-Weierstrass theorem. From the first result of theorem 1, it follows that η is non-increasing in an open neighborhood of θ' such that (NLIC) does not hold with equality to types in this neighborhood. Similarly, by lemma 5, there has to be an open neighborhood of $\hat{\theta}'$ such that (NLIC) holds with inequality from types in this neighborhood. The second result of theorem 1 implies that η is non-decreasing in this neighborhood.

Now turn to $\eta^-(\bar{\theta}) = 0$ (and therefore $q^-(\bar{\theta}) = q^{fb}(\bar{\theta})$) whenever $\Phi(\bar{\theta}, \hat{\theta}) > 0$ for all $\hat{\theta} \in \Theta \setminus \{\bar{\theta}\}$. Note that in this case $\Phi(\theta, \hat{\theta}) > 0$ for all $\theta \in (\bar{\theta} - \varepsilon, \bar{\theta}]$ and $\hat{\theta} \in \Theta \setminus \{\theta\}$ for some $\varepsilon > 0$: Otherwise, there would be a sequence $((\theta_n, \hat{\theta}_n))_{n=1}^{\infty}$ such that $\Phi(\theta_n, \hat{\theta}_n) = 0$ and $\lim_{n\to\infty} \theta_n = \bar{\theta}$. But then $\Phi(\bar{\theta}, \tilde{\theta}) = 0$ where $\tilde{\theta} < \bar{\theta}$ is the limit of a convergent subsequence of $(\hat{\theta}_n)_{n=1}^{\infty}$.

Clearly, $q(\theta)$ of $\theta \in (\bar{\theta} - \varepsilon, \bar{\theta}]$ does not affect non-local incentive constraints of other types, see figure 1b for an illustration. Consequently, the principal's payoff would be maximized by setting $q(\theta) = q^r(\theta)$ and therefore $\eta(\theta) = 0$ for $\theta \in (\bar{\theta} - \varepsilon, \bar{\theta}]$. Hence, the optimality of q is contradicted if it can be shown that the monotonicity constraint is not binding on $(\bar{\theta} - \varepsilon, \bar{\theta}]$. Suppose to the contrary that $q(\theta) = q^b > q^{fb}(\bar{\theta})$ holds for types $\theta \in [\theta', \bar{\theta})$ for some $\theta' < \bar{\theta}$.[22] By lemma 4, $\Phi(\theta, \hat{\theta}) > 0$ for $\theta \in (\theta', \bar{\theta}]$ and $\hat{\theta} \notin [\theta', \bar{\theta}]$. First, note that q has to be continuous at θ' as otherwise the principal's payoff could be

[21] By lemma 4, q can neither be discontinuous at $\hat{\theta}'$ nor at the limit of (a subsequence of) $(\hat{\theta}_n)_{n=1}^{\infty}$. Hence, Φ is continuous at these points.

[22] If types arbitrarily close to $\bar{\theta}$ are bunched on $q^b \leq q^{fb}(\bar{\theta})$, clearly $q^-(\bar{\theta}) \leq q^{fb}(\bar{\theta})$ which – together with $\eta \geq 0$ – implies $\eta^-(\bar{\theta}) = 0$.

increased by reducing q^b: As $q^b > q^{fb}(\bar{\theta}) \geq q^r(\theta)$ and the principal's objective in problem (SB) is concave in q with maximum at q^r, lowering q^b increases this objective. Given continuity of q at θ', $\Phi(\theta, \hat{\theta}) > 0$ for $\theta \in [\theta' - \varepsilon', \theta']$ and $\hat{\theta} < \theta$ for some small $\varepsilon' > 0$ by the same argument as in the proof of lemma 4.[23] Given that $q(\theta) > q^{fb}(\bar{\theta}) > q^r(\theta)$ for all $\theta \in [\theta' - \varepsilon', \bar{\theta})$, the principal's payoff could be increased by changing $q(\theta)$ to $q(\theta' - \varepsilon')$ for all $\theta \in [\theta' - \varepsilon', \bar{\theta}]$ (assuming $\varepsilon' > 0$ is small enough). This contradicts the optimality of q. Hence, the monotonicity constraint cannot bind and the result follows.

The part that $\eta^+(\underline{\theta}) = 0$ if $\Phi(\theta, \underline{\theta}) > 0$ for all $\theta > \underline{\theta}$ is similar: For the same argument as above, there exists an $\varepsilon > 0$ such that $\Phi(\theta, \hat{\theta}) > 0$ for any $\hat{\theta} \in [\underline{\theta}, \underline{\theta} + \varepsilon)$ and $\theta \in \Theta \setminus \{\hat{\theta}\}$. Reducing $q(\theta)$ to $\varepsilon' q^r(\theta) + (1 - \varepsilon')q(\theta)$ for all $\theta \in [\underline{\theta}, \underline{\theta} + \varepsilon/2)$ increases the principal's payoff and cannot violate the monotonicity constraint as $q(\theta) \geq q^r(\theta)$ and $q^r_\theta > 0$ by lemma 2. For $\varepsilon' > 0$ small enough, (NLIC) cannot be violated by the change.

Next, let q be constant on (θ_1, θ_2), i.e. $q(\theta) = q^b$ for all $\theta \in (\theta_1, \theta_2)$. Start with the simple case in which q is continuous at θ_1 and θ_2 and $\theta_2 < \bar{\theta}$. Suppose contrary to the theorem that $\eta(\theta_1) > \eta(\theta_2)$. By lemma 4, (NLIC) cannot hold with equality from types in (θ_1, θ_2) to types $\hat{\theta} \notin (\theta_1, \theta_2)$. Furthermore, there has to be an $\varepsilon > 0$ such that $\Phi(\theta, \hat{\theta}) > 0$ for $\hat{\theta} \neq \theta$ and $\theta \in (\theta_1 - \varepsilon, \theta_1) \cup (\theta_2, \theta_2 + \varepsilon)$: Otherwise, $\Phi(\frac{\theta_1 + \theta_2}{2}, \hat{\theta}) < 0$ by (C1) and $c_{q\theta\theta} > 0$.

Note that η is continuous at θ_1 and θ_2 (as q is) and therefore there exists $\tilde{\eta}$ such that $\eta(\theta) > \tilde{\eta}$ for $\theta \in [\theta' - \varepsilon, \theta' + \varepsilon]$ and $\eta(\theta) < \tilde{\eta}$ for $\theta \in [\theta_2 - \varepsilon, \theta_2 + \varepsilon]$ for $\varepsilon > 0$ small enough. Consider the following changed decision

$$q^c(\theta, \varepsilon') = \begin{cases} \varepsilon' q(\theta_1 - \varepsilon) + (1 - \varepsilon')q(\theta) & \text{for } \theta \in [\theta_1 - \varepsilon, \theta_1 + \varepsilon) \\ \delta(\varepsilon')q(\theta_2 + \varepsilon) + (1 - \delta(\varepsilon'))q(\theta) & \text{for } \theta \in [\theta_2 - \varepsilon, \theta_2 + \varepsilon] \\ q(\theta) & \text{else.} \end{cases}$$

where $\delta(\varepsilon')$ is chosen such that $\int_{\theta_1 - \varepsilon}^{\theta_1 + \varepsilon} \int_{q^c(\theta, \varepsilon')}^{q(\theta)} c_{q\theta}(y, x)\, dy\, dx = \int_{\theta_2 - \varepsilon}^{\theta_2 + \varepsilon} \int_{q(\theta)}^{q^c(\theta, \varepsilon')} c_{q\theta}(y, x)\, dy\, dx$. Following the same steps as in the proof of the second statement of this theorem, it is straightforward to show that q^c increases the principal's payoff and is implementable. This contradicts the optimality of q.

[23] In a nutshell, (C1) and $c_{q\theta\theta} > 0$ would imply that $\Phi(\bar{\theta}, \hat{\theta}) < 0$ for some $\hat{\theta}$ if (NLIC) was holding with equality from types arbitrarily close to θ'.

If q is discontinuous at θ_1, let q^c be $q^c(\theta) = q^b - \varepsilon'$ for $\theta \in [\theta_1, \theta_1 + \varepsilon)$ and q^c as in the last paragraph for $\theta \in [\theta_2 - \varepsilon, \theta_2 + \varepsilon]$ (and $q(\theta)$ for all other types). The same argument as above applies for $\varepsilon > 0$ and $\varepsilon' > 0$ small enough and δ chosen such that $\int_{\theta_1}^{\theta_1+\varepsilon} \int_{q^c(\theta,\varepsilon')}^{q(\theta)} c_{q\theta}(y,x)\, dy\, dx = \int_{\theta_2-\varepsilon}^{\theta_2+\varepsilon} \int_{q(\theta)}^{q^c(\theta,\varepsilon')} c_{q\theta}(y,x)\, dy\, dx$. If q is discontinuous at θ_2 or if $\theta_2 = \bar{\theta}$, the changed decision around θ_2 has to be $q^c(\theta) = q^b + \varepsilon'$ for $\theta \in [\theta_2 - \varepsilon, \theta_2]$ (while q^c around θ_1 as before). In both cases, this change increases the payoff of the principal if $\eta^+(\theta_1) > \eta^-(\theta_2)$. Since this contradicts optimality, $\eta^+(\theta_1) \leq \eta^-(\theta_2)$ as had to be shown.

Last, let q be discontinuous at θ', i.e. $q^-(\theta') < q^+(\theta')$. If q is such that there is no type $\tilde{\theta} \neq \theta'$ such that either $q^+(\theta') = q(\tilde{\theta})$ or $q^-(\theta') = q(\tilde{\theta})$, then lemma 4 implies that $\Phi(\theta, \theta') > 0$ for all $\theta \neq \theta'$. The same argument as in lemma 4 implies that there exists an open neighborhood of θ' such that $\Phi(\theta, \hat{\theta}) > 0$ for all $\hat{\theta} \neq \theta$. If q is strictly increasing in an open neighborhood of θ', the first part of theorem 1 implies that η is non-increasing and therefore $\eta^-(\theta') \geq \eta^+(\theta')$.

Hence, only the following cases remain for this proof: (i) q is constant on an interval (θ_1, θ'), (ii) q is constant on an interval (θ', θ_2), (iii) both (i) and (ii) are true and (iv) $q(\theta) = q^b$ for $\theta \in (\theta_i, \tilde{\theta}_i)$ and θ_i or $\tilde{\theta}_i$ can be chosen arbitrarily close to θ' but $q^+(\theta') \neq q^b \neq q^-(\theta')$. Define $q^a = \frac{q^-(\theta')+q^+(\theta')}{2}$. Suppose, contrary to the theorem, that $\eta^-(\theta') < \eta^+(\theta')$. As both one sided limits of η at θ' exist, there exists an $\varepsilon > 0$ such that $\eta(\theta) < \frac{\eta^-(\theta')+\eta^+(\theta')}{2}$ for all $\theta \in (\theta' - \varepsilon, \theta')$ and $\eta(\theta) > \frac{\eta^-(\theta')+\eta^+(\theta')}{2}$ for $\theta \in (\theta', \theta' + \varepsilon)$. Consider now the following changed decision

$$q^c(\theta, \varepsilon') = \begin{cases} \varepsilon' q^a + (1 - \varepsilon') q(\theta) & \text{for } \theta \in [\theta' - \varepsilon, \theta') \\ \delta(\varepsilon') q^a + (1 - \delta(\varepsilon')) q(\theta) & \text{for } \theta \in [\theta', \theta' + \varepsilon) \\ q(\theta) & \text{else} \end{cases}$$

where $\delta(\varepsilon')$ is chosen such that (8) holds. Following the same steps as above, it is straightforward that there exist $\varepsilon > 0$ and $\varepsilon' > 0$ such that the principal's payoff is higher under q^c than under q.

It remains to show that q^c is incentive compatible in each of the three cases. Note that only incentive constraints to types in $[\theta' - \varepsilon, \theta' + \varepsilon)$ have to be checked as all other incentive constraints are not affected or even relaxed. If there exists an open neighborhood of θ' such that for all $\hat{\theta}$ in this neighborhood $\Phi(\theta, \hat{\theta}) > 0$ for all θ such that $q(\theta) \neq q(\hat{\theta})$, then

(NLIC) will be satisfied in the changed decision for $\varepsilon' > 0$ small enough: As $q^c > s$, there exists an $\varepsilon'' > 0$ such that $\Phi(\theta, \hat{\theta}) > 0$ for all types such that $0 < \theta - \hat{\theta}^m(\hat{\theta}) < \varepsilon''$ where $\hat{\theta}^m(\hat{\theta}) = max\{\theta : \ q^-(\theta) = q(\hat{\theta})\}$. If a neighborhood of θ' as described exists, then $\min_{\theta \in [\hat{\theta}^m(\hat{\theta})+\varepsilon'', \bar{\theta}], \hat{\theta} \in [\theta'-\varepsilon, \theta'+\varepsilon]} \Phi(\theta, \hat{\theta}) > 0$ for $\varepsilon > 0$ small enough. This implies that the changed decision is implementable for $\varepsilon' > 0$ small enough because $\Phi^c = \Phi$ for $\varepsilon' = 0$ and Φ^c is continuous in ε'. Hence, it remains to check whether $\Phi(\theta, \hat{\theta}) = 0$ for $\hat{\theta}$ arbitrarily close to θ' and θ such that $q(\theta) \neq q(\hat{\theta})$ and whether the changed decision is implementable if this is the case.

In case (iv), it was already established (3 paragraphs above) that there is an open neighborhood of θ' such that $\Phi(\theta, \hat{\theta}) > 0$ for all $\hat{\theta}$ in this neighborhood and θ such that $q(\theta) \neq q(\hat{\theta})$.

In case (i), (NLIC) cannot hold with equality for $\hat{\theta} = \theta' < \theta$: If $\Phi(\theta, \theta') = 0$, then $\int_{\theta'}^{\theta} c_{q\theta}(q(\theta'), x)\, dx \geq 0$ as otherwise $\Phi(\theta, \theta' + \varepsilon) < 0$. But then $\Phi(\theta, \theta' - \varepsilon) = \Phi(\theta, \theta') - \int_{\theta'}^{\theta} \int_{q^-(\theta')}^{q^+(\theta')} c_{q\theta}(y, x)\, dy\, dx < 0$ by $c_{qq\theta} < 0$ which contradicts incentive compatibility. For the same reason, (NLIC) will not hold with equality to types $\hat{\theta} \in (\theta', \theta' + \varepsilon)$ for $\varepsilon > 0$ small. Hence, in case (i), (NLIC) has to be checked only for $\theta > \theta'$ and $\hat{\theta} \in [\theta' - \varepsilon, \theta')$.

Let ε be small enough such that $q(\theta' - \varepsilon) = q^-(\theta')$ and let $\Phi(\theta, \theta' - \varepsilon) = 0$. Then $\int_{\theta'}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx > 0$: Otherwise $\Phi(\theta, \theta') = \Phi(\theta, \theta' - \varepsilon) - \int_{\theta'}^{\theta} \int_{q^-(\theta')}^{q^+(\theta')} c_{q\theta}(y, x)\, dy\, dx < 0$ because $c_{qq\theta} < 0$ implies $\int_{\theta'}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx > \int_{\theta'}^{\theta} c_{q\theta}(y, x)\, dx$ for $y > q^-(\theta')$ and $\Phi(\theta, \theta' - \varepsilon)$ is 0 by assumption. Now let $\varepsilon > 0$ be small enough such that $\int_{\theta'-\varepsilon}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx > 0$. The changed contracts for types $\hat{\theta} \in [\theta' - \varepsilon, \theta']$ is then less attractive for θ: $\Phi^c(\theta, \hat{\theta}) = \Phi^c(\theta, \theta' - \varepsilon) + \int_{\theta'-\varepsilon}^{\theta} \int_{q(\theta'-\varepsilon)}^{min\{q^c(x), q^c(\hat{\theta})\}} c_{q\theta}(y, x)\, dy\, dx = \Phi(\theta, \theta' - \varepsilon) + \int_{\theta'-\varepsilon}^{\theta} \int_{q(\theta'-\varepsilon)}^{min\{q^c(x), q^c(\hat{\theta})\}} c_{q\theta}(y, x)\, dy\, dx > 0$ for $\varepsilon' > 0$ small because $\int_{\theta'-\varepsilon}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx > 0$ implies $\int_{\theta'-\varepsilon}^{\theta} c_{q\theta}(y, x)\, dx > 0$ for $y \in (q(\theta' - \varepsilon), min\{q^c(x), q^c(\hat{\theta})\})$. Hence, the change relaxes (NLIC). Cases (ii) and (iii) are similar to (i) and therefore relegated to the webappendix. □

Proof of proposition 1: Existence is shown in the webappendix. Here I show that any incentive compatible decision function q which imposes decisions below s for some type is dominated by the following changed decision

$$q^c(\theta) = \begin{cases} q(\theta) & \text{if } q(\theta) \geq s(\theta) \\ q^s(q(\theta), \theta) & \text{if } q(\theta) < s(\theta). \end{cases} \tag{9}$$

This changed decision is shown to be above s, monotonically increasing and incentive compatible.

First, it is shown that the principal's payoff is higher under q^c than under q: The principal maximizes the expected value of $u(q,\theta) - c(q,\theta) + [(1-F(\theta))/f(\theta)]c_\theta(q,\theta)$. If $q^s(q(\theta),\theta) \leq q^r(\theta)$, the principal's objective increases due to the change because of the concavity of (RP) and $q^r(\theta) > s(\theta)$ (recall that q^r is the pointwise maximizer of the principal's objective). If $q^s(q(\theta),\theta) > q^r(\theta)$, then the same conclusion follows from $q^v(q(\theta),\theta) \geq q^s(q(\theta),\theta) > q^r(\theta)$ and the concavity of (RP).

Second, the changed decision q^c is monotonically increasing: Suppose otherwise, i.e. suppose there are $\theta' < \theta''$ with $q^c(\theta') > q^c(\theta'')$. Then for any $\varepsilon > 0$ there have to be θ'_ε and θ''_ε such that $\theta'_\varepsilon < \theta''_\varepsilon < \theta'_\varepsilon + \varepsilon$ and $q^c(\theta'_\varepsilon) > q^c(\theta''_\varepsilon)$. Let $t : \Theta \to \mathbb{R}$ denote the transfer function implementing q. Then incentive compatibility requires $c(q(\theta''_\varepsilon),\theta'_\varepsilon) - c(q(\theta'_\varepsilon),\theta'_\varepsilon) \geq t(\theta''_\varepsilon) - t(\theta'_\varepsilon) \geq c(q(\theta''_\varepsilon),\theta''_\varepsilon) - c(q(\theta'_\varepsilon),\theta''_\varepsilon)$. Combining the two inequalities yields

$$\int_{\theta'_\varepsilon}^{\theta''_\varepsilon} \int_{q(\theta''_\varepsilon)}^{q(\theta'_\varepsilon)} c_{q\theta}(y,x)\, dy\, dx \geq 0. \tag{10}$$

Denote by $\tilde{\Theta}$ the set $\{\theta : q(\theta) \geq s(\theta)\}$. Inequality (10) cannot be satisfied if $\theta'_\varepsilon < \theta''_\varepsilon$ and $q(\theta'_\varepsilon) > q(\theta''_\varepsilon) \geq s(\theta''_\varepsilon)$ as the integrand is then negative for all arguments. For $\varepsilon > 0$ small enough, inequality (10) is also violated if both θ'_ε and θ''_ε are in $\Theta \setminus \tilde{\Theta}$ as then $q(\theta'_\varepsilon) < q(\theta''_\varepsilon) < s(\theta''_\varepsilon)$ is implied by $q^c(\theta'_\varepsilon) > q^c(\theta''_\varepsilon)$ and $c_{q\theta\theta} > 0$. Therefore, one of the two types θ'_ε and θ''_ε has to be in $\tilde{\Theta}$ and one has to be in $\Theta \setminus \tilde{\Theta}$. Say $\theta'_\varepsilon \notin \tilde{\Theta}$ and $\theta''_\varepsilon \in \tilde{\Theta}$. Then $\int_{q(\theta'_\varepsilon)}^{q(\theta''_\varepsilon)} c_{q\theta}(y,\theta'_\varepsilon)\, dy = \int_{q^c(\theta''_\varepsilon)}^{q^c(\theta'_\varepsilon)} -c_{q\theta}(y,\theta'_\varepsilon)\, dy > 0$ as $q^c(\theta'_\varepsilon) > q^c(\theta''_\varepsilon) = q(\theta''_\varepsilon) \geq s(\theta''_\varepsilon) > s(\theta'_\varepsilon)$. By $c_{q\theta\theta} > 0$, this implies $\int_{q(\theta'_\varepsilon)}^{q(\theta''_\varepsilon)} c_{q\theta}(y,x)\, dy > 0$ for any $x \in [\theta'_\varepsilon, \theta''_\varepsilon]$. But then inequality (10) is violated. A similar argument can be made for $\theta'_\varepsilon \in \tilde{\Theta}$ and $\theta''_\varepsilon \notin \tilde{\Theta}$. Therefore, inequality (10) and thereby incentive compatibility is violated in any case contradicting $q^c(\theta'_\varepsilon) > q^c(\theta''_\varepsilon)$. Hence, q^c is increasing.

Third, the changed decision q^c is incentive compatible: Since q^c is monotonically increasing, only downward misrepresentation has to be considered (lemma 3). Note that the rent function π is not affected by the change from q to q^c because of the definition of q^s and lemma 1. Therefore, one has only to check whether any type wants to misrepresent as a lower type $\hat{\theta}$ at which $q(\hat{\theta}) < s(\hat{\theta})$. Since $\pi(\theta)$ is unchanged, (NLIC) under the

changed decision can be written as

$$\begin{aligned}
\Phi^c(\theta,\hat{\theta}) &= \pi(\theta)-\pi(\hat{\theta})-c(q^c(\hat{\theta}),\hat{\theta})+c(q^c(\hat{\theta}),\theta)=\int_{\hat{\theta}}^{\theta}[-c_\theta(q(x),x)+c_\theta(q^c(\hat{\theta}),x)]\,dx \\
&= -\int_{\hat{\theta}}^{\theta}\int_{q^c(\hat{\theta})}^{q(x)} c_{q\theta}(y,x)\,dy\,dx = -\int_{\hat{\theta}}^{\theta}\int_{q^c(\hat{\theta})}^{q(\hat{\theta})} c_{q\theta}(y,x)\,dy\,dx - \int_{\hat{\theta}}^{\theta}\int_{q(\hat{\theta})}^{q(x)} c_{q\theta}(y,x)\,dy\,dx \\
&= \int_{\hat{\theta}}^{\theta}\int_{q(\hat{\theta})}^{q^c(\hat{\theta})} c_{q\theta}(y,x)\,dy\,dx + \Phi(\theta,\hat{\theta}) > 0
\end{aligned}$$

where the inequality follows from $\int_{q(\hat{\theta})}^{q^c(\hat{\theta})} c_{q\theta}(y,\hat{\theta})\,dy = 0$ by the definition of q^s and $c_{q\theta\theta} > 0$. □

Proof of proposition 2: Theorem 1 states that a solution could only be discontinuous at a type θ' where $\eta^-(\theta') \geq \eta^+(\theta')$. Given that q is increasing, $q^-(\theta') < q^+(\theta')$ at a hypothetical discontinuity type θ'. Using the definition of η in (4), one can calculate the change in $\eta(\theta')$ at the discontinuity type

$$\begin{aligned}
\eta^+(\theta')-\eta^-(\theta') &= \frac{[u_q(q^+(\theta'),\theta')-c(q^+(\theta'),\theta')]f(\theta')+(1-F(\theta'))c_{q\theta}(q^+(\theta'),\theta')}{c_{q\theta}(q^+(\theta'),\theta')} \\
&\quad - \frac{[u_q(q^-(\theta'),\theta')-c(q^-(\theta'),\theta')]f(\theta')+(1-F(\theta'))c_{q\theta}(q^-(\theta'),\theta')}{c_{q\theta}(q^-(\theta'),\theta')} \\
&= \int_{q^-(\theta')}^{q^+(\theta')} \frac{(u_{qq}-c_{qq})fc_{q\theta}+(1-F)c_{qq\theta}c_{q\theta}-(u_q-c_q)fc_{qq\theta}-(1-F)c_{q\theta}c_{qq\theta}}{c_{q\theta}^2}\,dx
\end{aligned}$$

where the functions in the integrand are evaluated at (x,θ') and θ' respectively and the last equality uses the fundamental theorem of calculus. Note that the integrand is positive for $x \leq q^{fb}(\theta')$. If $x > q^{fb}(\theta')$, the integrand can be written as

$$\frac{f(u_q-c_q)}{c_{q\theta}}\left(\frac{u_{qq}-c_{qq}}{u_q-c_q}-\frac{c_{qq\theta}}{c_{q\theta}}\right)$$

which is also positive due to the condition of the proposition. Hence, $\eta^-(\theta') < \eta^+(\theta')$ which contradicts theorem 1. □

Proof of proposition 3: The optimal decision is continuous at types where it is below q^{fb}: The proof of proposition 2 shows that a discontinuity at a type θ where $q^+(\theta) \leq q^{fb}(\theta)$ contradicts theorem 1.

Next it has to be shown that the decision is strictly monotone when it is below first best. This will be done in two steps. The first step is to show that $q_\theta(\theta) > 0$ at types θ

where η is differentiable and $\eta_\theta(\theta) \geq 0$. The second step is to show that there are types θ at which $\eta_\theta(\theta) \geq 0$ in any hypothetical interval of types with the same decision. This yields a contradiction together with the first step.

First, the decision $q(\theta)$ has to satisfy the first order condition $[u_q - c_q] + \frac{1-F-\eta}{f} c_{q\theta} = 0$ by theorem 1. For now, assume that q (and therefore η) are continuously differentiable in a neighborhood of θ. Then, given η, the sign of $q_\theta(\theta)$ can be determined from the implicit function theorem. Note that $q(\theta) \leq q^{fb}(\theta)$ implies $1 - F(\theta) - \eta(\theta) \geq 0$. Hence, the partial derivative of the left hand side of the first order condition with respect to q is negative. The sign of $q_\theta(\theta)$ is, therefore, the sign of the partial derivative of the left had side of the first order condition with respect to θ. Denoting $(1 - F(\theta) - \eta(\theta))$ by $\lambda(\theta)$ this derivative is

$$u_{q\theta}(q(\theta), \theta) - c_{q\theta}(q(\theta), \theta) + \frac{\lambda(\theta)}{f(\theta)} c_{q\theta\theta} + \frac{\partial\ (\lambda(\theta)/f(\theta))}{\partial\theta} c_{q\theta}(q(\theta), \theta).$$

The first three terms are clearly positive as $q(\theta) \leq q^{fb}(\theta)$ implies $\lambda(\theta) \geq 0$. The fourth term is positive if $\eta_\theta(\theta) \geq 0$ as then

$$\frac{\partial\ (\lambda(\theta)/f(\theta))}{\partial\theta} = \frac{-f^2(\theta) - f_\theta(\theta)(1 - F(\theta))}{f^2(\theta)} - \frac{\eta_\theta(\theta)}{f(\theta)} + \frac{f_\theta(\theta)\eta(\theta)}{f^2(\theta)} \leq 0$$

where the inequality stems from the monotone hazard rate assumption if $f_\theta(\theta) \leq 0$. If $f_\theta(\theta) > 0$, then $q^{fb}(\theta) \geq q(\theta)$ implies $\lambda(\theta) \geq 0$ which ensures the inequality above.

Now turn to the second step. Suppose contrary to the proposition that an interval (θ_1, θ_2) exists such that $q(\theta) = q^b$ for all $\theta \in (\theta_1, \theta_2)$. This implies that q – and therefore also η – are differentiable on (θ_1, θ_2). By theorem 1, $\eta^+(\theta_1) \leq \eta^-(\theta_2)$. Consequently, there has to be some type $\theta \in (\theta_1, \theta_2)$ such that $\eta_\theta(\theta) \geq 0$. But then the first step shows that $q_\theta(\theta) > 0$ which contradicts $q(\theta) = q^b$ for all $\theta \in (\theta_1, \theta_2)$.

The last part: q is below q^{fb} if q^m increases and there is no distortion at the top. The proof is by contradiction. Suppose the optimal decision $q(\theta)$ was above the first best decision for some types. Since there is no distortion at the top by assumption and since q is increasing, there has to be a type θ' such that $q(\theta') = q^{fb}(\theta')$ and $q(\theta) > q^{fb}(\theta)$ for $\theta \in (\theta' - \varepsilon, \theta')$ for some $\varepsilon > 0$.

Note that the arguments for continuity and strict monotonicity above are not tight at q^{fb}. Hence, for $\varepsilon > 0$ small enough, q has to be continuous and strictly increasing

on $(\theta' - \varepsilon, \theta')$. This implies that η is also continuous on this interval. Recall that $q(\theta) > q^{fb}(\theta)$ if and only if $\eta(\theta) > 1 - F(\theta)$. Since $1 - F(\theta)$ is decreasing, it follows that $\eta(\theta') = 0$ and there does not exist an $\varepsilon' > 0$ such that η is non-decreasing on $(\theta' - \varepsilon', \theta')$.

Therefore, for each $n \in \mathbb{N}$ there has to be a type θ_n with $\theta' - \frac{1}{n} < \theta_n < \theta'$ and $\Phi(\theta_n, \hat{\theta}_n) = 0$ for some $\hat{\theta}_n < \theta_n$ (theorem 1). By lemma 5, the sequence $(\hat{\theta}_n)_{n=1}^{\infty}$ is decreasing and therefore has a limit $\hat{\theta}'$. $\Phi(\theta', \hat{\theta}') = 0$ as Φ is continuous at $(\theta', \hat{\theta}')$ (following the argument of lemma 4, Φ cannot be discontinuous at $\hat{\theta}'$). As (C1) has to hold for each $(\theta_n, \hat{\theta}_n)$ and $q(\theta_n) > q^{fb}(\theta_n)$, $q(\hat{\theta}_n) < q^m(\theta_n)$ for each n. Hence, $q(\hat{\theta}_n) < q^m(\theta')$ as q^m is increasing. In the limit, however, $q(\hat{\theta}') = q^m(\theta')$ has to hold by (C1) as $q(\theta') = q^{fb}(\theta')$. Hence, $\hat{\theta}' < \hat{\theta}_n$ but $q(\hat{\theta}') > q(\hat{\theta}_n)$ which contradicts that q is increasing. □

Proofs of section 5

Some preliminary results are used repeatedly in the following proofs and therefore stated separately here. They are shown under the assumptions of section 5, i.e. that the sufficient conditions for continuity and strict monotonicity are satsified.

Lemma 6. *Let q be a continuous solution to problem (SB). The following four sets are closed: The sets $\{\theta : \Phi(\theta, \hat{\theta}) = 0$ for some $\hat{\theta} \neq \theta\}$ and $\{\hat{\theta} : \Phi(\theta, \hat{\theta}) = 0$ for some $\theta \neq \hat{\theta}\}$, the set of types from which (NLIC) binds and the set to which (NLIC) binds.*

Proof: Let the sequence $(\theta_n)_{n=1}^{\infty}$ converge to θ'. Assume that for each θ_n there exists a $\hat{\theta}_n$ such that $\Phi(\theta_n, \hat{\theta}_n) = 0$. By the Bolzano Weierstrass theorem, there exists a convergent subsequence of $(\hat{\theta}_n)_{n=1}^{\infty}$. Call its limit $\hat{\theta}'$. By the continuity of Φ and as $\Phi(\theta_n, \hat{\theta}_n) = 0$ for all n, $\Phi(\theta', \hat{\theta}') = 0$. If (NLIC) binds from each θ_n, η cannot be constant in any open neighborhood of any θ_n. Hence, η cannot be constant in any open neighborhood of θ' while being non-increasing in a neighborhood of θ' by theorem 1. Hence, (NLIC) binds from θ'. The proof for $\{\hat{\theta} : \Phi(\theta, \hat{\theta}) = 0$ for some $\theta \neq \hat{\theta}\}$ and the set of $\hat{\theta}$ from which (NLIC) binds is analogous. □

Lemma 7. *(NLIC) binds from θ' if for some $\varepsilon > 0$ either (i) $\eta(\theta) > \eta(\theta')$ for all $\theta \in (\theta' - \varepsilon, \theta')$ and $\eta(\theta) \leq \eta(\theta')$ for all $\theta \in (\theta', \theta' + \varepsilon)$ or (ii) $\eta(\theta) \geq \eta(\theta')$ for all $\theta \in (\theta' - \varepsilon, \theta')$ and $\eta(\theta) < \eta(\theta')$ for all $\theta \in (\theta', \theta' + \varepsilon)$.*
(NLIC) binds to θ' if either (i) $\eta(\theta) < \eta(\theta')$ for all $\theta \in (\theta' - \varepsilon, \varepsilon')$ and $\eta(\theta) \geq \eta(\theta')$ for all $\theta \in (\theta', \theta' + \varepsilon)$ or (ii) $\eta(\theta) \leq \eta(\theta')$ for all $\theta \in (\theta' - \varepsilon, \varepsilon')$ and $\eta(\theta) > \eta(\theta')$ for all $\theta \in (\theta', \theta' + \varepsilon)$.

Proof: Let the conditions of the first statement be satisfied. Then η is not non-decreasing in any neighborhood of θ'. By theorem 1, (NLIC) can then not bind to θ'. Also η cannot be constant in any open neighborhood of θ'. Hence, (NLIC) cannot be slack at θ' as lemma 6 implies that the set of types at which (NLIC) is slack is open. Consequently, (NLIC) binds from θ'. The second statement is proven similarly. □

Lemma 8. *If (NLIC) binds from $\bar{\theta}$, then $\eta(\bar{\theta}) = \eta(\hat{\theta}^m)$ where $\hat{\theta}^m = \max \hat{\Theta}^m$ and $\hat{\Theta}^m = \{\hat{\theta} :$ (NLIC) binds from $\bar{\theta}$ to $\hat{\theta}\}$. Furthermore, η is non-decreasing on $[\min \hat{\Theta}^m, \max \hat{\Theta}^m]$. If (NLIC) binds to $\underline{\theta}$, $\eta(\underline{\theta}) = \eta(\theta^m)$ where $\theta^m = \min \Theta^m$ and $\Theta^m = \{\theta :$ (NLIC) binds from θ to $\underline{\theta}\}$ Furthermore, η is non-increasing on $[\min \Theta^m, \max \Theta^m]$.*

Proof: Take $\hat{\theta}_1$, $\hat{\theta}_2 \in [\min \hat{\Theta}^m, \max \hat{\Theta}^m]$ such that $\hat{\theta}_2 > \hat{\theta}_1$. By lemma 5, (NLIC) cannot bind from any type in $(\hat{\theta}_1, \hat{\theta}_2)$. Hence, theorem 1 implies that η is non-decreasing on $(\hat{\theta}_1, \hat{\theta}_2)$. By the continuity of η, $\eta(\hat{\theta}_1) \leq \eta(\hat{\theta}_2)$.

If (NLIC) is slack from and to all types in $(\hat{\theta}^m, \bar{\theta})$, then $\eta(\bar{\theta}) = \eta(\hat{\theta}^m)$ by theorem 1 as η is constant on $(\hat{\theta}^m, \bar{\theta})$. If (NLIC) binds from a $\theta' \in (\hat{\theta}^m, \bar{\theta})$ to $\hat{\theta}'$ then $\hat{\theta}' \in (\hat{\theta}^m, \bar{\theta})$ by lemma 5 (and similarly if (NLIC) binds to $\hat{\theta}' \in (\hat{\theta}^m, \bar{\theta})$ to θ', then $\theta' \in (\hat{\theta}^m, \bar{\theta})$). For any such pair $(\theta', \hat{\theta}')$, (C3) implies $\eta(\theta') = \eta(\hat{\theta}')$. Let Θ' ($\hat{\Theta}'$) be the set of types in $(\hat{\theta}^m, \bar{\theta})$ from (to) which (NLIC) binds. By (C3), lemma 5 and the continuity of η, $\eta(\sup \Theta') = \eta(\inf \hat{\Theta}')$. By theorem 1, lemma 5 and the continuity of η, $\eta(\bar{\theta}) = \eta(\sup \Theta')$ and $\eta(\inf \hat{\Theta}') = \eta(\hat{\theta}^m)$. Therefore, $\eta(\bar{\theta}) = \eta(\hat{\theta}^m)$. The proof for $\underline{\theta}$ is analogous. □

Lemma 9. *Let q be a continuous solution to problem (SB). Let (NLIC) bind from θ' to $\hat{\theta}'$. $\eta(\theta) \geq \eta' \equiv \min(\eta(\theta'), \eta(\hat{\theta}'))$ for all $\theta \in [\hat{\theta}', \theta']$.*

Proof: First, let $\eta' = \eta(\theta') = \eta(\hat{\theta}')$ (which is implied by (C3) if $\hat{\theta}'$ and θ' are interior). Suppose, to the contrary, there was a type $\tilde{\theta} \in [\hat{\theta}', \theta']$ with $\eta(\tilde{\theta}) < \eta'$. Denote $\tilde{\theta}' = sup\{\theta : \eta(\theta) \leq \frac{\eta' + \eta(\tilde{\theta})}{2}$ and $\theta \in [\hat{\theta}', \theta']\}$ and note that $\eta(\tilde{\theta}') = \frac{\eta' + \eta(\tilde{\theta})}{2}$ as η is continuous (by the continuity of q). Then, (NLIC) binds to $\tilde{\theta}'$; see lemma 7. The type $\tilde{\theta}''$ from which (NLIC) binds to $\tilde{\theta}'$ has to have $\eta(\tilde{\theta}'') = \frac{\eta' + \eta(\tilde{\theta})}{2}$ by (C3) and therefore, by the definition of $\tilde{\theta}'$, $\tilde{\theta}'' > \theta'$ (note that by lemma 8 $\tilde{\theta}'' \neq \theta'$ even if $\theta' = \bar{\theta}$ as η would not be increasing on $\hat{\Theta}^m$ then). But then the binding (NLIC) between $(\theta', \hat{\theta}')$ and $(\tilde{\theta}'', \tilde{\theta}')$ overlap which contradicts lemma 5.

Second, let $\theta' = \bar{\theta}$. By lemma 8, $\eta(\bar{\theta}) = \eta(\hat{\theta}^m)$ where $\hat{\theta}^m$ is defined in lemma 8. The first part of the proof implies $\eta(\theta) \geq \eta(\bar{\theta})$ for all $\theta \in (\hat{\theta}^m, \bar{\theta})$. As lemma 8 establishes

that η is non-decreasing on $[\min \hat{\Theta}^m, \max \hat{\Theta}^m]$, the result also holds for other $\hat{\theta}$ to which (NLIC) binds from $\bar{\theta}$. The proof for the case $\hat{\theta}' = \underline{\theta}$ is analogous. □

Proof of theorem 2: Take two types θ' and $\hat{\theta}'$ such that (NLIC) is binding from θ' to $\hat{\theta}'$ under the optimal decision q. By (C3), $\eta(\theta') = \eta(\hat{\theta}') \equiv \eta'$ if $\hat{\theta}'$ and θ' are interior.

As described in the text, $(\theta', \hat{\theta}')$ is a local minimizer of $\Phi^{\eta'}$: By (C1) and (C2), the first order condition of minimizing Φ^{η} are satisfied. By lemmas 8 and 9 and the fact that $q(\cdot, \eta)$ is increasing in η, this also holds true if $\hat{\theta}'$ or θ' are boundary types. To verify that $(\theta', \hat{\theta}')$ is a minimizer, note that $q(\theta) \geq q(\theta, \eta')$ for $\theta \in (\hat{\theta}', \hat{\theta}' + \varepsilon) \cup (\theta' - \varepsilon, \theta')$ and $q(\theta) \leq q(\theta, \eta')$ for $\theta \in (\hat{\theta} - \varepsilon, \hat{\theta}) \cup (\theta', \theta' + \varepsilon)$ for $\varepsilon > 0$ small enough because η is non-increasing (non-decreasing) in a neighborhood of θ' (of $\hat{\theta}'$); see theorem 1. Take $\theta'' \in (\theta', \theta' + \varepsilon)$ and $\hat{\theta}'' \in (\hat{\theta}', \hat{\theta}' + \varepsilon)$. As q and $q(\cdot, \eta')$ are continuous, there exists a $\hat{\theta}''' \in (\hat{\theta}', \hat{\theta}'')$ such that $q(\hat{\theta}''') = q(\hat{\theta}'', \eta')$. Then,

$$\Phi^{\eta}(\theta'', \hat{\theta}'') - \Phi^{\eta}(\theta', \hat{\theta}') = \Phi(\theta'', \hat{\theta}''') - \Phi(\theta', \hat{\theta}') + \int_{\hat{\theta}'''}^{\hat{\theta}''} \int_{q(x,\eta')}^{q(\hat{\theta}''')} -c_{q\theta}(y, x)\, dy\, dx$$
$$+ \int_{\hat{\theta}'}^{\hat{\theta}'''} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx + \int_{\theta'}^{\theta''} \int_{q(x)}^{q(x,\eta')} -c_{q\theta}(y, x)\, dy\, dx \geq \Phi(\theta'', \hat{\theta}''') - \Phi(\theta', \hat{\theta}') \geq 0$$

where the first inequality follows from $q(x) > s(x)$ and $q(x, \eta') > s(x)$. The second inequality follows from $\Phi(\theta'', \hat{\theta}''') \geq 0 = \Phi(\theta', \hat{\theta}')$ by incentive compatibility. Similar inequalities hold for $\theta'' \in (\theta' - \varepsilon, \theta')$ or $\hat{\theta}'' \in (\theta' - \varepsilon, \hat{\theta}')$ or both; see the webappendix for details. Hence, $(\theta', \hat{\theta}')$ minimizes Φ^{η} locally.

Now, suppose that $(\theta', \hat{\theta}')$ does not minimize $\Phi^{\eta'}(\theta, \hat{\theta})$ on $[\hat{\theta}', \theta']$ and call the minimizer $(\theta'', \hat{\theta}'')$. Since q is continuous and $\eta(\theta) \geq \eta'$ for all $\theta \in [\hat{\theta}', \theta']$ (lemma 9), there has to be a $\hat{\theta}''' \in [\hat{\theta}', \hat{\theta}'']$ such that $q(\hat{\theta}''') = q(\hat{\theta}'', \eta')$. Then,

$$\Phi(\theta', \hat{\theta}') = \Phi(\theta'', \hat{\theta}''') + \Phi^{\eta'}(\theta', \hat{\theta}') - \Phi^{\eta'}(\theta'', \hat{\theta}'') + \int_{\theta''}^{\theta'} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx$$
$$+ \int_{\hat{\theta}'}^{\hat{\theta}'''} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx + \int_{\hat{\theta}'''}^{\hat{\theta}''} \int_{q(x,\eta')}^{q(\hat{\theta}''')} -c_{q\theta}(y, x)\, dy\, dx$$
$$\geq \Phi(\theta'', \hat{\theta}''') + \Phi^{\eta'}(\theta', \hat{\theta}') - \Phi^{\eta'}(\theta'', \hat{\theta}'') > 0$$

where the first inequality stems from $s(x) < q(x, \eta') \leq q(x)$ as $\eta(x) \geq \eta'$ for $x \in [\hat{\theta}', \theta']$. The second inequality is true by incentive compatibility between θ'' and $\hat{\theta}'''$ and the

assumption that $(\theta'', \hat{\theta}'')$ minimize $\Phi^{\eta'}$ on $[\hat{\theta}', \theta']$. As $\Phi(\theta', \hat{\theta}') = 0$ by assumption, the desired contradiction emerges. Hence, $(\theta', \hat{\theta}')$ minimize $\Phi^{\eta'}$ on $[\hat{\theta}', \theta']$. It was already shown above that $(\theta', \hat{\theta}')$ are local minimzers of $\Phi^{\eta'}$. Taking these two results together, the first claim in the proposition follows.

For the last claim, suppose contrary to the proposition that $\Phi^{\eta'}(\theta', \hat{\theta}') > 0$ and suppose (NLIC) was binding from θ' to $\hat{\theta}'$. Then $\eta(\theta) \geq \eta'$ for all $\theta \in [\hat{\theta}', \theta']$ (see lemma 9). This implies $q(\theta) \geq q(\theta, \eta')$ for all $\theta \in [\hat{\theta}', \theta']$ with equality at $\hat{\theta}'$ and θ'. Consequently, $\Phi(\theta', \hat{\theta}') \geq \Phi^{\eta'}(\theta', \hat{\theta}') > 0$, i.e. (NLIC) cannot bind from θ' to $\hat{\theta}'$. □

Proof of proposition 4: The proof starts with four preliminary results and proceeds then in three further steps.

Preliminary result 1: If – in the solution to problem (SB) – there exists a type θ_0 such that $\eta(\theta_0) = \eta' > 0$, then there exists a type θ' with $\eta(\theta') = \eta'$ such that (NLIC) binds either to or from θ'.

Let $\theta' = sup\{\theta : \eta(\theta) = \eta'\}$. As η is continuous by the contiuity of q, $\eta(\theta') = \eta'$.

Consider $\theta' < \bar{\theta}$: By the definition of θ' and the continuity of η, either (i) $\eta(\theta) < \eta'$ for all $\theta > \theta'$ or (ii) $\eta(\theta) > \eta'$ for all $\theta > \theta'$. In case (i), $\eta(\theta) \geq \eta'$ for $\theta \in (\theta' - \varepsilon, \theta')$ for some $\varepsilon > 0$: Suppose otherwise. Then η is neither non-decreasing nor non-increasing in any open neighborhood of θ'. By theorem 1, (NLIC) can therefore not bind from or to θ'. Then, by lemma 6, there has to be an open neighborhood of θ' such that (NLIC) is slack for all types in this neighborhood. As η is not constant in any open neighborhood of θ', this is impossible which is the desired contradiction. In case (ii), a similar argument applies. In both cases, the conditions of lemma 7 are satsified.

If $\theta' = \bar{\theta}$, then by theorem 1 there exists a $\hat{\theta}'$ such that $\Phi(\theta', \hat{\theta}') = 0$ and (NLIC) binds from θ' to $\hat{\theta}'$.

Preliminary result 2: For $\eta \leq \bar{\eta}$, $\theta(\eta)$ is decreasing and $\hat{\theta}(\eta)$ is increasing in η. If $\theta(\eta) < \bar{\theta}$ (respectively $\hat{\theta}(\eta) > \underline{\theta}$) for $\eta \in (\eta'', \eta')$ with $0 \leq \eta'' < \eta' \leq \bar{\eta}$, the monotonicity of $\theta(\eta)$ (respectively $\hat{\theta}(\eta)$) is strict on (η'', η').

Take $\eta' > \eta'' \geq 0$ with $\eta' \leq \bar{\eta}$ and let $\theta' = \theta(\eta')$ and $\hat{\theta}' = \hat{\theta}(\eta')$. Note that $\Phi_{\theta}^{\eta''}(\theta', \hat{\theta}') < 0$ and $\Phi_{\hat{\theta}}^{\eta''}(\theta', \hat{\theta}') > 0$ because (i) $(\theta', \hat{\theta}(\eta'))$ satisfy the first order condition for minimizing $\Phi^{\eta'}$ and (ii) $q(\theta, \eta') > q(\theta, \eta'')$ for all types and $c_{qq\theta} < 0$. This implies that there has to be a local minimizer $(\theta^m, \hat{\theta}^m)$ of $\Phi^{\eta''}$ such that $\theta^m \geq \theta'$ and $\hat{\theta}^m \leq \hat{\theta}'$ (and strict inequalities if $\theta' < \bar{\theta}$, respectively $\hat{\theta}' > \underline{\theta}$): Consider the auxiliary minimization problem $\Phi^{\eta''}$ on

$[\theta', \tilde{\theta}] \times [\underline{\theta}, \hat{\theta}']$ where $\tilde{\theta} > \theta'$ is the highest type in the domain of the minimization of $\Phi^{\eta''}$ in step 2 of the algorithm, i.e. either $\tilde{\theta} = \bar{\theta}$ or $\tilde{\theta}$ solves $1 - F(\tilde{\theta}) = \eta''$. The auxiliary minimization problem has a solution $(\theta^m, \hat{\theta}^m)$ as $\Phi^{\eta''}$ is continuous. As $\Phi^{\eta''}_{\theta}(\theta', \hat{\theta}) \leq \Phi^{\eta''}_{\theta}(\theta', \hat{\theta}') < 0$ for all $\hat{\theta} \leq \hat{\theta}'$, $\theta^m \neq \theta'$ unless $\theta' = \bar{\theta}$. Similarly, $\hat{\theta}^m \neq \hat{\theta}'$ unless $\hat{\theta}' = \underline{\theta}$ because $\Phi^{\eta''}_{\hat{\theta}}(\theta, \hat{\theta}') \geq \Phi^{\eta''}_{\hat{\theta}}(\theta', \hat{\theta}') > 0$ for all $\theta \geq \theta'$. This implies that $(\theta^m, \hat{\theta}^m)$ is also a local minimizer when $\Phi^{\eta''}$ is minimized over the full domain. By assumption, the only local minimizer in this problem is $(\theta(\eta''), \hat{\theta}(\eta''))$ which implies $\hat{\theta}^m = \hat{\theta}(\eta'') < \hat{\theta}(\eta') < \theta(\eta') < \theta(\eta'') = \theta^m$ and establishes the result.

Preliminary result 3: In step 2 of the algorithm, $\min_{\theta,\hat{\theta}} \Phi^{\eta}(\theta, \hat{\theta})$ is strictly increasing in η for $\eta \leq \bar{\eta}$.

Let $\bar{\eta} \geq \eta' > \eta'' \geq 0$ and note from preliminary result 2 that $\hat{\theta}(\eta'') \leq \hat{\theta}(\eta') < \theta(\eta') \leq \theta(\eta'')$. For brevity, let $\theta' = \theta(\eta')$ and $\hat{\theta}' = \hat{\theta}(\eta')$. Then, $\Phi^{\eta'}(\theta', \hat{\theta}') = \Phi^{\eta''}(\theta', \hat{\theta}') + \int_{\hat{\theta}'}^{\theta'} \int_{q(\hat{\theta}', \eta'')}^{q(\hat{\theta}', \eta')} c_{q\theta}(y, x)\, dy\, dx - \int_{\hat{\theta}'}^{\theta'} \int_{q(x, \eta'')}^{q(x, \eta')} c_{q\theta}(y, x)\, dy\, dx > \Phi^{\eta''}(\theta', \hat{\theta}') + \int_{\hat{\theta}'}^{\theta'} \int_{q(\hat{\theta}', \eta'')}^{q(\hat{\theta}', \eta')} c_{q\theta}(y, x)\, dy\, dx > \Phi^{\eta''}(\theta', \hat{\theta}') > \min_{\theta,\hat{\theta}} \Phi^{\eta''}(\theta, \hat{\theta})$ where the first inequality holds as $q(x, \eta') > q(x, \eta'') > s(\theta)$. The second inequality follows from $c_{qq\theta} < 0$ and the necessary first order condition for minimizing $\Phi^{\eta'}$: $\int_{\hat{\theta}'}^{\theta'} c_{q\theta}(q(\hat{\theta}', \eta'), x)\, dx \geq 0$.

Preliminary result 4: $\bar{\eta} = \max \eta(\theta)$.

Let $\eta^{max} = \max \eta(\theta)$. Preliminary result 1 states that there have to be θ' and $\hat{\theta}'$ such that (NLIC) binds from θ' to $\hat{\theta}'$ and η attains its maximum at one of the two types. In fact, $\eta(\theta') = \eta(\hat{\theta}') = \eta^{max}$: If both types are interior, this is implies by (C3). For boundary types, this is implied by lemma 8 and the definition of η^{max}. Hence, $\Phi^{\eta^{max}}(\theta', \hat{\theta}') \leq 0$ by theorem 2. This has to hold with equality as – by the definition of η^{max} and the monotonicity of $q(\cdot, \eta)$ in η – $\Phi^{\eta^{max}}(\theta', \hat{\theta}') \geq \Phi(\theta', \hat{\theta}') = 0$.

Step 1: The algorithm assigns the optimal decision to all types from which (NLIC) binds in the optimal mechanism.

If θ is an interior type from which (NLIC) binds, this follows immediately from theorem 2 and the assumption that Φ^{η} has a unique local minimizer. This leaves the case where (NLIC) is binding from $\bar{\theta}$. The algorithm assigns the highest $\eta \leq \bar{\eta}$ for which $\bar{\theta}$ is part of the Φ^{η} minimizer. Lemma 9 states that $\eta(\bar{\theta}) = \eta(\hat{\theta}^m)$ where $\hat{\theta}^m = max\{\hat{\theta} :$ (NLIC) binds from $\bar{\theta}$ to $\hat{\theta}\}$. Then, $(\bar{\theta}, \hat{\theta}^m)$ are the unique local mininizer of $\Phi^{\eta(\bar{\theta})}$ by theorem 2. Also by theorem 2, $\Phi^{\eta(\bar{\theta})}(\bar{\theta}, \hat{\theta}^m) \leq 0$ which implies $\eta(\bar{\theta}) \leq \bar{\eta}$ by preliminary result 3. Therefore, what remains to be shown is the following: If $\Phi^{\eta(\bar{\theta})}(\bar{\theta}, \hat{\theta}^m) < 0$, then

$\bar{\theta}$ is not part of a local minimizer of $\Phi^{\eta'}$ for any $\eta' \in (\eta(\bar{\theta}), \bar{\eta}]$.

By the continuity of η and preliminary result 4, there is a type θ_0 such that $\eta(\theta_0) = \eta'$. By preliminary result 1, there is a θ' with $\eta(\theta') = \eta'$ and (NLIC) binds to of from θ'. If (NLIC) binds from θ'' to θ', then $\theta'' \neq \bar{\theta}$ as $\eta(\bar{\theta}) < \eta(\theta')$ would contradict lemma 8. As (θ'', θ') minimize $\Phi^{\eta'}$ by theorem 2 (and lemma 8 in case $\theta' = \underline{\theta}$), $\bar{\theta}$ will then not be part of the unique $\Phi^{\eta'}$ minimizer. If (NLIC) binds from θ' to $\hat{\theta}'$, then $\hat{\theta}' \neq \bar{\theta}$ as $\hat{\theta}' < \theta'$. Again by theorem 2 (and lemma 8 in case $\hat{\theta} = \underline{\theta}$), $(\theta', \hat{\theta}')$ minimize $\Phi^{\eta'}$. Hence, $\bar{\theta}$ is not part of the unique $\Phi^{\eta'}$ minimizer.

Step 2: The algorithm assigns the optimal decision to all types to which (NLIC) binds in the optimal mechanism. The proof is analogous to the proof of step 1.

Step 3: The algorithm assigns the optimal decision to all types where (NLIC) is slack in the optimal mechanism.

If a type θ' where (NLIC) is slack is not assigned a decision in step 4a, the algorithm assigns the optimal decision because theorem 1 ensures that η is constant between θ' and the type closest to θ' from or to which (NLIC) binds. Hence, it is sufficient to show that θ' does not minimize $\Phi^{\eta'}$ for $\eta' \in [0, \bar{\eta}]$. Suppose otherwise for some η'. If there is a type θ_0 such that $\eta(\theta_0) = \eta'$, then preliminary result 1 implies that there is a type θ'' with $\eta(\theta'') = \eta'$ such that (NLIC) binds to or from θ''. Given theorem 2 and the assumption that there is a unique local minimizer of $\Phi^{\eta'}$, this would contradict that (NLIC) is slack at θ_0. (By lemma 8, this is also true if θ'' or θ_0 are boundary types.)

If $\eta(\theta) > \eta'$ for all types, then (NLIC) binds from $\bar{\theta}$ to $\underline{\theta}$ and $(\bar{\theta}, \underline{\theta})$ are local minimizers of $\Phi^{\eta''}$ for some $\eta'' > \eta'$; i.e. $\int_{\underline{\theta}}^{\bar{\theta}} c_{q\theta}(q(\underline{\theta}, \eta''), x)\, dx \geq 0$ and $\int_{q(\underline{\theta},\eta'')}^{q(\bar{\theta},\eta'')} c_{q\theta}(y, \bar{\theta})\, dy \geq 0$. As $q(\cdot, \eta)$ is increasing in η, this implies that $(\bar{\theta}, \underline{\theta})$ is also a local minimizer of $\Phi^{\eta'}$. By assumption, the $\Phi^{\eta'}$ minimizer is unique. As (NLIC) is slack at θ', $\underline{\theta} \neq \theta' \neq \bar{\theta}$ which contradicts that θ' is part of the $\Phi^{\eta'}$ minimizer. Finally, $\eta(\theta) < \eta'$ for all types is impossible: By the definition of $\bar{\eta}$ and preliminary result 3, incentive compatibility would be violated in this case. $\square$

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Webappendix

Adverse selection without single crossing

Christoph Schottmüller

This webappendix contains supplementary material. First, there are several solved numerical examples. I give some details to the example of the main text and then turn to two other examples. The first example illustrates the use of the algorithm to numerically solve complicated problems where non-local incentive constraints bind from an interval of interior types. The second example features an optimal contract menu in which some types are bunched. Also the third derivative $c_{q\theta\theta}$ is negative in the relevant range leading to a downward sloping sign switching function s. As the third derivative does not change sign in the relevant range of q, the results from the paper still apply. The *Mathematica* (or *Python*) codes containing detailed calculations of all examples can be found online (`www.schottmueller.me`).

Second, the webappendix contains proofs that have been used in the same (or a very similar) way before. Existence of an optimal contract menu and optimality of non-stochastic contracts is shown similarly to Jullien (2000). The variational condition (C3) is shown as in Araujo and Moreira (2001) and Araujo and Moreira (2010).

A. Numerical Examples

A.1. Addendum to the example in the main text

I skipped showing that $(\bar{\theta}, \underline{\theta})$ are the unique minimizers of all relevant Φ^η. Showing that $(\bar{\theta}, \underline{\theta})$ satisfy (6) and (7) is easy: When plugging in $\theta = 1/4$ and $\hat{\theta} = 1/2$, (6) becomes

$$0 \leq \frac{9}{128} - \frac{6}{1280(2-\eta)} - \frac{126}{25600(2-\eta)^2}$$

which is true for $\eta \leq 1.7$ and therefore for all $\eta \leq \bar{\eta} = 1.67$. Similarly, (7) becomes

$$0 \leq \frac{1}{160}\left(10 - \frac{1}{2-\eta}\right)$$

which holds for all $\eta \leq 1.9$.

It remains to verify that $(\underline{\theta}, \bar{\theta})$ is the unique minimizer of Φ^η for all relevant η. First, I show that it is impossible that both (6) and (7) hold with equality, i.e. there is no pair of interior types minimizing Φ^η. This is done by contradiction.

Suppose there are $1/4 < \hat{\theta}'' < \theta'' < 1/2$ that minimize Φ^η locally. As $q(\theta, \eta) > s(\theta)$ and since $\Phi^\eta(\theta, \theta) = 0$, there has to be a $(\hat{\theta}', \theta')$ with $\hat{\theta}'' < \hat{\theta}' < \theta' < \theta''$ such that $(\theta', \hat{\theta}')$ locally maximize Φ^η: To see this, consider the maximization problem $\max_{\theta,\hat{\theta}} \Phi^\eta(\theta, \hat{\theta})$ over the domain $[\hat{\theta}'', \theta''] \times [\hat{\theta}'', \theta]$ (note the constraint $\hat{\theta} \leq \theta$). By the Weierstrass theorem, this problem has a solution $(\theta', \hat{\theta}')$. In this solution, $\hat{\theta}' < \theta'$ because $q(\theta, \eta) > s(\theta)$ and therefore $\Phi^\eta(\theta, \theta - \varepsilon) > \Phi^\eta(\theta, \theta) = 0$. Clearly, $(\theta', \hat{\theta}') \neq (\theta'', \hat{\theta}'')$ because $(\theta'', \hat{\theta}'')$ is a local minimizer of Φ^η. Given that $\hat{\theta}' \neq \hat{\theta}''$, it is easy to see that $\theta' \neq \theta''$ because the strict monotonicity of $q(\cdot, \eta)$ implies that $\int_{q(\hat{\theta}')}^{q(\theta'')} c_{q\theta}(y, \theta'')\, dy \neq 0$ for any $\hat{\theta}' \neq \hat{\theta}''$, i.e. (7) cannot hold with equality for θ'' and $\hat{\theta}' \neq \hat{\theta}''$. Similarly, $\hat{\theta}'' \neq \hat{\theta}'$ given that $\theta' \neq \theta''$ as (8) cannot hold for $\hat{\theta}''$ and $\theta' \neq \theta''$ with equality. Summing up, there exists a local Φ^η maximizer $(\theta', \hat{\theta}')$ such that $\hat{\theta}'' < \hat{\theta}' < \theta' < \theta''$.

As $(\hat{\theta}', \theta')$ is a local maximizer of Φ^η, $(\hat{\theta}', \theta')$ also satisfy (6) and (7). Consider the function $g(\hat{\theta}) = \int_{\hat{\theta}}^{\theta''} c_{q\theta}(q(\hat{\theta}, \eta), x)\, dx$. Clearly, $g(\theta'') = 0$ and, by (7), $g(\hat{\theta}'') = 0$. As (7) is satisfied with equality for $(\hat{\theta}', \theta')$ and as $\theta' < \theta''$, we must have $g(\hat{\theta}') > 0$. Because $q(\theta'', \eta) > s(\theta'')$, $g(\theta'' - \varepsilon) < 0$ for $\varepsilon > 0$ small enough. Since g is continuous, there has to exist some $\hat{\theta}''' \in (\hat{\theta}', \theta'')$ such that $g(\hat{\theta}''') = 0$. Consequently, g is zero at θ'', $\hat{\theta}'''$ and $\hat{\theta}''$. I will now show that g is convex for the relevant range of η which implies that g can attain each value at most twice. This will be the desired contradiction. Straightforward calculation gives

$$g''(\hat{\theta}) = \frac{1}{\theta''} \left(\frac{12\hat{\theta} - 4\theta''}{5(2-\eta)} + 2 \right)$$

which is positive for $\eta < 2$.

Second, suppose that $\bar{\theta}$ is the solution for $\theta(\eta)$. Then it has to be shown that there is no interior minimizer $\hat{\theta}$ solving (7) with equality. Plugging in $\theta = 1/2$ into (7) gives after some rearranging

$$\frac{1}{2} - (1 - \hat{\theta})2\hat{\theta} - (1 - 2\hat{\theta})\frac{2\hat{\theta}^2}{5(2-\eta)} = 0.$$

Note that the first order condition is obviously satisfied for $\hat{\theta} = 1/2$ and that the left hand side of the last equation is strictly convex on $[1/4, 1/2]$ for $\eta < 2$, i.e. there is at

most one interior solution. This solution will however be a Φ^η maximizer. This can be readily seen from the graph or the fact that $\Phi^\eta(1/2, 1/2-\varepsilon) > 0$ as $q^\eta > s$. Hence, there is no interior $\hat{\theta}(\eta)$.

Third, given that $\hat{\theta}(\eta) = \underline{\theta}$ it has to be shown that there cannot be an interior $\theta(\eta)$ solving (6) with equality. Plugging in $\hat{\theta} = 1/4$, (6) becomes

$$\frac{32\theta^5}{25(2-\eta)^2} - 8\theta^3 + \theta + \frac{\theta}{10(2-\eta)} - \frac{1}{32} - \frac{1}{160(2-\eta)} - \frac{1}{3200(2-\eta)^2} = 0.$$

The left hand side of the last equation is strictly concave on the relevant range for η. Along the lines of the argument in the previous paragraph, this is sufficient for showing that there is no interior Φ^η minimizer.

A.2. Non-local constraints binding in the interior

This example deals with optimal non-linear pricing by a monopolist. Its main purpose is to illustrate how to use the algorithm to solve the problem numerically in the more complicated case in which non-local incentive constraints are binding from an interval of interior types. Roughly speaking, the private information of the consumers is their demand elasticity.[24] More specifically, their utility function is

$$v(q, \theta) = q^\theta + \frac{3}{10}\theta.$$

The cross-derivative $v_{q\theta} = q^{\theta-1}(1 + \theta ln(q))$ changes sign at $q = e^{-1/\theta}$. Hence, marginal utility is increasing in type for quantities $q > e^{-1/\theta}$ but decreasing in type for $q < e^{-1/\theta}$, i.e. SC is violated. Types are uniformly distributed over $[0.45, 0.95]$. The monopolist has constant marginal costs of 0.96. In this example, (CVR) does not hold everywhere. However, it is easy to verify numerically that discontinuous jumps are irrelevant.[25] The relevant first order condition for using the algorithm is

$$2\theta q^{\theta-1} - 2 * 0.96 - (1 - 2(\theta - 0.45) - \eta)(\theta q^{\theta-1} ln(q) + q^{\theta-1}) = 0.$$

[24] The price elasticity of demand is here defined as the relative demand change caused by a 1% increase in the marginal price. Using the first order condition $p'(q) = \theta q^{\theta-1}$, the elasticity for the utility function below can be derived as $|1/(\theta - 1)|$.

[25] The way this is done in the Python file is that for every given type it is verified that (4) defines a monotone one-to-one relationship between q and η as long as $q < 1$. As (5) holds, there is no distortion at the top. Therefore, the numerical check and theorem 5, which states that $\eta^-(\theta') \geq \eta^+(\theta')$ at discontinuity types θ', imply that the monotonicity constraint would be violated at the highest hypothetical discontinuity type.

Although the problem is too complicated for a algebraic solution, solutions can be calculated numerically. I describe briefly how this is done and refer to a Python file where the computation for this example is done.

For a given η, the equation can be solved for a fine grid of types. The rent function under $q(\eta, \theta)$ can be approximated using the types on the grid for which the decision is calculated: As the slope of the rent function is $-c_\theta$, it can be calculated at every grid type. If the grid is fine enough, one can approximate the slope of the rent function for types not on the grid using the slope of close types that are on the grid. Consequently, the rents of all types can be (approximately) calculated. This allows to write the $\Phi^\eta(\theta, \hat{\theta})$ function according to (NLIC). This function can then be minimized numerically. This is done repeatedly for increasing values of η (starting from zero) until the minimum of $\Phi^\eta(\theta, \hat{\theta})$ is zero. This gives $\bar{\eta}$. In the process, one has obtained already the minimizing types for $\eta < \bar{\theta}$ between which non-local incentive constraints will bind. By theorem 1, η is constant at all other types which allows to compute the optimal decision for those types.

In this example, the solution in figure WA.1 emerges. The non-local incentive constraints are binding from an interval of interior types to the lowest type. Non-local incentive constraints are slack for the highest types. Therefore, the optimal decision coincides with the relaxed decision for those types.

A.3. Bunching and differently signed third derivatives

I want to discuss the assumptions on third derivatives, i.e. $c_{qq\theta} < 0$ and $c_{q\theta\theta} > 0$. The fact that these derivatives do not change sign ensures that the cross derivative $c_{q\theta}$ changes sign only once for any given θ (or q). While this property is important for the analysis (e.g. for lemma 3), it is immaterial which sign the third derivatives have as long as the sign is the same for all relevant decisions and types. To illustrate this (and also to show an example where the monotonicity constraint binds) consider the following setup: Types are distributed uniformly on $[2, 3]$ and the principal's objective is the expected value of $q(\theta) - t(\theta)$. The agent's utility is given by

$$\pi(q, \theta) = t(\theta) - \frac{(q - \sigma/\theta)^2}{\theta^2} + \gamma(3 - \theta).$$

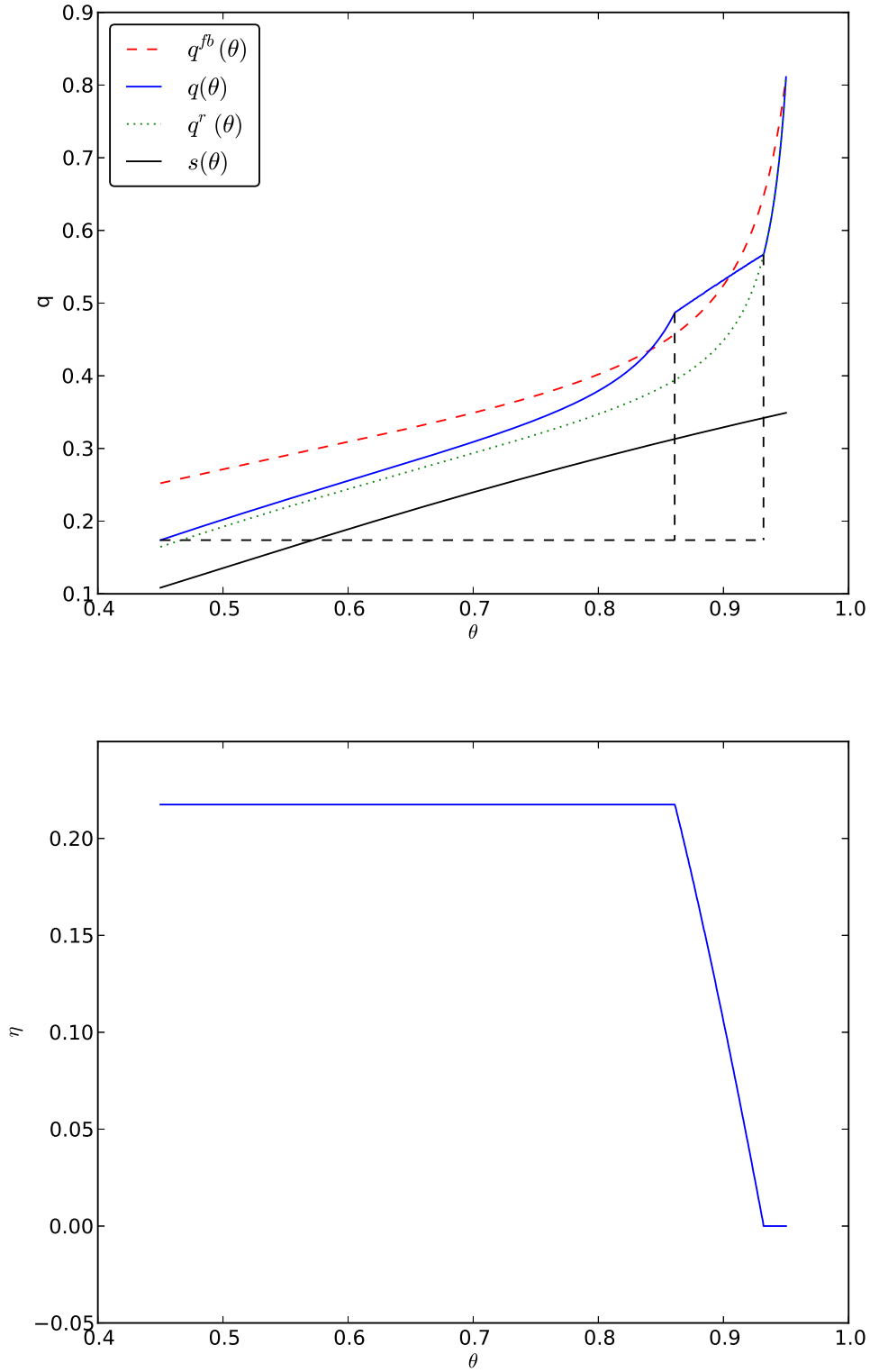

Figure WA.1: non-local constraints bind from an interval of interior types to the lowest type

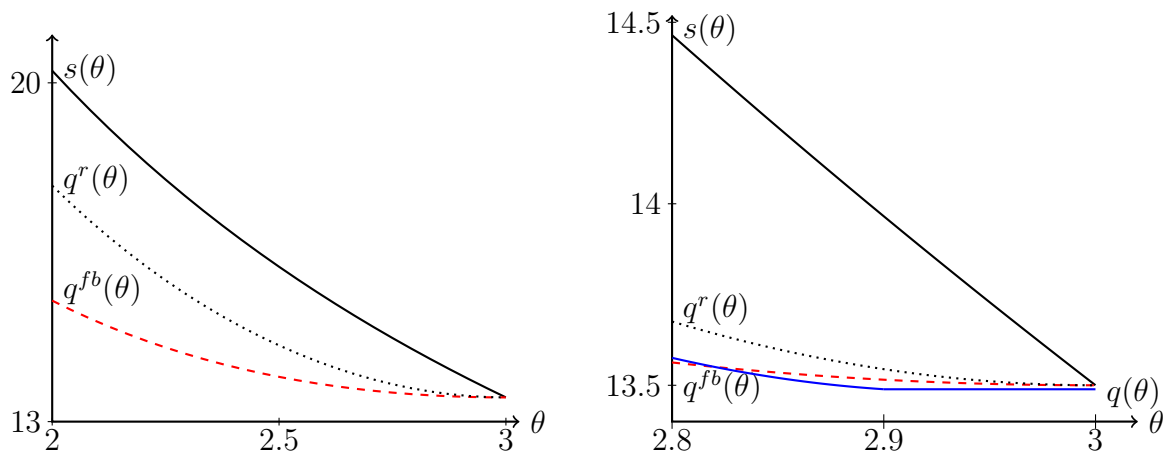

Figure WA.2: numerical example: bunching

The story behind this example is in the context of compensation of workers.[26] The principal is the owner of a firm and the agent a worker the firm wants to hire. For the quality of the worker, talent and effort are relevant, e.g. talent is what the worker produces in a regular working time like the 40 hours week and effort is the additional time he is willing to invest. Assume the worker creates value $q = e\theta + T$ where T is his talent, e is the unobservable effort and θ is his type. The owner of the firm observes a public signal, e.g. education, which is a mix of talent and productivity (he does not observe T and θ directly). To be precise, assume that the signal is $\sigma = \theta * T$. Given this signal, a more productive worker will have lower talent and vice versa. The production function of the manager for a given signal is $q = e\theta + \sigma/\theta$ where q is the quantity/value produced by the worker. If costs of effort are $e^2 + \gamma(\theta)$ and the worker's preferences are quasilinear in money, his utility function is the one above (with t denoting the wage).

Here the parameter values $\sigma = 27$ and $\gamma = 12$ are used. In this case, third derivatives have the following signs in the relevant range of the decision: $c_{qq\theta} < 0$ and $c_{q\theta\theta} < 0$. Consequently, the sign switching decision $s(\theta)$ is downward sloping. As depicted in figure WA.2a, first best decision and relaxed decision are also downward sloping.

Note that (4) is linear in q. It is therefore easy to verify that (4) defines a strictly monotonically one-to-one relationship between η and q (for a given θ) in the relevant range. Following the arguments in the proof of proposition 2, the solution will therefore be continuous.

[26] A related example can be found in Araujo and Moreira (2010).

The example has an additional feature which is that the monotonicity constraint will be binding: q^r has slope 0 at the top type $\bar{\theta} = 3$. For $\eta > 0$, $q(\cdot, \eta)$ is increasing on some interval $(\theta'(\eta), \bar{\theta}]$. Hence, the monotonicity constraint will have to bind for types in the top end of the type distribution if (NLIC) binds. It is easy to check that (NLIC) cannot bind to these high types (i.e. $\Phi^0(\theta, \hat{\theta}) > 0$ for $\hat{\theta}$ high enough). Since $q(\cdot, \eta)$ satisfies the monotonicity constraint for all types at the lower end to which (NLIC) can bind, i.e. $\Phi^0(\bar{\theta}, \underline{\theta}) < 0$ and the boundary types minimize Φ^0, theorem 2 implies that (NLIC) will bind at types minimizing Φ^η.

It turns out that also in this example $(\bar{\theta}, \underline{\theta})$ is the unique local minimizer of $\Phi^\eta(\theta, \hat{\theta})$ for all relevant $\eta \geq 0$. Therefore only the non-local incentive constraint from the highest to the lowest type is binding. However, the monotonicity constraint is binding for the highest types. For each $q(\cdot, \eta)$, the optimal bunching interval $[\theta_s(\eta), \bar{\theta}]$ is determined by the condition

$$\int_{\theta_s(\eta)}^{\bar{\theta}} [u_q(q(\theta, \eta), \theta) - c_q(q(\theta, \eta), \theta)] f(\theta) + (1 - F(\theta) - \eta(\theta)) c_{q\theta}(q(\theta, \eta), \theta) \, d\theta = 0.$$

For each η, this gives the bunching interval, i.e. $q(\cdot, \eta)$ is the normal $q(\cdot, \eta)$ on $[\underline{\theta}, \theta_s(\eta)]$ and $q(\theta_s(\eta), \eta)$ for all types in $(\theta_s(\eta), \bar{\theta}]$. $\bar{\eta}$ is now determined such that $\Phi^{\bar{\eta}}(3, 2) = 0$ where Φ^η is calculated using the ironed out solution. Here, $\bar{\eta}$ turns out to be approximately 0.18 and the solution for the highest types is depicted in figure WA.2b. The solution exhibits bunching of types in $[2.9, 3]$.

B. Proofs and derivations

I prove the third point of lemma 1 first as an intermediate result.

Lemma WA1. *Every implementable decision q is bounded from above.*

Proof: Take any two types θ' and θ'' with $q(\theta') > s(\bar{\theta})$ and $q(\theta'') > s(\bar{\theta})$ and let $\theta'' > \theta'$.[27] Incentive compatibility implies

$$c(q(\theta''), \theta') - c(q(\theta'), \theta') \geq t(\theta'') - t(\theta') \geq c(q(\theta''), \theta'') - c(q(\theta'), \theta'').$$

[27] If two such types do not exist, there is nothing to prove: Either there is exactly one type θ with $q(\theta) > s(\bar{\theta})$ which means that q is bounded from above by $q(\theta)$ or q is bounded from above by $s(\bar{\theta})$.

Combining the two inequalities gives

$$\int_{\theta'}^{\theta''} \int_{q(\theta'')}^{q(\theta')} c_{q\theta}(q,\theta)\, dq\, d\theta \geq 0. \tag{WA.1}$$

As $q(\theta') > s(\bar{\theta})$ and $q(\theta'') > s(\bar{\theta})$, the last inequality can only be satisfied if $q(\theta'') \geq q(\theta')$. Hence, q is monotonically increasing on the set of types for which $q(\theta) > s(\bar{\theta})$. Now suppose that $q(\theta)$ was not bounded from above. Then there exists exactly one limit type θ'' such that for any $q' \in \mathbb{R}$ and any $\varepsilon > 0$ there exists a $\theta' \in (\theta'' - \varepsilon, \theta'')$ with $q(\theta') > q'$. Since $q(\theta'')$ is finite and $c_{qq\theta} < 0$, (WA.1) cannot hold for θ' and θ'' for q' high enough.[28] This contradicts that q is not bounded from above. □

Proof of lemma 1: Boundedness of q is proven in lemma WA1. As the range of q is bounded by lemma WA1, the envelope theorem of corollary 4 in Milgrom and Segal (2002) applies. Monotonicity is proven in lemma 1 of Araujo and Moreira (2010). For completeness both proofs are sketched here:

Recall that $\pi(\theta) \equiv t(\theta) - c(q(\theta),\theta)$. Incentive compatibility between θ and $\hat{\theta} < \theta$ implies

$$\frac{c(q(\theta),\hat{\theta}) - c(q(\theta),\theta)}{\theta - \hat{\theta}} \geq \frac{\pi(\theta) - \pi(\hat{\theta})}{\theta - \hat{\theta}} \geq \frac{c(q(\hat{\theta}),\hat{\theta}) - c(q(\hat{\theta}),\theta)}{\theta - \hat{\theta}}.$$

As q is bounded and c is C^3, the inequalities imply that π is a Lipschitz function. It is, therefore, almost everywhere differentiable with $\pi_\theta(\theta) = -c_\theta(q(\theta),\theta)$ for almost all types. The envelope condition follows by the fundamental theorem of calculus.

To show monotonicity, take a θ' with $q(\theta') > s(\theta')$ and let q be right continuous with left-hand side limit at θ'. By right continuity, there exists a type θ'' such that $c_{q\theta}(q(\theta_1),\theta_2) < 0$ for all $\theta_1, \theta_2 \in (\theta', \theta'')$.

First, it is shown that $q^-(\theta') \leq q^+(\theta')$. Suppose otherwise. Then – by right continuity of q at θ' – there exists a $\delta > 0$ such that $q(\theta) < q^-(\theta')$ for all $\theta \in (\theta', \theta' + \delta)$. But then $\Phi(\theta' + \delta, \theta' - \varepsilon) < 0$ for some $\varepsilon > 0$. This contradicts NLIC and proves the claim $q^-(\theta') \leq q^+(\theta')$.

Second, q is increasing on (θ', θ''). This follows directly from (WA.1) and $c_{q\theta}(q(\theta_1),\theta_2) < 0$ for all $\theta_1, \theta_2 \in (\theta', \theta'')$.

To complete the monotonicity proof, now take a θ' such that $q(\theta') < s(\theta')$ and let

[28] This argument does not require $q(\theta'')$ to be above $s(\bar{\theta})$: Because of $c_{qq\theta} < 0$, $\lim_{q'\to\infty} \int_{q(\theta'')}^{q'} c_{q\theta}(q,\theta)\, dq = -\infty$.

q be right continuous with left-hand side limit at θ'. By right continuity, there exists a type $\theta'' > \theta'$ such that $c_{q\theta}(q(\theta_1), \theta_2) > 0$ for all $\theta_1, \theta_2 \in (\theta', \theta'')$.

First, it is shown that $q^-(\theta') \geq q^+(\theta')$. Suppose otherwise. Then – by right continuity – there exists a $\delta > 0$ such that $q(\theta) > q^-(\theta')$ for all $\theta \in (\theta', \theta' + \delta)$. But then $\Phi(\theta' + \delta, \theta' - \varepsilon) < 0$ for some $\varepsilon > 0$ which contradicts (NLIC).

Second, q is decreasing on (θ', θ''). This follows directly from (WA.1) and $c_{q\theta}(q(\theta_1), \theta_2) > 0$ for all $\theta_1, \theta_2 \in (\theta', \theta'')$. □

Proof of lemma 2: By $u_{qq} - c_{qq} < 0$, q^{fb} is implicitly defined by the first order condition $u_q(q, \theta) - c_q(q, \theta) = 0$. The implicit function theorem shows that $sgn\{q^{fb}_\theta(\theta)\} = sgn\{u_{q\theta}(q^{fb}(\theta), \theta) - c_{q\theta}(q^{fb}(\theta), \theta)\}$. Assumption 2 implies that $c_{q\theta}(q^{fb}(\theta), \theta) < 0$ and therefore $q^{fb}_\theta(\theta) > 0$ for all $\theta \in \Theta$.

The concavity of the relaxed program follows directly from assumption 1 as $u_{qq} - c_{qq} + (1 - F)c_{qq\theta} < 0$. Therefore, q^r is defined by the first order condition (3). We immediately get $q^r(\bar{\theta}) = q^{fb}(\bar{\theta}) > s(\bar{\theta})$ where the inequality follows from assumption 2. The first order condition (3) implies that no type can have $q^r(\theta) = s(\theta)$: If this was the case, (3) would imply that $q^r(\theta) = q^{fb}(\theta)$ but this would contradict assumption 2 which says $q^{fb}(\theta) > s(\theta)$. As the continuity and differentiability assumptions on u and c imply that q^r is continuous, we get that $q^r(\bar{\theta}) > s(\bar{\theta})$ implies $q^r(\theta) > s(\theta)$ for all types. Therefore, $c_{q\theta}(q^r(\theta), \theta) < 0$ and $q^r \leq q^{fb}(\theta)$ by (3) where the inequality is strict for all types but $\bar{\theta}$.

Using the implicit function theorem on (3) yields together with assumption 1 that $q^r_\theta > 0$. □

B.1. Existence of an optimal contract menu

This subsection shows that an optimal contract menu exists and therefore the characterization done in the paper is meaningful. Before showing existence, one intermediate result is derived which implies that the optimal decision is bounded from above by a specific value $\bar{q}$ defined below.

Define $\tilde{q} > s(\bar{\theta})$ such that $\int_0^{\tilde{q}} c_{q\theta}(y, \bar{\theta})\, dy = 0$. As $\tilde{q} > s(\bar{\theta})$, the integral is strictly decreasing in $\tilde{q}$. By $c_{qq\theta} < 0$, the integral is concave in $\tilde{q}$ which implies $\lim_{\tilde{q}\to\infty} \int_0^{\tilde{q}} c_{q\theta}(y, \bar{\theta})\, dy = -\infty$. Hence, $\tilde{q}$ exists and is unique and therefore properly defined.

Lemma WA2. *Any incentive compatible contract menu with a decision $q(\theta)$ above*

$\bar{q} = max\{q^{fb}(\bar{\theta}), \tilde{q}\}$ for some type $\theta < \bar{\theta}$ is dominated by a contract menu consisting of decision

$$q^c(\theta) = min\{q(\theta), \bar{q}\}$$

and transfers such that $\pi(\underline{\theta}) = 0$ and $\pi_\theta(\theta) = \int_{\underline{\theta}}^{\theta} -c_\theta(q^c(x), x)\, dx$.

Proof. By lemma WA1, the optimal decision is bounded. The boundedness of q and the assumption that c is three times continuously differentiable imply that the envelope theorem by Milgrom and Segal (2002) applies; see lemma 1. The principal's problem can, therefore, be written as maximization of the virtual valuation subject to NLIC.

Note that incentive compatibility of $q^c(\theta)$ is obvious if $q(\theta) > \bar{q}$ for all θ. Now define $\theta^m = inf\{\theta : \ q(\theta) > \bar{q}\}$. Note that incentive compatibility from θ^m to any lower type is not affected by the change from q to q^c since $\Phi(\theta^m, \hat{\theta})$ does not change.

The next step is to see that $q(\theta) > \bar{q}$ for all $\theta > \theta^m$. By way of contradiction, suppose there exists a type $\theta'' > \theta^m$ with $q(\theta'') \leq \bar{q}$. By the definition of θ^m, there exists a type $\theta' \in [\theta^m, \theta'')$ with $q(\theta') > \bar{q}$. By the definition of $\bar{q}$ and $c_{q\theta\theta} > 0$, (WA.1) is violated for θ' and θ'': $\int_{q(\theta'')}^{q(\theta')} c_{q\theta}(y, \theta)\, dy < 0$ for any θ as $0 \leq q(\theta'') < \bar{q} < q(\theta')$. This contradicts that q is incentive compatible. Consequently, $q(\theta) > \bar{q}$ for all $\theta > \theta^m$.

Therefore, all types above θ^m will have $\bar{q}$ as their changed decision. It follows from lemma 3 that only incentive compatibility from types above θ^m to types below θ^m has to be checked. Take an arbitrary $\theta > \theta^m$ and some $\hat{\theta} < \theta^m$. Then $\Phi(\theta, \hat{\theta}) = \Phi(\theta^m, \hat{\theta}) - \int_{\theta^m}^{\theta} \int_{q(\hat{\theta})}^{\bar{q}} c_{q\theta}(y, x)\, dy\, dx > 0$ where the inequality follows from the incentive compatibility between θ^m and $\hat{\theta}$ under $q(\theta)$, the definition of $\tilde{q}$ and $c_{q\theta\theta} > 0$.

The concavity of the virtual valuation implies that the principal's payoff under $q^c(\theta)$ is higher than under $q(\theta)$. In fact, it is strictly higher if $\theta^m < \bar{\theta}$ because the set of types with a changed decision has measure $1 - F(\theta^m) > 0$ in this case. □

Next, the existence proof. It is assumed that $q^v(q, \theta) \geq q^s(q, \theta)$ for all $q \in [0, s(\theta)]$ and all $\theta \in [\underline{\theta}, \bar{\theta}]$ and therefore every q that is not monotonically increasing is dominated by a decision that is monotonically increasing (see the first part of the proof of proposition 1 in the main text). Given this and the previous two results, the existence proof in Jullien (2000) applies. For completeness, I replicate the proof briefly. The problem of

the principal is the program

$$\max_{q(\theta)} \int_{\underline{\theta}}^{\bar{\theta}} (u(q(\theta),\theta) - c(q(\theta),\theta))f(\theta) + (1-F(\theta))c_\theta(q(\theta),\theta)\, d\theta$$

subject to

$$\begin{aligned} \Phi(\theta,\hat{\theta}) &\geq 0 \quad \text{for all} \quad \theta,\hat{\theta} \in [\underline{\theta},\bar{\theta}] \\ 0 &\leq q(\theta) \leq \bar{q} \quad \text{for all} \quad \theta \in [\underline{\theta},\bar{\theta}]. \end{aligned}$$

Let W^* be the supremum value of the program. Take a sequence of decision functions such that q^n leads to an objective value larger than $W^* - \frac{1}{n}$ and each q^n is implementable. Because of (9) and the argument there, the sequence can be chosen such that each q^n is an increasing function. Because of lemma WA2, each q^n can be chosen such that $q^n \leq \bar{q}$. Then Helly's selection theorem, see Billingsley (1986) Thm. 25.9, yields that there exists a non-decreasing function q which is the limit of a subsequence q^{n_k} at every point of continuity of q and therefore almost everywhere on $[\underline{\theta},\bar{\theta}]$. Lebesgue's dominated convergence theorem, see Billingsley (1986) Thm. 16.4, yields that the principal's payoff under q is W^*. By lemma WA3 below, q is implementable and therefore an optimal contract menu exists.

Lemma WA3. *Take a sequence of implementable, increasing decision functions $q^n \leq \bar{q}$, $n = 1, 2 \ldots$. Let q be an increasing decision such that q is the pointwise limit of the sequence $(q^n)_{n=1}^{\infty}$ at all points of continuity of q. Then, q is implementable.*

Proof. As q is increasing, only downward incentive compatibility has to be checked; see lemma 3. Take two arbitrary types $\hat{\theta}$ and θ such that $\hat{\theta} < \theta$.

First, assume that q is continuous at $\hat{\theta}$. Define $\tilde{c}_{q\theta} = \max_{q\in[0,q(\bar{\theta})+1],\, \theta\in[\underline{\theta},\bar{\theta}]} |c_{q\theta}(q,\theta)|$. Since $[0, q(\bar{\theta})+1] \times [\underline{\theta},\bar{\theta}]$ is compact and $c_{q\theta}(\cdot)$ is continuous by assumption, $\tilde{c}_{q\theta}$ exists.

Now suppose contrary to the lemma that $\Phi(\theta,\hat{\theta}) = -\varepsilon$ for and some $\varepsilon > 0$ and therefore incentive compatibility is violated under q. From convergence of $(q^n)_{n=1}^{\infty}$, for each $\delta > 0$ there exists an N_δ such that $|q^n(\theta) - q(\theta)| \leq \delta$ for all types at which q is continuous and all $n > N_\delta$. Note that by the monotonicity of q, q is continuous almost

everywhere. Therefore,

$$\Phi(\theta,\hat{\theta}) = \int_{\hat{\theta}}^{\theta} \int_{q(\hat{\theta})}^{q(x)} -c_{q\theta}(y,x)\, dy\, dx \geq \int_{\hat{\theta}}^{\theta} \int_{q^n(\hat{\theta})}^{q^n(x)} -c_{q\theta}(y,x)\, dy\, dx - \int_{\hat{\theta}}^{\theta} 2\delta\tilde{c}_{q\theta}\, dx$$

for an arbitrary $n > N_\delta$. But then choosing a $\delta < \frac{\varepsilon}{2\tilde{c}_{q\theta}(\bar{\theta}-\underline{\theta})}$ shows that $\Phi(\theta,\hat{\theta}) > -\varepsilon$ as $\Phi^{q^n}(\theta,\hat{\theta}) \geq 0$ where Φ^{q^n} denotes Φ under q^n. This contradicts the definition of ε and therefore $q(\theta)$ is incentive compatible.

Second, let q be discontinuous at $\hat{\theta}$. There are two cases: Either (i) $\int_{\hat{\theta}}^{\theta} c_{q\theta}(q(\hat{\theta}),x)\, dx \geq 0$ or (ii) $\int_{\hat{\theta}}^{\theta} c_{q\theta}(q(\hat{\theta}),x)\, dx < 0$. If (i), then

$$\Phi(\theta,\hat{\theta}) = \int_{\hat{\theta}}^{\theta} \int_{q(\hat{\theta})}^{q(x)} -c_{q\theta}(y,x)\, dy\, dx \geq \int_{\hat{\theta}}^{\theta} \int_{q^-(\hat{\theta})}^{q(x)} -c_{q\theta}(y,x)\, dy\, dx$$

because of (i) and $c_{qq\theta} < 0$. By the first step, incentive compatibility is satisfied for all types at which q is continuous and therefore

$$\int_{\hat{\theta}}^{\theta} \int_{q^-(\hat{\theta})}^{q(x)} -c_{q\theta}(y,x)\, dy\, dx \geq 0$$

which then implies $\Phi(\theta,\hat{\theta}) \geq 0$.

If (ii), then

$$\Phi(\theta,\hat{\theta}) = \int_{\hat{\theta}}^{\theta} \int_{q(\hat{\theta})}^{q(x)} -c_{q\theta}(y,x)\, dy\, dx \geq \int_{\hat{\theta}}^{\theta} \int_{q^+(\hat{\theta})}^{q(x)} -c_{q\theta}(y,x)\, dy\, dx$$

because of (ii) and $c_{qq\theta} < 0$. As q is incentive compatible at all points of continuity of q, the last inequality implies $\Phi(\theta,\hat{\theta}) \geq 0$. □

B.2. Addenda proof theorem 1

Skipped cases when showing that η is non-increasing at discontinuity type θ'
The main text skipped the proofs of incentive compatibility in cases (ii) and (iii). Both are provided here. What remains to be shown is that the changed decision is incentive compatible in these two cases. Case (ii) is considered first.

First, take $\hat{\theta} < \theta'$. As described in the main text, the continuity of Φ^c in ε' implies that it is sufficient to show that $\Phi(\theta,\hat{\theta}) > 0$ for all $\hat{\theta} \in (\theta' - \varepsilon, \theta')$ and $\theta \geq \theta'$ for some $\varepsilon > 0$. Suppose, contrary to this, that there are types θ such that $\Phi(\theta,\hat{\theta}) = 0$ for $\hat{\theta} < \theta'$

arbitrarily close to θ'. Then there exists a type $\theta > \theta'$ that is indifferent between his contract and the limit contract associated with $q^-(\theta')$. This implies $\int_{\theta'}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx \leq 0$ as otherwise $\Phi(\theta, \theta' - \varepsilon) < 0$ which would violate incentive compatibility. But then $\Phi(\theta, \theta') = \Phi(\theta, \theta'^-) + \int_{\theta'}^{\theta} \int_{q^-(\theta')}^{q(\theta')} c_{q\theta}(y, x)\, dy\, dx < 0$ by $c_{qq\theta} < 0$ and $\int_{\theta'}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx \leq 0$. Hence, for $\varepsilon > 0$ small enough $\Phi(\theta, \hat{\theta}) > 0$ for all $\theta \neq \hat{\theta}$ and $\hat{\theta} \in (\theta' - \varepsilon, \theta')$.

Second, take $\hat{\theta} \in [\theta', \theta' + \varepsilon)$. Choose $\varepsilon > 0$ small enough such that $q(\theta) = q(\theta')$ for all $\theta \in [\theta', \theta' + \varepsilon)$. Suppose that $\Phi(\theta, \theta') = 0$ for some θ with $q(\theta) > q(\theta')$. Note that $\int_{\theta'}^{\theta} c_{q\theta}(q(\theta'), x)\, dx < 0$ as otherwise $\Phi(\theta, \theta' - \varepsilon'') \leq \Phi(\theta, \theta') - \int_{\theta'}^{\theta} \int_{q^-(\theta')}^{q^+(\theta')} c_{q\theta}(y, x)\, dy\, dx < 0$ for $\varepsilon'' > 0$ small enough. Now let $\varepsilon > 0$ be small enough such that $\int_{\theta'+\varepsilon}^{\theta} c_{q\theta}(q(\theta'), x)\, dx < 0$. The changed contract for type $\hat{\theta} \in [\theta', \theta' + \varepsilon)$ is then less attractive for θ: $\Phi^c(\theta, \hat{\theta}) = \Phi(\theta, \theta') - \int_{\hat{\theta}}^{\theta} \int_{q^c(\hat{\theta})}^{min(q^c(x), q(\theta'))} c_{q\theta}(y, x)\, dy\, dx > 0$ for $\varepsilon' > 0$ small enough because $\int_{\theta'+\varepsilon}^{\theta} c_{q\theta}(q(\theta'), x)\, dx < 0$ implies $\int_{\theta'+\varepsilon}^{\theta} c_{q\theta}(y, x)\, dx < 0$ for $y \in [q^c(\hat{\theta}), min(q^c(x), q(\theta'))]$. Hence, the change relaxes (NLIC). If there is no type θ such that $\Phi(\theta, \theta') = 0$, then – by the definition of case (ii) and the result of the previous paragraph – there is an open neighborhood of θ' such that $\Phi(\theta, \hat{\theta}) > 0$ for all $\hat{\theta}$ in this neigborhood and θ such that $q(\theta) \neq q(\hat{\theta})$. This implies that q^c is incentive compatible for $\varepsilon' > 0$ small enough (as shown in the main text).

Case (iii) is a combination of case (i) and case (ii). Choose $\varepsilon > 0$ small enough such that $q(\theta' - \varepsilon) = q^-(\theta')$ and $q(\theta') = q(\theta' + \varepsilon)$. Analogous to the two cases it is then shown that the change relaxes those non-local incentive constraints that could hold with equality. If there is a type θ such that $\Phi(\theta, \theta') = 0$, then $\int_{\theta'}^{\theta} c_{q\theta}(q(\theta'), x)\, dx < 0$ as shown in case (ii). Taking $\varepsilon > 0$ small enough such that $\int_{\theta'+\varepsilon}^{\theta} c_{q\theta}(q(\theta'), x)\, dx < 0$. Then for $\hat{\theta} \in [\theta', \theta' + \varepsilon)$ the incentive constraint is relaxed: $\Phi^c(\theta, \hat{\theta}) = \Phi(\theta, \theta') - \int_{\hat{\theta}}^{\theta} \int_{q^c(\hat{\theta})}^{min(q^c(x), q(\theta'))} c_{q\theta}(y, x)\, dy\, dx > 0$ for $\varepsilon' > 0$ small enough because $\int_{\theta'+\varepsilon}^{\theta} c_{q\theta}(q(\theta'), x)\, dx < 0$ implies $\int_{\theta'+\varepsilon}^{\theta} c_{q\theta}(y, x)\, dx < 0$ for $y \in [q^c(\hat{\theta}), q(\theta')]$.

Analogously to case (i), say there is a type $\theta > \theta'$ such that $\Phi(\theta, \hat{\theta}) = 0$ for $\hat{\theta} \in (\theta' - \varepsilon, \theta')$. Then $\int_{\theta'}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx > 0$ as shown in case (i) and we can choose $\varepsilon > 0$ small enough such that $\int_{\theta'-\varepsilon}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx > 0$. The changed contract of types $\hat{\theta} \in (\theta' - \varepsilon, \theta')$ is then less attractive for θ: $\Phi^c(\theta, \hat{\theta}) = \Phi(\theta, \theta' - \varepsilon) + \int_{\theta'-\varepsilon}^{\theta} \int_{q(\theta'-\varepsilon)}^{min(q^c(x), q^c(\hat{\theta}))} c_{q\theta}(y, x)\, dy\, dx > 0$ for $\varepsilon' > 0$ small enough as $\int_{\theta'-\varepsilon}^{\theta} c_{q\theta}(q^-(\theta'), x)\, dx > 0$ implies $\int_{\theta'-\varepsilon}^{\theta} c_{q\theta}(y, x)\, dx > 0$ for $y \in (q(\theta' - \varepsilon), min(q^c(x), q^c(\hat{\theta})))$.

B.3. Binding monotonicity constraint

This subsection shows how the properties of theorem 1 extend if the monotonicity constraint is explicitly taken into account. Instead of (4) the solution is then characterized by

$$\nu_\theta(\theta) = (u_q(q(\theta),\theta) - c_q(q(\theta),\theta))f(\theta) + (1 - F(\theta) - \eta(\theta))c_{q\theta}(q(\theta),\theta) \tag{WA.2}$$

where $\nu(\theta)q_\theta(\theta) = 0$ for all $\theta \in \Theta$, i.e. $\nu(\theta)$ is the Lagrange parameter of the monotonicity constraint. If the start and ending type of an interval on which the monotonicity constraint is binding are denoted by θ_s^b and θ_e^b, then obviously $\int_{\theta_s^b}^{\theta_e^b} \nu_\theta(\theta)\, d\theta = 0$. As described in the existing literature on ironing, see Guesnerie and Laffont (1984) or the exposition in Fudenberg and Tirole (1991, ch. 7), the interval is characterized by this last condition and the endpoint conditions $\nu(\theta_s^b) = \nu(\theta_e^b) = 0$. Lemma WA4 formally shows that η in (WA.2) has the properties of theorem 1 also for types at which the monotonicity constraint binds.

Lemma WA4. *If the monotonicity constraint binds for types in (θ_s^b, θ_e^b) in the optimal solution, then there exists a function $\eta(\theta)$ which satisfies the properties of theorem 1 also for types in (θ_s^b, θ_e^b). In particular, η is non-decreasing on (θ_s^b, θ_e^b) and constant if $\Phi(\theta, \hat{\theta}) > 0$ for all $\hat{\theta} \in (\theta_s^b, \theta_e^b)$ and $\theta > \theta_e^b$. Furthermore, η satisfies (i) $\eta(\theta) = \eta(\hat{\theta})$ if $\Phi(\theta, \hat{\theta}) = 0$ and (C1) as well as (C2) hold, (ii) $\int_{\theta_s^b}^{\theta_e^b} \nu_\theta(\theta)\, d\theta = 0$ with $\nu_\theta(\theta)$ defined as in (WA.2).*

Proof. From lemma 4, $\Phi(\theta, \hat{\theta}) > 0$ for all $\theta \in [\theta_s^b, \theta_e^b)$ and $\hat{\theta} < \theta_s^b$. To satisfy similar properties as in theorem 1, $\eta(\theta)$ has therefore to be non-decreasing on (θ_s^b, θ_e^b).

Let η be defined by (4) for all types not in (θ_s^b, θ_e^b). Define η for types in (θ_s^b, θ_e^b) using the following two step procedure: First, all $\hat{\theta} \in (\theta_s^b, \theta_e^b)$ such that there exists a θ with $\Phi(\theta, \hat{\theta}) = 0$ and (C1) as well as (C2) are satisfied are assigned $\eta(\hat{\theta}) = \eta(\theta)$. Second, types in $\theta \in (\theta_s^b, \theta_e^b)$ who are not assigned a value for $\eta(\theta)$ in step 1 are assigned the same η as the highest type $\theta' < \theta$ that was already assigned a value $\eta(\theta')$.

Now it is shown that the constructed η is non-decreasing on (θ_s^b, θ_e^b): Say, there are two types $\hat{\theta}_1, \hat{\theta}_2 \in (\theta_s^b, \theta_e^b)$ with $\hat{\theta}_2 > \hat{\theta}_1$ which are assigned an η in the first step. Then (C2) implies that $\theta_1 > \theta_2$. From theorem 1 and the structure of the solution as depicted

in figure 4, it follows that $\eta(\theta_2) \geq \eta(\theta_1)$. Therefore, $\eta(\hat{\theta}_2) \geq \eta(\hat{\theta}_1)$. The second step does not change the monotonicity of $\eta(\theta)$ which proves that $\eta(\theta)$ is non-decreasing on (θ_s^b, θ_e^b).

If $\Phi(\theta, \hat{\theta}) > 0$ for all $\hat{\theta} \in (\theta_s^b, \theta_e^b)$ and $\theta > \theta_e^b$, no type is assigned a value for η in step 1. Consequently, η is constant on (θ_s^b, θ_e^b).

Last, it is necessary to show that – with the above constructed η on (θ_s^b, θ_e^b) – there is no upward jump of η at θ_e^b. If no type is assigned an η in the first step of the procedure above, this is obvious from theorem 1. Therefore, take the case where some type in (θ_s^b, θ_e^b) is assigned a value $\eta(\theta)$ in the first step of the procedure. Then the claim follows from theorem 1: Say, $\eta^-(\theta_e^b) = \eta(\theta_1)$ for some type θ_1 such that $\Phi(\theta_1, \hat{\theta}) = 0$ for $\hat{\theta} \in (\theta_s^b, \theta_e^b]$. Lemma 5 and theorem 1 imply that $\eta^+(\theta_e^b) = \eta^-(\theta_1)$.[29] Since η is non-increasing[30] at θ_1 according to theorem 1, it follows that $\eta^-(\theta_1) \geq \eta^+(\theta_1)$ and therefore $\eta^-(\theta_e^b) \geq \eta^+(\theta_e^b)$. A similar argument holds for θ_s^b.

It remains to show $\int_{\theta_s^b}^{\theta_e^b} \nu_\theta(\theta)\, d\theta = 0$. But this follows directly from $\nu(\theta_s^b) = \nu(\theta_e^b) = 0$. □

B.4. Stochastic contracts

The paper concentrated on deterministic contracts. Although hardly observed in practice, one could think of stochastic contracts. In the framework of the paper, this would mean that a type θ is assigned a probability distribution over the decision q instead of one deterministic decision $q(\theta)$. The idea behind a stochastic contract is to relax incentive constraints. Intuitively, this could work if different types have different degrees of risk aversion. See Rochet (2009) for an example where random contracts are optimal. The following proposition gives a sufficient condition under which deterministic decisions are optimal.[31]

Proposition 5. *The optimal decision is deterministic if the assumptions of proposition 1, (CVR) and*

$$\frac{\partial \frac{c_{q\theta\theta}}{c_{q\theta}}}{\partial q} \geq 0 \tag{WA.3}$$

[29] If non-local incentive constraints bind from types $\theta' \in (\theta_e^b, \theta_1)$ to types $\hat{\theta}' \in (\theta_e^b, \theta_1)$, this holds still true because of the necessary condition (C3). Also discontinuities at $\theta'' \in (\theta_e^b, \theta_1)$ do not matter as by lemma 4 and theorem 1 $\eta(\theta)$ is non-increasing at θ''. If there are several bunching intervals, the argument holds for the highest interval and given this, it holds for the second highest etc..

[30] Note that q is strictly increasing at θ_1 if it is continuous there (lemma 4). If q is discontinuous at θ_1, theorem 1 states that $\eta^-(\theta_1) \geq \eta^+(\theta_1)$.

[31] To illustrate: In example 1, $\frac{\partial \frac{c_{q\theta\theta}}{c_{q\theta}}}{\partial q} = \frac{4\theta^3}{(\theta^3 - 2q\theta)^2} \geq 0$ and condition (WA.3) holds.

hold.

Condition (CVR) and (WA.3) differ from the conditions for non-stochastic contracts in Maskin and Riley (1984). In Maskin and Riley (1984), only local incentive constraints bind and they bind "downward". It is then shown that assigning the expected decision increases the principal's payoff and relaxes local incentive constraints if risk aversion is decreasing in type. Decreasing risk aversion is therefore a sufficient condition for the optimality of deterministic contracts. This reasoning is flawed in case non-local incentive constraints are binding: Assigning the expected decision decreases the slope of the rent function $\pi(\theta)$ because $-c_\theta$ is convex in q. Hence, profit differences between θ and $\hat{\theta}$ are smaller under the expected decision compared to the stochastic contract. Non-local incentive constraints are, therefore, harder to satisfy.

Strausz (2006) points out that stochastic mechanisms are not optimal whenever the principal's problem can be written as a pointwise maximization of the virtual valuation. However, this is not the case if SC is violated because the virtual valuation maximization is subject to (NLIC).

Proposition 5 takes therefore another way known from Jullien (2000). When rewriting the principal's optimization problem as an optimization over rent profiles π (instead of over decision functions q) condition (CVR) ensures that the resulting program is concave. Condition (WA.3) ensures that the set of implementable utility profiles is convex. These two properties imply that a stochastic decision is worse for the principal than a deterministic decision implementing the same utility profile. The conditions of proposition 1 allow to focus on decisions above s, i.e. monotone solutions.

Proof of proposition5. First, note that under the conditions of proposition 1 one can focus on decisions above $s(\theta)$: If some $q(\theta) < s(\theta)$ was used in a stochastic contract menu with positive probability, the principal could do better by assigning $q^s(q(\theta), \theta)$ instead of $q(\theta)$. The proof is equivalent to the one of proposition 1.[32]

Second, suppose the optimal contract menu was stochastic and denote by $G(q, \theta)$ the distribution of q at type θ. Consider now an alternative deterministic contract menu

[32] Admittedly, it is not obvious whether lemma 3 holds for stochastic contract menus. Therefore, I show here explicitly that upward incentive constraints are relaxed if all $q(\theta) < s(\theta)$ are substituted by $q^s(q(\theta), \theta)$. Take $\hat{\theta} > \theta$, then the incentive constraint can be written as $\int_\theta^{\hat{\theta}} \int_q c_\theta(q, x)\, dG(q, x) - \int_q c_\theta(q, x)\, dG(q, \hat{\theta})\, dx \geq 0$. Changing $q(\theta) < s(\theta)$ to $q^s(q(\theta), \theta)$ does not change the first term. But since $\int_{q(\theta)}^{q^s(q(\theta),\theta)} c_{q\theta}(q, x)\, dq < 0$ for all $x < \hat{\theta}$, the change relaxes the incentive constraint through the second term.

$q^*(\theta)$ where $q^*(\theta) \geq s(\theta)$ is determined such that $c_\theta(q^*(\theta), \theta) = \int_q c_\theta(q, \theta)\, dG(q, \theta)$. In short, the slope of the rent function $\pi(\theta)$ and therefore the rent of each type remains the same under both menus.[33] It will be shown that under the assumptions of proposition 5 this change increases the principal's payoff and relaxes incentive compatibility.

Since only $q(\theta) \geq s(\theta)$ have to be considered, there is a one to one relationship between q and $-c_\theta(q, \theta)$. Define $h(z, \theta) \geq s(\theta)$ as the decision corresponding to $-c_\theta$ being z, i.e. $z = -c_\theta(h(z, \theta), \theta)$. Then the principal's objective can be written as

$$W = \int_{\underline{\theta}}^{\bar{\theta}} ([u(h(z, \theta), \theta) - c(h(z, \theta), \theta)]f(\theta) - [1 - F(\theta)]z)\ d\theta. \tag{WA.4}$$

The next step is to show that W is concave in z. This implies that the deterministic decision q^* increases the principal's payoff. The last step will then be to show that this deterministic decision is also incentive compatible.

Using $h_z(z, \theta) = \frac{1}{-c_{q\theta}(h,\theta)}$, which follows from the definition of $h(z, \theta)$, it is straightforward to derive

$$\frac{\partial^2 W}{\partial z^2} = \int_{\underline{\theta}}^{\bar{\theta}} \left(\frac{c_{qq\theta}(h, \theta)}{c_{q\theta}^2(h, \theta)} \left[\frac{u_{qq}(h, \theta) - c_{qq}(h, \theta)}{c_{qq\theta}} - \frac{u_q(h, \theta) - c_q(h, \theta)}{c_{q\theta}(h, \theta)} \right] f(\theta) \right)\ d\theta.$$

By condition (CVR), the integrand is negative and therefore W is concave in z.

Incentive compatibility of q^* means that for arbitrary types θ and $\hat{\theta}$

$$\Phi^*(\theta, \hat{\theta}) \equiv \int_{\hat{\theta}}^{\theta} \Big(c_\theta(q^*(\hat{\theta}), x) - c_\theta(q^*(x), x) \Big)\ dx \geq 0.$$

To verify this, it is useful to see that (WA.3) implies that $-c_{\theta\theta}(h(z, \theta), \theta)$ is convex in z:

$$\frac{d^2\{-c_{\theta\theta}(h(z, \theta), \theta)\}}{dz^2} = \frac{-1}{c_{q\theta}} \frac{\partial \frac{c_{q\theta\theta}}{c_{q\theta}}}{\partial q} \geq 0$$

This convexity implies

$$\frac{d}{d\theta} \left\{ \int_q -c_\theta(q, \theta)\, dG(q, \hat{\theta}) + c_\theta(q^*(\hat{\theta}), \theta) \right\} = \int_q -c_{\theta\theta}(q, \theta)\, dG(q, \hat{\theta}) + c_{\theta\theta}(q^*(\hat{\theta}), \theta) \geq 0.$$

The last inequality implies that $\int_{\hat{\theta}}^{\theta} \int_q -c_\theta(q, x)\, dG(q, \hat{\theta}) + c_\theta(q^*(\hat{\theta}), x)\, dx$ is a convex func-

[33] It is straightforward to check that local incentive compatibility requires the slope of the rent function under the stochastic menu to be $\int_q c_\theta(\tilde{q}, \theta)\, dG(q, \theta)$.

tion of θ with a minimum at $\theta = \hat{\theta}$ where the function value is 0. Consequently,

$$\int_{\hat{\theta}}^{\theta} c_\theta(q^*(\hat{\theta}), x)\, dx \geq \int_{\hat{\theta}}^{\theta} \int_q c_\theta(q, x)\, dG(q, \hat{\theta})\, dx.$$

But then,

$$\Phi^*(\theta, \hat{\theta}) = \int_{\hat{\theta}}^{\theta} \Big(c_\theta(q^*(\hat{\theta}), x) - c_\theta(q^*(x), x) \Big)\, dx \geq \int_{\hat{\theta}}^{\theta} \left(\int_q c_\theta(q, x)\, dG(q, \hat{\theta}) - c_\theta(q^*(x), x) \right)\, dx \geq 0$$

where the last inequality follows from the incentive compatibility of the stochastic menu $G(q, \theta)$ and the definition of q^*. □

B.5. Algorithm: Addenda and extensions

This subsection consists of two addenda to the main text. Afterwards the possibility of several Φ^η minimizer and its implications for the algorithm are discussed.

Lemma WA5. *The optimal decision q is strictly increasing if (CVR) holds and $q(\cdot, \eta)$ is strictly increasing for all $\eta \geq 0$.*

Suppose, to the contrary, that $q(\theta) = q^b$ for all $\theta \in [\theta_1, \theta_2]$. By theorem 1 and the continuity of q, $\eta(\theta_1) \leq \eta(\theta_2)$. But then $q(\theta_1) = q(\theta_1, \eta(\theta_1)) < q(\theta_2, \eta(\theta_1)) \leq q(\theta_2, \eta(\theta_2)) = q(\theta_2)$ where the strict inequality follows from the assumption that $q(\cdot, \eta)$ is strictly increasing in type. This contradicts $q(\theta_1) = q(\theta_2) = q^b$. □

Addendum to the proof of theorem 2: The proof of the main text showed that $\Phi^\eta(\theta', \hat{\theta}') \leq \Phi^\eta(\theta'', \hat{\theta}'')$ for either $\theta'' \in (\theta', \theta' + \varepsilon)$ and $\hat{\theta}'' \in (\hat{\theta}', \hat{\theta}' + \varepsilon)$ or $\theta'' \in (\theta' - \varepsilon, \theta')$ and $\hat{\theta}'' \in (\hat{\theta}', \hat{\theta}' + \varepsilon)$. Here I present the argument for the remaining two cases which is, however, very similar to the main text.

First, take $\theta'' \in (\theta', \theta' + \varepsilon)$ and $\hat{\theta}' \in (\hat{\theta}' - \varepsilon, \hat{\theta}')$. By the continuity of q and $q(\cdot, \eta')$, there exists a $\hat{\theta}''' \in (\hat{\theta}'', \hat{\theta}')$ such that $q(\hat{\theta}''') = q(\hat{\theta}'', \eta')$. Then,

$$\Phi^\eta(\theta'', \hat{\theta}'') - \Phi^\eta(\theta', \hat{\theta}') = \Phi(\theta'', \hat{\theta}''') - \Phi(\theta', \hat{\theta}') + \int_{\hat{\theta}''}^{\hat{\theta}'''} \int_{q(\hat{\theta}''')}^{q(x,\eta')} -c_{q\theta}(y, x)\, dy\, dx$$
$$+ \int_{\hat{\theta}'''}^{\hat{\theta}'} \int_{q(x)}^{q(x,\eta')} -c_{q\theta}(y, x)\, dy\, dx + \int_{\theta'}^{\theta''} \int_{q(x)}^{q(x,\eta')} -c_{q\theta}(y, x)\, dy\, dx \geq \Phi(\theta'', \hat{\theta}''') - \Phi(\theta', \hat{\theta}') \geq 0$$

where the first inequality follows from $q(x) > s(x)$ and $q(x, \eta') > s(x)$. The second inequality follows from $\Phi(\theta'', \hat{\theta}''') \geq 0 = \Phi(\theta', \hat{\theta}')$ by incentive compatibiltiy and the

assumption that (NLIC) binds from θ' to $\hat{\theta}'$.

Second, let $\theta'' \in (\theta' - \varepsilon, \theta')$ and $\hat{\theta}' \in (\hat{\theta}' - \varepsilon, \hat{\theta}')$. By the continuity of q and $q(\cdot, \eta')$, there exists a $\hat{\theta}''' \in (\hat{\theta}'', \hat{\theta}')$ such that $q(\hat{\theta}''') = q(\hat{\theta}'', \eta')$. Then,

$$\Phi^\eta(\theta'', \hat{\theta}'') - \Phi^\eta(\theta', \hat{\theta}') = \Phi(\theta'', \hat{\theta}''') - \Phi(\theta', \hat{\theta}') + \int_{\hat{\theta}''}^{\hat{\theta}'''} \int_{q(\hat{\theta}''')}^{q(x,\eta')} -c_{q\theta}(y, x)\, dy\, dx$$
$$+ \int_{\hat{\theta}'''}^{\hat{\theta}'} \int_{q(x)}^{q(x,\eta')} -c_{q\theta}(y, x)\, dy\, dx - \int_{\theta''}^{\theta'} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx \geq \Phi(\theta'', \hat{\theta}''') - \Phi(\theta', \hat{\theta}') \geq 0$$

where the first inequality follows from $q(x) > s(x)$ and $q(x, \eta') > s(x)$. The second inequality follows from $\Phi(\theta'', \hat{\theta}''') \geq 0 = \Phi(\theta', \hat{\theta}')$ by incentive compatibiltiy and the assumption that (NLIC) binds from θ' to $\hat{\theta}'$. □

B.5.1. Several Φ^η minimizers

Theorem 2 states that (NLIC) binds only between minimizers of Φ^η. If – for a given η – there are several such minimizers it is not a priori clear which minimizer is the one where (NLIC) binds. I will go through several configurations of Φ^η minimizers and show that incentive compatibility puts enough structure on the problem to determine which Φ^η minimizer is "relevant" (in the sense that (NLIC) binds at this minimzer). These results take η as given, that is, Φ^η has several minimizers for a given η and the question is which is/are the minimizer(s) at which (NLIC) binds. While the configurations of minimizers discussed below do not cover all possible cases, they nevertheless illustrate how the results in the main text can be extended if necessary.

Throughout I keep the assumption that either (CVR) is satisfied and $q(\cdot, \eta)$ is strictly increasing for all $\eta \geq 0$ or that the conditions of proposition 3 are satisfied.

Case 1 (separated minimizers). Let $(\theta_i(\eta), \hat{\theta}_i(\eta))$, $i = 1 \ldots n$, be the local minimizers of Φ^η. Assume that $\hat{\theta}_i < \theta_i < \hat{\theta}_{i+1} < \theta_{i+1}$ for $i = 1 \ldots n-1$. This case turns out to be a simple extension of the argument in the main text. The solution can consist of several "brackets" of type pairs between which (NLIC) binds; see figure 4 for an illustration.

Lemma WA6. *Given the assumptions of case 1, (NLIC) binds in the solution of (SB) from $\theta_i(\eta)$ to $\hat{\theta}_i(\eta)$ if and only if $\Phi^\eta(\theta_i(\eta), \hat{\theta}_i(\eta)) \leq 0$.*

Lemma WA6 implies that the algorithm has to be adjusted in case 1 such that only those minimizers $(\theta_i(\eta), \hat{\theta}_i(\eta))$ are assigned a decision in 4a for which $\Phi^\eta(\theta_i(\eta), \hat{\theta}_i(\eta)) \leq 0$.

The algorithm will then assign the optimal decision to all types from or to which (NLIC) binds. By theorem 1, the algorithm then also assigns the optimal decision to all other types.

Proof of lemma WA6: As (NLIC) cannot bind from $\theta_i(\eta')$ to $\hat{\theta}_i(\eta')$ if $\Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta')) > 0$ (theorem 2), $\Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta')) \leq 0$ for $i = 1, \ldots, n$ is assumed for notational convenience in this proof. Because of theorem 2, no relevant $\Phi^{\eta'}$ minimizer is lost by this assumption.

First, it is shown that $\eta(\theta) \geq \eta'$ for all $\theta \in [\hat{\theta}_i(\eta'), \theta_i(\eta')]$. To start with, suppose $\eta(\theta) < \eta'$ for all $\theta \in [\hat{\theta}_i(\eta'), \theta_i(\eta')]$. Then $q(\theta) \leq q(\theta, \eta')$ for all $\theta \in [\hat{\theta}_i(\eta'), \theta_i(\eta')]$. Let $\hat{\theta}'$ be the type where $q(\hat{\theta}') = q(\hat{\theta}_i(\eta'), \eta')$.[34] Then (NLIC) between $\theta_i(\eta')$ and $\hat{\theta}'$ is violated as $\Phi(\theta_i(\eta'), \hat{\theta}') < \Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta')) \leq 0$. Next, suppose there are types θ_0 and θ_1 both in $(\hat{\theta}_i(\eta'), \theta_i(\eta'))$ such that $\eta(\theta_0) < \eta' \leq \eta(\theta_1)$. Suppose $\theta_0 < \theta_1$ and define $\theta' = sup\{\theta : \eta(\theta) < \eta' \text{ and } \theta \leq \theta_1\}$. By the continuity of η, θ' exists and $\eta(\theta') = \eta'$. (NLIC) binds to θ' because of lemma 7. Then theorem 2 implies that (θ'', θ') are local minimizer of $\Phi^{\eta'}$ for some θ''. But by the definition of case 1 this is impossible. A similar argument applies if $\theta_0 > \theta_1$, i.e. then there is a type $\theta' \in (\theta_1, \theta_0)$ with $\eta(\theta') = \eta'$ from which (NLIC) binds which is again impossible by the definition of case 1. Hence, $\eta(\theta) \geq \eta'$ for all $\theta \in [\hat{\theta}_i(\eta'), \theta_i(\eta')]$.

Second, the claim of the lemma holds for $(\theta_n(\eta'), \hat{\theta}_n(\eta'))$.

Define $\theta'_n = inf\{\theta : \eta(\theta) < \eta' \text{ and } \theta \geq \theta_n(\eta')\}$. If θ'_n exists, $\eta(\theta'_n) = \eta'$ by the continuity of η. If $\theta'_n > \theta_n(\eta')$, (NLIC) would have to bind from θ'_n (lemma 7). By theorem 2, θ'_n has then to be part of a minimizer of $\Phi^{\eta'}$. This would contradict the definition of θ_n as highest type minimzing $\Phi^{\eta'}$. Hence, if θ'_n exists, then $\theta'_n = \theta_n(\eta')$ and (NLIC) binds from θ'_n to $\hat{\theta}'_n$, i.e. the lemma holds in this case.

If θ'_n does not exist, $\eta(\bar{\theta}) \geq \eta'$ (by the first step and continuity of η) and (NLIC) binds from $\bar{\theta}$ (theorem 1). Hence, $(\bar{\theta}, \hat{\theta}'')$ is a local minimizer of $\Phi^{\eta''}$ for some $\hat{\theta}''$ and some $\eta'' \geq \eta'$ by theorem 2. If $\eta'' = \eta'$, then $\bar{\theta} = \theta_n(\eta')$ and the claim of the lemma holds. If $\eta'' > \eta'$, the fact that $(\bar{\theta}, \hat{\theta}'')$ is a local minimizer of $\Phi^{\eta''}$ implies that $\int_{\hat{\theta}''}^{\bar{\theta}} c_{q\theta}(q(\hat{\theta}'', \eta''), x)\, dx \geq 0$ and $\int_{q(\hat{\theta}'', \eta'')}^{q(\bar{\theta}, \eta'')} c_{q\theta}(y, \bar{\theta})\, dy \geq 0$. As $q(\cdot, \eta)$ increases in η (and because of $c_{qq\theta} < 0$ and $c_{q\theta\theta} > 0$), these two inequalities hold for η' strictly (given $\eta' < \eta''$), i.e. $\int_{\hat{\theta}''}^{\bar{\theta}} c_{q\theta}(q(\hat{\theta}'', \eta'), x)\, dx > 0$

[34] If no such type exists, incentive compatibility is clearly violated as $\Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta')) \leq 0$ by assumption.

and $\int_{q(\hat{\theta}'',\eta')}^{q(\bar{\theta},\eta')} c_{q\theta}(y,\bar{\theta})\,dy > 0$. Hence, $(\bar{\theta},\hat{\theta}''')$ is, for some $\hat{\theta}''' \leq \theta''$, a local minimizer of $\Phi^{\eta'}$ which means $\theta_n(\eta') = \bar{\theta}$ and $\hat{\theta}''' = \hat{\theta}_n(\eta')$.

If $\theta_n(\eta') = \bar{\theta}$ and $\eta(\bar{\theta}) > \eta'$, consider $\hat{\theta}' = sup\{\theta : \eta(\theta) = \eta'\}$. If $\hat{\theta}'$ exists, then (NLIC) has to bind to $\hat{\theta}'$ from a higher type which implies $\hat{\theta}' = \hat{\theta}_n(\eta')$ and the type from which (NLIC) binds is $\theta_n(\eta') = \bar{\theta}$. If $\hat{\theta}'$ does not exist, then $\eta(\theta) > \eta'$ for all types which implies that (NLIC) binds from $\bar{\theta}$ to $\underline{\theta}$. Hence, $\int_{\underline{\theta}}^{\bar{\theta}} c_{q\theta}(q(\underline{\theta}),x)\,dx \geq 0$ and $\int_{q(\underline{\theta})}^{q(\bar{\theta})} c_{q\theta}(y,\bar{\theta})\,dy \geq 0$. As $q(\theta) > q(\theta,\eta')$ for all types, this implies $\int_{\underline{\theta}}^{\bar{\theta}} c_{q\theta}(q(\underline{\theta},\eta'),x)\,dx > 0$ and $\int_{q(\underline{\theta},\eta')}^{q(\bar{\theta},\eta')} c_{q\theta}(y,\bar{\theta})\,dy > 0$, i.e. $(\bar{\theta},\underline{\theta})$ is a local minimizer of $\Phi^{\eta'}$ which by the structure of case 1 implies that $n = 1$ and $\theta_1(\eta') = \bar{\theta}$ and $\hat{\theta}_1(\eta') = \underline{\theta}$. Clearly, the lemma holds in this case by proposition 4.

Third, the lemma holds for $i < n$ by induction: Let the result be true for all $j > i$.

As a first case, let $\Phi^{\eta'}(\theta_i(\eta'),\hat{\theta}_i(\eta')) < 0$. Then $\eta(\theta) \geq \eta'$ for all $\theta \in (\hat{\theta}_i(\eta'),\theta_i(\eta'))$: This was shown in the first step. Given that $\Phi^{\eta'}(\theta_i(\eta'),\hat{\theta}_i(\eta')) < 0$ and that there is no $\Phi^{\eta'}$ minimizer within $(\hat{\theta}_i(\eta'),\theta_i(\eta'))$, it follows that $\eta(\theta) > \eta'$ for all $\theta \in (\hat{\theta}_i(\eta'),\theta_i(\eta'))$.

Define $\theta_i' = inf\{\theta : \eta(\theta) = \eta' \text{ and } \theta \geq \theta_i(\eta')\}$. Note that $\eta(\hat{\theta}_{i+1}(\eta')) = \eta'$ by the induction hypothesis and therefore θ_i' exists and $\theta_i' \leq \hat{\theta}_{i+1}$. By the definition of θ_i' and the fact that $\eta(\theta) > \eta'$ for $\theta \in (\hat{\theta}_i(\eta'),\theta_i(\eta'))$, (NLIC) binds from θ_i' (lemma 7). Hence, $\hat{\theta}_{i+1}(\eta') \neq \theta_i'$ by lemma 5 and the induction hypothesis. Theorem 2 implies then that θ_i' has to be part of a local $\Phi^{\eta'}$ minimizer which implies $\theta_i' = \theta_i(\eta')$ (because there is no other $\Phi^{\eta'}$ minimizer between $\theta_i(\eta')$ and $\hat{\theta}_{i+1}(\eta')$ by the assumptions of case 1). Hence, (NLIC) binds from $\theta_i(\eta')$ to $\hat{\theta}_i'(\eta')$.

The second subcase is $\Phi^{\eta'}(\theta_i,\hat{\theta}_i) = 0$ and $\eta(\theta) < \eta'$ for some $\theta \in (\theta_i,\hat{\theta}_{i+1})$. Define $\theta_i' = inf\{\theta : \eta(\theta) < \eta' \text{ and } \theta \geq \theta_i(\eta')\}$. As $\eta(\theta_i) \geq \eta'$ by the first step, $\theta_i' \geq \theta_i$. By lemma 7, (NLIC) has to bind from θ_i' which means θ_i' has to be part of a $\Phi^{\eta'}$ minimizer by theorem 2. Therefore, $\theta_i' = \theta_i$ and again (NLIC) binds from $\theta_i(\eta')$ to $\hat{\theta}_i'(\eta')$.

It remains to show that a third case, i.e. $\Phi^{\eta'}(\theta_i,\hat{\theta}_i) = 0$ and $\eta(\theta) = \eta'$ for all $\theta \in (\theta_i,\hat{\theta}_{i+1})$ is impossible.[35] Indeed, I will show that such a decision function is not optimal.[36] Note that (NLIC) is slack for all types in $(\theta_i,\hat{\theta}_{i+1})$ in this case as η is constant.

[35] Other cases cannot occur: By the induction hypothesis $\eta(\hat{\theta}_{i+1}) = \eta'$ and η is non-decreasing around $\hat{\theta}_{i+1}$. Therefore, $\eta(\theta) \leq \eta'$ for all $\theta \in (\theta_i,\hat{\theta}_{i+1})$: If there was a $\theta' \in (\theta_i,\hat{\theta}_{i+1})$ with $\eta(\theta') > \eta'$, there would have to be a type $\theta'' \in (\theta',\hat{\theta}_{i+1})$ with $\eta(\theta'') = \eta'$ from which (NLIC) binds (lemma 7). But by assumption there is no $\Phi^{\eta'}$ minimizer there.

[36] Here I use an argument that is based on optimality not only on incentive compatibility. However, this case is actually not a problem for the algorithm anyway: If the algorithm assigns the shadow value

Therefore, the following changed decision is implementable for $\varepsilon > 0$ small enough

$$q^c(\theta) = \begin{cases} (1-\varepsilon)q(\theta) + \varepsilon q(\frac{\theta_i+\hat{\theta}_{i+1}}{2}) & \text{if } \theta \in [\frac{\theta_i+\hat{\theta}_{i+1}}{2}, \frac{\theta_i+3\hat{\theta}_{i+1}}{4}] \\ q(\theta) & \text{else.} \end{cases}$$

This decision clearly increases the principal's payoff as the changed decision is closer to q^r and the principal's objective is strictly concave (lemma 2). The changed decision is implementable for $\varepsilon > 0$ small enough: This is true by the continuity of Φ in ε and because $\Phi(\theta, \hat{\theta}) > 0$ for all $\theta \neq \hat{\theta}$ if $\hat{\theta} \in (\theta_i, \hat{\theta}_{i+1})$.[37] □

Case 2a (simply nested minimizers). Assume that the local minimizers of Φ^η are nested, i.e. $\hat{\theta}_i(\eta) < \hat{\theta}_{i+1}(\eta) < \theta_{i+1}(\eta) < \theta_i(\eta)$ for $i = 1, \ldots, n-1$. The following lemma states that the global minimizer of Φ^η is relevant in case of nested minimizers.

Lemma WA7. *If $\Phi^\eta(\theta_i(\eta), \hat{\theta}_i(\eta)) < \Phi^\eta(\theta_j(\eta), \hat{\theta}_j(\eta))$ for all $j = 1, \ldots, n$, $j \neq i$ and $\Phi^\eta(\theta_i(\eta), \hat{\theta}_i(\eta)) \leq 0$, then (NLIC) binds from $\theta_i(\eta)$ to $\hat{\theta}_i(\eta)$ and does not bind from θ_j to $\hat{\theta}_j$ for $j \neq i$.*
If Φ^η has several minimizer, i.e. $\Phi^\eta(\theta_i, \hat{\theta}_i) = \min_{\theta,\hat{\theta}} \Phi^\eta(\theta, \hat{\theta}) \leq 0$ for all $i \in I \subseteq \{1, \ldots, n\}$, then (NLIC) binds from $\theta_{\min I}(\eta)$ to $\hat{\theta}_{\min I}(\eta)$. If and only if $\min_{\theta,\hat{\theta}} \Phi^\eta(\theta, \hat{\theta}) < 0$, (NLIC) also binds from $\theta_{\max I}(\eta)$ to $\hat{\theta}_{\max I}(\eta)$. (NLIC) does not bind from $\theta_i(\eta)$ to $\hat{\theta}_i(\eta)$ if $i \neq \max I$ and $i \neq \min I$.

It should be noted that lemma WA6 and lemma WA7 can be combined, i.e. if there are several separated nests then lemma WA7 applies for each nest.

Proof of lemma WA7: Take an arbitrary $\eta' \geq 0$. Theorem 2 already established that (NLIC) cannot bind from $\theta_i(\eta')$ to $\hat{\theta}_i(\eta')$ if $\Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta')) > 0$. Let $(\theta_i(\eta'), \hat{\theta}_i(\eta'))$ be a global $\Phi^{\eta'}$ minimizer and denote by $(\theta_j(\eta'), \hat{\theta}_j(\eta'))$ a local $\Phi^{\eta'}$ minimizer such that $0 \geq \Phi^{\eta'}(\theta_j(\eta'), \hat{\theta}(\eta')) > \Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta'))$. For now, assume that $(\theta_i(\eta'), \hat{\theta}_i(\eta'))$ is the unique global minimizer of $\Phi^{\eta'}$. For notational simplicity, I drop the argument "(η')" in the remainder of the proof.

η' to $\hat{\theta}_i$ and θ_i, this will lead to the correct solution even if (NLIC) is not binding from θ_i to $\hat{\theta}_i$ in the strict definition I use in the paper.

[37] In fact, $min_{\hat{\theta} \in [\frac{\theta_i+\hat{\theta}_{i+1}}{2}, \frac{\theta_i+3\hat{\theta}_{i+1}}{4}], \theta \geq \hat{\theta}+\varepsilon'} \Phi(\theta, \hat{\theta}) > 0$. If it was not, some type in $[\frac{\theta_i+\hat{\theta}_{i+1}}{2}, \frac{\theta_i+3\hat{\theta}_{i+1}}{4}]$ would have to be a local minimizer of $\Phi^{\eta'}$ but by assumption there is no local minimizer of $\Phi^{\eta'}$ in $(\theta_i, \hat{\theta}_{i+1})$. Choosing $\varepsilon' > 0$ small enough (NLIC) can – under the changed decision – also not bind from θ' to $\hat{\theta}'$ with $\theta' - \hat{\theta}' < \varepsilon'$ because $q^c(\theta) > q^r(\theta) > s(\theta)$ for $\theta \in (\theta_i, \hat{\theta}_{i+1})$ and $\varepsilon > 0$ small enough.

I start with the case where $\eta(\theta) = \eta'$ for some type θ. In this case, (NLIC) has to bind for (at least) one of the $\Phi^{\eta'}$ minimizers because of theorem 2 and the preliminary result in the proof of proposition 4.

Suppose, contrary to the lemma, that (NLIC) binds from θ_j to $\hat{\theta}_j$ in the solution to problem (SB). There are two possibilities:

First, suppose $\hat{\theta}_j < \hat{\theta}_i < \theta_i < \theta_j$. But this violates theorem 2 which states that $(\theta_j, \hat{\theta}_j)$ has to minimize $\Phi^{\eta'}$ on $[\hat{\theta}_j, \theta_j]$ if (NLIC) binds from θ_j to $\hat{\theta}_j$.

Second, let $\hat{\theta}_i < \hat{\theta}_j < \theta_j < \theta_i$. Note that $\eta(\theta) \leq \eta'$ for all $\theta \in (\hat{\theta}_i, \hat{\theta}_j) \cup (\theta_j, \theta_i)$: To see this, suppose there was a type $\theta' \in (\theta_j, \theta_i)$ such that $\eta(\theta') > \eta'$ and define $\theta'' = sup\{\theta : \eta(\theta) = \eta' \text{ and } \theta < \theta'\}$. (NLIC) binds to θ'' by lemma 7. As (NLIC) cannot bind from and to θ_j (see lemma 5 and the discussion below this lemma), $\theta'' > \theta_j$. By theorem 2, θ'' is part of a $\Phi^{\eta'}$ minimizer, i.e. $\theta'' = \hat{\theta}_k(\eta')$ for some $k \in \{1, 2, \ldots, n\}$. By the nested structure of case 2, this is impossible as $\hat{\theta}_k(\eta') < \theta_j$ for all $k = 1, \ldots, n$. A similar argument applies for types in $(\hat{\theta}_i, \hat{\theta}_j)$. Hence, $\eta(\theta) \leq \eta'$ for all $\theta \in (\hat{\theta}_i, \hat{\theta}_j) \cup (\theta_j, \theta_i)$.

This implies $q(\theta) \leq q(\theta, \eta')$ for all $\theta \in (\hat{\theta}_i, \hat{\theta}_j) \cup (\theta_j, \theta_i)$ and therefore

$$\Phi(\theta_i, \hat{\theta}_i) = \Phi(\theta_j, \hat{\theta}_j) + \Phi^{\eta'}(\theta_i, \hat{\theta}_i) - \Phi^{\eta'}(\theta_j, \hat{\theta}_j) + \int_{\hat{\theta}_i}^{\hat{\theta}_j} \int_{q(x)}^{q(x,\eta')} c_{q\theta}(y, x)\, dy\, dx$$
$$+ \int_{\theta_j}^{\theta_i} \int_{q(x)}^{q(x,\eta')} c_{q\theta}(y, x)\, dy\, dx \leq \Phi^{\eta'}(\theta_i, \hat{\theta}_i) - \Phi^{\eta'}(\theta_j, \hat{\theta}_j) < 0.$$

This contradicts incentive compatibility of the optimal decision function.

Hence, (NLIC) binds for a global $\Phi^{\eta'}$ minimizer if $\eta(\theta) = \eta'$ for some type.

If $\eta(\theta) < \eta'$ for all types then $q(\theta) < q(\theta, \eta')$ for all types and therefore $\Phi(\theta_i, \hat{\theta}_i) = \Phi^{\eta'}(\theta_i, \hat{\theta}_i) + \int_{\hat{\theta}_i}^{\theta_i} \int_{q(\hat{\theta}_i)}^{q(\hat{\theta}_i,\eta')} -c_{q\theta}(y, x)\, dy\, dx - \int_{\hat{\theta}_i}^{\theta_i} \int_{q(x)}^{q(x,\eta')} -c_{q\theta}(y, x)\, dy\, dx < \Phi^{\eta'}(\theta_i, \hat{\theta}_i) \leq 0$ where the first inequality follows from $c_{qq\theta} < 0$ and $\int_{\hat{\theta}_i}^{\theta_i} c_{q\theta}(q(\hat{\theta}_i, \eta'), x)\, dx \geq 0$ because $(\theta_i, \hat{\theta}_i)$ locally minimize $\Phi^{\eta'}$. This violates incentive compatibility. Hence, $\eta(\theta) < \eta'$ for all types is impossible.

If $\eta(\theta) > \eta' \geq 0$ for all types then (NLIC) binds from $\bar{\theta}$ to $\underline{\theta}$. Hence, $\int_{\underline{\theta}}^{\bar{\theta}} c_{q\theta}(q(\underline{\theta}), x)\, dx \geq 0$ and $\int_{q(\underline{\theta})}^{q(\bar{\theta})} c_{q\theta}(y, \bar{\theta})\, dx \geq 0$. This implies by $q(\theta) > q(\theta, \eta')$ that $\int_{\underline{\theta}}^{\bar{\theta}} c_{q\theta}(q(\underline{\theta}, \eta'), x)\, dx > 0$ and $\int_{q(\underline{\theta},\eta')}^{q(\bar{\theta},\eta')} c_{q\theta}(y, \bar{\theta})\, dx > 0$. Hence, $\hat{\theta}_1(\eta') = \underline{\theta}$ and $\theta_1(\eta') = \bar{\theta}$ and $(\bar{\theta}, \underline{\theta})$ is a local $\Phi^{\eta'}$ minimizer. If $(\bar{\theta}, \underline{\theta})$ was not the unique global $\Phi^{\eta'}$ minimizer, the same argument as in the first step would establish that there is a θ_i and $\hat{\theta}'_i$ with $\Phi(\theta'_i, \hat{\theta}'_i) < 0$. Hence, $(\bar{\theta}, \underline{\theta})$ is

the unique global $\Phi^{\eta'}$ minimizer in this case and the lemma holds.

It remains to analyze the case where there are several global $\Phi^{\eta'}$ minimizers. In this case, (NLIC) will bind for the most exterior and most interior global $\Phi^{\eta'}$ minimizer, i.e. θ_i to $\hat{\theta}_i$ for the highest and lowest i among the global minimizers. To see this, let $\theta_j = max\{\theta : \eta(\theta) = \eta'\}$. Then (NLIC) binds from θ_j to $\hat{\theta}_j = min\{\theta : \eta(\theta) = \eta'\}$ and $\eta(\theta) < \eta'$ for all $\theta \in [\underline{\theta}, \hat{\theta}_j) \cup (\theta_j, \bar{\theta}]$. Suppose there existed a global $\Phi^{\eta'}$ minimizer $(\theta_i, \hat{\theta}_i)$ such that $\hat{\theta}_i < \hat{\theta}_j < \theta_j < \theta_i$. As $q(\theta) < q(\theta, \eta')$ for $\theta \in [\underline{\theta}, \hat{\theta}_j) \cup (\theta_j, \bar{\theta}]$, $\Phi(\theta_i, \hat{\theta}_i) < \Phi(\theta_j, \hat{\theta}_j) + \Phi^{\eta'}(\theta_i, \hat{\theta}_i) - \Phi^{\eta'}(\theta_j, \hat{\theta}_j) \leq 0$ contradicting incentive compatibility. Hence, there is no global $\Phi^{\eta'}$ minimizer $(\theta_i, \hat{\theta}_i)$ such that $\hat{\theta}_i < \hat{\theta}_j < \theta_j < \theta_i$, i.e. (NLIC) binds for the most exterior global $\Phi^{\eta'}$ minimizer.

To show the same for the most interior, let θ_j be the lowest type such that (i) $\eta(\theta) = \eta'$ and (ii) (NLIC) binds from θ_j. Assume $\Phi^{\eta'}(\theta_j, \hat{\theta}_j) < 0$ which implies that $\eta(\theta) > \eta'$ for all $\theta \in (\hat{\theta}_j, \theta_j)$ as everything else contradicts either the definition of θ_j or incentive compatibility between θ_j and $\hat{\theta}_j$. Suppose there was a global $\Phi^{\eta'}$ minimizer $(\theta_i, \hat{\theta}_i)$ such that $\hat{\theta}_j < \hat{\theta}_i < \theta_i < \theta_j$. Define $\hat{\theta}'$ by $q(\hat{\theta}') = q(\hat{\theta}_i, \eta')$. Then, because $q(\theta) > q(\theta, \eta')$ for all $\theta \in (\hat{\theta}_j, \hat{\theta}_i) \cup (\theta_i, \theta_j)$, $\Phi(\theta_i, \hat{\theta}') < \Phi(\theta_j, \hat{\theta}_j) - \Phi^{\eta'}(\theta_j, \hat{\theta}_j) + \Phi^{\eta'}(\theta_i, \hat{\theta}_i) = 0$ which contradicts incentive compatibility. Hence, there cannot be such an $(\hat{\theta}_i, \theta_i)$ which means (NLIC) binds for the most interior global $\Phi^{\eta'}$ minimizer.

If the most interior global $\Phi^{\eta'}$ minimizer is denoted by $(\theta_i, \hat{\theta}_i)$ and the most exterior $\Phi^{\eta'}$ minimizer is denoted by $(\theta_e, \hat{\theta}_e)$, then $\eta(\theta') = \eta'$ for all $\theta \in (\hat{\theta}_e, \hat{\theta}_i) \cup (\theta_i, \theta_e)$: As $0 = \Phi(\theta_e, \hat{\theta}_e) - \Phi(\theta_i, \hat{\theta}_i) = \Phi^{\eta'}(\theta_e, \hat{\theta}_e) - \Phi^{\eta'}(\theta_i, \hat{\theta}_i) + \int_{\hat{\theta}_e}^{\hat{\theta}_i} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx + \int_{\theta_i}^{\theta_e} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx = \int_{\hat{\theta}_e}^{\hat{\theta}_i} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx + \int_{\theta_i}^{\theta_e} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx$, η cannot be above (or below) η' for all types in (θ_i, θ_e) (because of (C3) this would imply that η is above (below) η' for all types in $(\hat{\theta}_e, \hat{\theta}_i)$). If there was a type $\theta' \in (\theta_i, \theta_e)$ such that $\eta(\theta') > \eta'$, then (NLIC) would have to bind to $\hat{\theta}' = sup\{\theta : \eta(\theta) = \eta' \text{ and } \theta < \theta'\}$. As $\hat{\theta}' \geq \theta_i$, this is impossible as there is no $\Phi^{\eta'}$ minimizer $(\theta_k, \hat{\theta}_k)$ with $\hat{\theta}_k \geq \theta_i$ by the assumption of case 2a. □

Case 2b (multiply nested minimizers). Assume that $\hat{\theta}_1(\eta) < \hat{\theta}_2(\eta) < \theta_2(\eta) < \cdots < \hat{\theta}_n(\eta) < \theta_n(\eta) < \theta_1(\eta)$, i.e. several separated Φ^{η} minimizers are nested within another minimizer. By theorem 2, (NLIC) can only bind from $\theta_i(\eta)$ to $\hat{\theta}_i(\eta)$ if $\Phi^{\eta}(\theta_i(\eta), \hat{\theta}_i(\eta)) \leq 0$. A Φ^{η} minimizer not satisfying this inequality is irrelevant for determining the solution. To avoid cluttering notation by case distinction I will therefore not consider such minimizers

and assume $\Phi^{\eta}(\theta_i(\eta), \hat{\theta}_i(\eta)) \leq 0$ for $i = 1, \ldots, n$.

Lemma WA8. *Given the assumptions of case 2b, (NLIC) binds from θ_1 to $\hat{\theta}_1$ if and only if $\Phi^{\eta}(\theta_1(\eta), \hat{\theta}_1(\eta)) \leq \sum_{i=2}^{n} \Phi^{\eta}(\theta_i(\eta), \hat{\theta}_i(\eta))$. For $i \geq 2$, (NLIC) binds from θ_i to $\hat{\theta}_i$ if and only if $\Phi^{\eta}(\theta_1(\eta), \hat{\theta}_1(\eta)) \geq \sum_{i=2}^{n} \Phi^{\eta}(\theta_i(\eta), \hat{\theta}_i(\eta))$ and $\Phi^{\eta}(\theta_i(\eta), \hat{\theta}_i(\eta)) \leq 0$ and one of these two inequalities holds strictly.*

Proof of lemma WA8: Take an $\eta' \geq 0$ at which the structure of local $\Phi^{\eta'}$ minimizers is as in case 2b. The proof that (NLIC) binds from some $\theta_i(\eta')$ to some $\hat{\theta}_i(\eta')$ if $\Phi^{\eta'}(\theta_j(\eta'), \hat{\theta}_j(\eta')) \leq 0$ for some $j = 1, \ldots, n$ is analogous to lemma WA7 and therefore not duplicated.

First, let $\Phi^{\eta'}(\theta_1(\eta'), \hat{\theta}_1(\eta')) < \sum_{i=2}^{n} \Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta'))$. Suppose (NLIC) was binding from $\theta_j(\eta')$ to $\hat{\theta}_j(\eta')$ for all $j \in J \subseteq \{2, \ldots, n\}$ (and possibly also from $\theta_1(\eta')$ to $\hat{\theta}_1(\eta')$). To shorten notation, I will omit the argument "(η')" for the remainder of the proof.

Let j and k be consecutive elements of J, i.e. $j, k \in J$, $j < k$ and $\nexists z \in J. j < z < k$. Then, $\eta(\theta) = \eta'$ for all $\theta \in [\theta_j, \hat{\theta}_k]$ and $\eta(\theta) \leq \eta'$ for $\theta \in [\hat{\theta}_1, \hat{\theta}_{\min J}] \cup [\theta_{\max J}, \theta_1]$. To show this, suppose there was a $\theta' \in [\theta_j, \hat{\theta}_k]$ with $\eta(\theta') > \eta'$. Define $\hat{\theta}' = sup\{\theta : \eta(\theta) = \eta' \text{ and } \theta < \theta'\}$. (NLIC) has to bind to $\hat{\theta}'$ (lemma 7). Note that $\hat{\theta}' \geq \theta_j$ as $\eta(\theta_j) = \eta'$ because (NLIC) binds from θ_j. Actually, $\hat{\theta}' > \theta_j$ by lemma 5. But then, by theorem 2, $\hat{\theta}'$ has to be part of a $\Phi^{\eta'}$ minimizer although there is none (to which (NLIC) binds) in $(\theta_j, \hat{\theta}_k)$ by assumption. Hence, $\eta(\theta) = \eta'$ for all $\theta \in (\theta_j, \hat{\theta}_k)$ which implies $\eta(\hat{\theta}_k) = \eta'$. This implies $\eta(\theta_k) = \eta'$: If (NLIC) binds from θ_k to $\hat{\theta}_k$ this is implied by (C3). If (NLIC) does not bind, then $\eta(\theta) = \eta'$ for all $\theta \in (\hat{\theta}_k, \theta_k)$ as there is not $\Phi^{\eta'}$ minimizer in $(\hat{\theta}_k, \theta_k)$ by assumption. Given that $\eta(\theta_k) = \eta(\hat{\theta}_k) = \eta'$, one can iterate the previous argument for $k + 1$. All other cases are dealt with similarly.

The last paragraph implies that $q(\theta) \leq q(\theta, \eta')$ for $\theta \in \bigcup_{j \in J \setminus \{\max J\}} [\theta_j, \hat{\theta}_k] \cup [\hat{\theta}_1, \hat{\theta}_{\min J}] \cup [\theta_{\max J}, \theta_1]$. Therefore,

$$\begin{aligned} \Phi(\theta_1, \hat{\theta}_1) &\leq \Phi^{\eta'}(\theta_1, \hat{\theta}_1) - \left(\sum_{j \in J} \Phi^{\eta'}(\theta_j, \hat{\theta}_j) \right) + \sum_{j \in J} \Phi(\theta_j, \hat{\theta}_j) \qquad \text{(WA.5)} \\ &= \Phi^{\eta'}(\theta_1, \hat{\theta}_1) - \sum_{j \in J} \Phi^{\eta'}(\theta_j, \hat{\theta}_j) < 0 \end{aligned}$$

where the first inequality holds strictly if $\eta(\theta) < \eta'$ for some $\theta \in [\hat{\theta}_1, \hat{\theta}_{\min J}] \cup [\theta_{\max J}, \theta_1]$ and the second inequality holds strictly by $\sum_{j=2}^{k} \Phi^{\eta'}(\theta_j, \hat{\theta}_j) > \Phi^{\eta'}(\theta_1, \hat{\theta}_1)$. Hence, incentive

compatibility of the optimal decision is contradicted. Therefore, (NLIC) cannot bind from $\theta_j(\eta')$ to $\hat{\theta}_j(\eta')$ for $j = 2, \ldots, k \leq n$ if $\Phi^{\eta'}(\theta_1(\eta'), \hat{\theta}_1(\eta')) < \sum_{i=2}^{n} \Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta'))$. (NLIC) has then to bind from θ_1 to $\hat{\theta}_1$ only – as stated in the lemma.

Second, let $\sum_{j=2}^{k} \Phi^{\eta'}(\theta_j, \hat{\theta}_j) = \Phi^{\eta'}(\theta_1, \hat{\theta}_1)$. The same steps as above still hold but WA.5 holds with equality if $J = \{1, \ldots, n\}$. Hence, (NLIC) binds still from θ_1 to $\hat{\theta}_1$ as $\eta < \eta'$ for $\theta > \theta_1$ (and for $\theta < \hat{\theta}_1$): Otherwise, another $\Phi^{\eta'}$ minimizer $(\theta_0, \hat{\theta}_0)$ with $\theta_1 < \theta_0$ (or $\hat{\theta}_0 < \hat{\theta}_1$) would have to exist which was ruled out by assumption in case 2b.

Recall that (NLIC) binding from θ_1 to $\hat{\theta}_1$ implies $\eta(\theta) \geq \eta'$ for all $\theta \in (\hat{\theta}_1, \theta_1)$ (lemma 9). Therefore, (NLIC) does not bind from θ_j to $\hat{\theta}_j$ if $\Phi^{\eta'}(\theta_j, \hat{\theta}_j) = 0$; i.e. if $\Phi(\theta_j, \hat{\theta}_j) = 0$, then η is constant in neighborhoods of θ_j and $\hat{\theta}_j$.

However, (NLIC) binds from θ_i to $\hat{\theta}_i$ if $\Phi^{\eta'}(\theta_i, \hat{\theta}_i) < 0$: Suppose not. For incentive compatibility, $\eta(\theta) > \eta'$ for some $\theta \in (\hat{\theta}_i, \theta_i)$ and – using the same arguments as above – this implies $\eta(\theta) > \eta'$ for all $\theta \in (\hat{\theta}_i, \theta_i)$. If (NLIC) does not bind from θ_i to $\hat{\theta}_i$, this conclusion even holds true at θ_i and $\hat{\theta}_i$, i.e. $\eta(\theta) > \eta'$ for all $\theta \in [\hat{\theta}_i, \theta_i]$. Then,

$$
\begin{aligned}
\Phi(\theta_1, \hat{\theta}_1) &= \Phi^{\eta'}(\theta_1, \hat{\theta}_1) - \left(\sum_{j=2}^{n} \Phi^{\eta'}(\theta_j, \hat{\theta}_j) \right) + \sum_{j=2}^{n} \left(\Phi(\theta_j, \hat{\theta}_j) + \int_{\hat{\theta}_j}^{\theta_j} \int_{q(\hat{\theta}_j, \eta')}^{q(\hat{\theta}_j)} -c_{q\theta}(y, x)\, dy\, dx \right) \\
&\quad + \int_{\hat{\theta}_1}^{\hat{\theta}_2} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx + \int_{\theta_n}^{\theta_1} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx \\
&\quad + \sum_{j=2}^{n-1} \int_{\theta_j}^{\hat{\theta}_{j+1}} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx \\
&\geq \sum_{j=2}^{n} \left(\Phi(\theta_j, \hat{\theta}_j) + \int_{\hat{\theta}_j}^{\theta_j} \int_{q(\hat{\theta}_j, \eta')}^{q(\hat{\theta}_j)} -c_{q\theta}(y, x)\, dy\, dx \right) \\
&\geq \Phi(\theta_i, \hat{\theta}_i) + \sum_{j=2}^{n} \left(\int_{\hat{\theta}_j}^{\theta_j} \int_{q(\hat{\theta}_j, \eta')}^{q(\hat{\theta}_j)} -c_{q\theta}(y, x)\, dy\, dx \right) > 0
\end{aligned}
$$

where the first inequality follows from $q(x) \geq q(x, \eta') > s(x)$ and the assumption $\Phi^{\eta'}(\theta_1(\eta'), \hat{\theta}_1(\eta')) = \sum_{i=2}^{n} \Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta'))$. The second inequality follows from incentive compatibility. The third inequality holds as $(\theta_j, \hat{\theta}_j)$ is a local minimizer of $\Phi^{\eta'}$ and therefore the first order condition $\int_{\hat{\theta}_j}^{\theta_j} -c_{q\theta}(q(\hat{\theta}_j), x)\, dx = 0$ has to hold. Together with $c_{qq\theta} < 0$, this implies the third inequality which is strict as $q(\hat{\theta}_i) > q(\hat{\theta}_i, \eta')$ by $\eta(\hat{\theta}_i) > \eta'$. This contradicts that (NLIC) binds from θ_1 to $\hat{\theta}_1$. Hence, (NLIC) has to bind from θ_i to $\hat{\theta}_i$.

Third, let $\Phi^{\eta'}(\theta_1(\eta'), \hat{\theta}_1(\eta')) > \sum_{i=2}^{n} \Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta'))$. Suppose, contrary to the lemma, that (NLIC) was binding from θ_1 to $\hat{\theta}_1$. Then $\eta(\theta) \geq \eta'$ for all $\theta \in (\hat{\theta}_1, \theta_1)$ by lemma 9 which implies $q(\theta) \geq q(\theta, \eta')$ for these types. Then,

$$
\begin{aligned}
\Phi(\theta_1, \hat{\theta}_1) &= \Phi^{\eta'}(\theta_1, \hat{\theta}_1) - \left(\sum_{j=2}^{n} \Phi^{\eta'}(\theta_j, \hat{\theta}_j) \right) + \sum_{j=2}^{n} \left(\Phi(\theta_j, \hat{\theta}_j) + \int_{\hat{\theta}_j}^{\theta_j} \int_{q(\hat{\theta}_j, \eta')}^{q(\hat{\theta}_j)} -c_{q\theta}(y, x)\, dy\, dx \right) \\
&\quad + \int_{\hat{\theta}_1}^{\hat{\theta}_2} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx + \int_{\theta_n}^{\theta_1} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx \\
&\quad + \sum_{j=2}^{n-1} \int_{\theta_j}^{\hat{\theta}_{j+1}} \int_{q(x,\eta')}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx \\
&> \sum_{j=2}^{n} \left(\Phi(\theta_j, \hat{\theta}_j) + \int_{\hat{\theta}_j}^{\theta_j} \int_{q(\hat{\theta}_j, \eta')}^{q(\hat{\theta}_j)} -c_{q\theta}(y, x)\, dy\, dx \right) \\
&\geq \sum_{j=2}^{n} \int_{\hat{\theta}_j}^{\theta_j} \int_{q(\hat{\theta}_j, \eta')}^{q(\hat{\theta}_j)} -c_{q\theta}(y, x)\, dy\, dx > 0
\end{aligned}
$$

where the first inequality follows from $q(x) \geq q(x, \eta') > s(x)$ and the assumption $\Phi^{\eta'}(\theta_1(\eta'), \hat{\theta}_1(\eta')) > \sum_{i=2}^{n} \Phi^{\eta'}(\theta_i(\eta'), \hat{\theta}_i(\eta'))$. The second inequality follows from incentive compatibility. The third inequality holds as $(\theta_j, \hat{\theta}_j)$ is a local minimizer of $\Phi^{\eta'}$ and therefore the first order condition $\int_{\hat{\theta}_j}^{\theta_j} -c_{q\theta}(q(\hat{\theta}_j), x)\, dx = 0$ has to hold. Together with $c_{qq\theta} < 0$, this implies the third inequality. This contradicts that (NLIC) binds from θ_1 to $\hat{\theta}_1$. Having ruled this possibility out, lemma WA6 implies the result. □

Case 3 (overlapping minimizers). Assume that some local Φ^{η} minimizers overlap, e.g. $\hat{\theta}_1(\eta) < \hat{\theta}_2(\eta) < \theta_1(\eta) < \theta_2(\eta)$. It turns out that this situation is equivalent to a multiply nested situation as in case 2b.

Theorem 2 implies that (NLIC) cannot bind from $\theta_i(\eta)$ to $\hat{\theta}_i(\eta)$ if those two types do not minimize Φ^{η} on $[\hat{\theta}_i(\eta), \theta_i(\eta)]$. Hence, such Φ^{η} are not relevant and not taken into consideration from the outset. Using a similar argument as in the proof of lemma 5, the following result is obtained:

Lemma WA9. *For $i = 1, 2$, let $(\theta_i, \hat{\theta}_i)$ minimize Φ^{η} locally and let $(\theta_i, \hat{\theta}_i)$ minimize Φ^{η} on $[\hat{\theta}_i, \theta_i]$. Assume $\hat{\theta}_1 < \hat{\theta}_2 < \theta_1 < \theta_2$. Then there exists a $(\theta_0, \hat{\theta}_0)$ such that (i) $(\theta_0, \hat{\theta}_0)$ minimizes Φ^{η} locally, (ii) $\hat{\theta}_0 \leq \hat{\theta}_1$ and $\theta_0 \geq \theta_2$, (iii) $\Phi^{\eta}(\theta_0, \hat{\theta}_0) < min\{\Phi^{\eta}(\theta_1, \hat{\theta}_1), \Phi^{\eta}(\theta_2, \hat{\theta}_2)\}$.*

In words, if two Φ^{η} minimizers overlap, there exists a nesting Φ^{η} minimizer. Recall

lemma 5 which implies that (NLIC) cannot bind both from θ_1 to $\hat{\theta}_1$ and from θ_2 to $\hat{\theta}_2$. If there are no other Φ^η minimizers in $(\hat{\theta}_0, \theta_0)$, the same arguments as in lemma WA7 imply that (NLIC) binds from θ_0 to $\hat{\theta}_0$ and (NLIC) is slack between θ_i and $\hat{\theta}_i$, $i = 1, 2$, because $\Phi^\eta(\theta_0, \hat{\theta}_0) < min\{\Phi^\eta(\theta_1, \hat{\theta}_1), \Phi^\eta(\theta_2, \hat{\theta}_2)\}$.

With multiple overlapping Φ^η minimizers, i.e. $(\theta_i, \hat{\theta}_i)$ for $i = 1, \ldots, n$ such that $\hat{\theta}_i < \hat{\theta}_{i+1} < \theta_i < \theta_{i+1}$ for $i =, 1 \ldots, n-1$, the arguments of lemma WA9 still imply that there is a nesting Φ^η minimizer $(\theta_0, \hat{\theta}_0)$ such that $\hat{\theta}_0 \leq \hat{\theta}_1$, $\theta_0 \geq \theta_n$ and $\Phi^\eta(\theta_0, \hat{\theta}_0) < \min_i \Phi^\eta(\theta_i, \hat{\theta}_i)$. However, this does not immediately imply that (NLIC) binds from θ_0 to $\hat{\theta}_0$: If there is a set $J \subset \{1, 2, \ldots, n\}$ such that $\theta_j < \hat{\theta}_{j+1}$ for every $j \in J$ and $\sum_{j \in J} \Phi^\eta(\theta_j, \hat{\theta}_j) < \Phi^\eta(\theta_0, \hat{\theta}_0)$, then – following the arguments in the proof of lemma WA8 – (NLIC) will bind from θ_j to $\hat{\theta}_j$ for all $j \in J$ and (NLIC) will be slack between θ_0 and $\hat{\theta}_0$. If no such set J exists, (NLIC) binds from θ_0 to $\hat{\theta}_0$ and is slack between all θ_i and $\hat{\theta}_i$ within $(\hat{\theta}_0, \theta_0)$.

Proof of lemma WA9: First, note that $\Phi^\eta(\theta_1, \hat{\theta}_2) \geq max\{\Phi^\eta(\theta_1, \hat{\theta}_1), \Phi^\eta(\theta_2, \hat{\theta}_2)\}$: If this was not the case, e.g. say $\Phi^\eta(\theta_1, \hat{\theta}_2) < \Phi^\eta(\theta_2, \hat{\theta}_2)$, then $(\theta_2, \hat{\theta}_2)$ would not minimize Φ^η on $[\hat{\theta}_2, \theta_2]$ as assumed.

Second, similarly to the proof of lemma 5

$$\begin{aligned}
\Phi^\eta(\theta_2, \hat{\theta}_1) &= \Phi^\eta(\theta_1, \hat{\theta}_1) + \Phi^\eta(\theta_2, \hat{\theta}_2) - \Phi^\eta(\theta_1, \hat{\theta}_2) + \int_{\theta_1}^{\theta_2} \int_{q(\hat{\theta}_1, \eta)}^{q(\hat{\theta}_2, \eta)} -c_{q\theta}(y, x)\, dy\, dx \\
&< \Phi^\eta(\theta_1, \hat{\theta}_1) + \Phi^\eta(\theta_2, \hat{\theta}_2) - \Phi^\eta(\theta_1, \hat{\theta}_2) \leq min\{\Phi^\eta(\theta_1, \hat{\theta}_1), \Phi^\eta(\theta_2, \hat{\theta}_2)\}
\end{aligned}$$

where the first inequality uses $\int_{q(\hat{\theta}_1, \eta}^{q(\theta_1, \eta)} c_{q\theta}(y, \theta_1)\, dy = 0$ (because $(\theta_1, \hat{\theta}_1)$ is a local Φ^η minimizer), $q(\hat{\theta}_2, \eta) < q(\theta_1, \eta)$ and $c_{q\theta\theta} > 0$. The second inequality follows directly from the first paragraph of this proof.

As $(\theta_1, \hat{\theta}_1)$ and $(\theta_2, \hat{\theta}_2)$ locally minimize Φ^η, the first order conditions $\int_{q(\hat{\theta}_2, \eta)}^{q(\theta_2, \eta)} c_{q\theta}(y, \theta_2)\, dy \geq 0$ and $\int_{\hat{\theta}_1}^{\theta_1} c_{q\theta}(q(\hat{\theta}_1, \eta), x)\, dx \geq 0$ hold. Consequently, $\int_{q(\hat{\theta}_1, \eta)}^{q(\theta_2, \eta)} c_{q\theta}(y, \theta_2)\, dy > 0$ and $\int_{\hat{\theta}_1}^{\theta_2} c_{q\theta}(q(\hat{\theta}_1, \eta), x)\, dx > 0$. Hence, $\Phi^\eta(\theta, \hat{\theta})$ is decreasing in the first and increasing in the second argument at $(\theta_2, \hat{\theta}_1)$ which implies that there has to exist a local minimizer $(\theta_0, \hat{\theta}_0)$ with the properties of the lemma. □

Other cases. If the local minimizers of Φ^η have a different structure than in the cases 1 to 3, the previous lemmas can often still be used to determine the relevant minimizers. This is true because the proofs are based on incentive compatibility. To give an example, suppose the minimizers of Φ^η were ordered $\hat{\theta}_1 < \hat{\theta}_2 < \hat{\theta}_3 < \theta_3 < \theta_2 < \hat{\theta}_4 < \theta_4 < \theta_1$. The

minimizers $(\hat{\theta}_2, \theta_2)$ and $(\hat{\theta}_3, \theta_3)$ are nested. Assume $\Phi^\eta(\theta_2, \hat{\theta}_2) < \Phi^\eta(\theta_3, \hat{\theta}_3)$. In a nutshell, the proof of lemma WA7 establishes that (NLIC) from θ_2 to $\hat{\theta}_2$ would be violated if (NLIC) was binding from θ_3 to $\hat{\theta}_3$. This holds still true. Therefore (NLIC) cannot bind from θ_3 to $\hat{\theta}_3$ and $(\theta_3, \hat{\theta}_3)$ is an irrelevant minimizer. This leaves us with the multiply nested structure $\hat{\theta}_1 < \hat{\theta}_2 < \theta_2 < \hat{\theta}_4 < \theta_4 < \theta_1$ for which lemma WA8 establishes the relevant minimizer.

B.6. Variational condition

This subsection derives condition (C3) for strictly monotone solutions. The derivation follows Araujo and Moreira (2010) where (C1) does not play a role as only U-shaped decisions are analyzed. However, the result for monotone decisions is already derived in the working paper version Araujo and Moreira (2001). Starting from (C2), Araujo and Moreira (2010) derive the following condition (with $q = q(\theta) = q(\hat{\theta})$):

$$\frac{u_q(q,\theta) - c_q(q,\theta) + \frac{1-F(\theta)}{f(\theta)} c_{q\theta}(q,\theta)}{c_{q\theta}(q,\theta)} f(\theta) = \frac{u_q(q,\hat{\theta}) - c_q(q,\hat{\theta}) + \frac{1-F(\hat{\theta})}{f(\hat{\theta})} c_{q\theta}(q,\hat{\theta})}{c_{q\theta}(q,\hat{\theta})} f(\hat{\theta}) \tag{WA.6}$$

To derive a similar condition for $q(\theta) \neq q(\hat{\theta})$, let (NLIC) bind from θ to $\hat{\theta}$ with $\theta, \hat{\theta} \in (\underline{\theta}, \bar{\theta})$ and assume that $q(\cdot)$ is strictly monotone in neigborhoods of θ and $\hat{\theta}$ and continuous at θ and $\hat{\theta}$.

Given θ and $q(\theta)$, the equation $c_\theta(q(\theta), \theta) = c_\theta(q(\hat{\theta}), \theta)$ pins down a decision $q(\hat{\theta})$ where incentive compatibility could be binding. Given this, $q(\hat{\theta})$ as well as θ and $q(\theta)$, the equation $c_q(q(\hat{\theta}), \hat{\theta}) = c_q(q(\hat{\theta}), \theta)$ determines $\hat{\theta}$. Therefore, the critical $\hat{\theta}$ can be written as a function of θ and $q(\theta)$, i.e. $\hat{\theta} = \phi(\theta, q(\theta))$.

Differentiating the two conditions, the partial derivatives ϕ_θ and ϕ_q can be obtained as

$$\begin{aligned}
\phi_\theta(\theta, q) &= \frac{c_{q\theta}(\hat{q},\theta)}{c_{q\theta}(\hat{q},\hat{\theta})} + \frac{(c_{qq}(\hat{q},\theta) - c_{qq}(\hat{q},\hat{\theta}))(c_{\theta\theta}(q,\theta) - c_{\theta\theta}(\hat{q},\theta))}{c_{q\theta}(\hat{q},\hat{\theta}) c_{q\theta}(\hat{q},\theta)} \\
\phi_q(\theta, q) &= \frac{c_{q\theta}(q,\theta)[c_{qq}(\hat{q},\theta) - c_{qq}(\hat{q},\hat{\theta})]}{c_{q\theta}(\hat{q},\hat{\theta}) c_{q\theta}(\hat{q},\theta)}
\end{aligned}$$

where $\hat{q} = q(\hat{\theta})$ and $q = q(\theta)$.

Denote by h an implementable perturbation of the optimal solution q^* on some interval $[\theta_1, \theta_2]$ with $h(\theta_1) = h(\theta_2) = 0$. I denote in the remainder of this section the

optimal solution by $q^*(\theta)$ and the perturbed solution by $q(\theta) = q^*(\theta) + \varepsilon h(\theta)$. I will call a perturbation h implementable if there exists an $\varepsilon' > 0$ such that the decision $q^* + \varepsilon h$ is implementable for all $\varepsilon \in [0, \varepsilon']$. Implementability implies that $\hat{\theta} = \phi(\theta, q(\theta))$ if the incentive constraint binds from θ to $\hat{\theta}$ (assuming strict monotonicity and continuity of q for $\varepsilon \in [0, \varepsilon']$).

The idea of the variational argument is the following: I want to derive a necessary condition for a type θ such that $\Phi(\theta, \hat{\theta}) = 0$ for some $\hat{\theta}$. To do so, it is assumed that also under the perturbed decision the incentive constraint is binding for θ and some (other) $\hat{\theta}$. The type $\hat{\theta}$ to which the non-local incentive constraint binds depends on the perturbation and is given by $\phi(\theta, q(\theta))$. The way one should think about it is that incentive compatibility is binding from each $\theta \in (\theta_1, \theta_2)$ to some $\hat{\theta}$ in some interval $(\hat{\theta}_1, \hat{\theta}_2)$. The specific type $\hat{\theta}$ to which a non-local incentive constraint binds from a given θ depends on the perturbation h.

The part of the principal's objective function affected by the perturbation can be written as[38]

$$\begin{aligned} G(\varepsilon) &= \int_{\theta_1}^{\theta_2} g(q(\theta), \theta)\, d\theta + \int_{\phi(\theta_2, q(\theta_2))}^{\phi(\theta_1, q(\theta_1))} g(q(\theta), \theta)\, d\theta \\ &= \int_{\theta_1}^{\theta_2} \{g(q(\theta), \theta) - g(\hat{q}(\theta, q(\theta)), \phi(\theta, q(\theta)))\, [\phi_q(q(\theta), \theta) q_\theta(\theta) + \phi_\theta(q(\theta), \theta)]\}\, d\theta \end{aligned} \tag{WA.7}$$

where $g(q(\theta), \theta) = \left[u(q(\theta), \theta) - c(q(\theta), \theta) + \frac{1-F(\theta)}{f(\theta)} c_\theta(q(\theta), \theta)\right] f(\theta)$ is the virtual valuation weighted by the density. The second line is a normal change of variables where $\hat{q}(\theta, q)$ denotes the $\hat{q}$ solving $c_\theta(q, \theta) = c_\theta(\hat{q}, \theta)$ with $q \neq \hat{q}$. Note that $\partial \hat{q}/\partial q = c_{q\theta}(q, \theta)/c_{q\theta}(\hat{q}, \theta)$.

Differentiating (WA.7) gives

$$G'(0) = \int_{\theta_1}^{\theta_2} \{g_q h - \hat{g}((\phi_{qq} q_\theta^* + \phi_{q\theta}) h + \phi_q h_\theta) - (\hat{g}_q \hat{q}_q + \hat{g}_\theta \phi_q)(\phi_q q_\theta^* + \phi_\theta) h\}\, d\theta = 0$$

where arguments are omitted and a hat denotes evaluation at $(\hat{\theta}, q^*(\hat{\theta}))$ and $\hat{\theta} = \phi(\theta, q^*(\theta))$.

[38] It follows from lemma 5 that $\phi(\theta_1, q(\theta_1)) > \phi(\theta_2, q(\theta_2))$.

Integrating $\int_{\theta_1}^{\theta_2}(\hat{g}\phi_q)h_\theta\, d\theta$ by parts and substituting yields for the previous equation

$$\int_{\theta_1}^{\theta_2}\{g_q - \hat{g}_q\hat{q}_q\phi_\theta + \hat{g}_q\hat{q}_\theta\phi_q\}\, h\, d\theta = \int_{\theta_1}^{\theta_2}\left\{g_q - \hat{g}_q\frac{c_{q\theta}(q(\theta),\theta)}{c_{q\theta}(q(\hat{\theta}),\hat{\theta})}\right\} h\, d\theta = 0.$$

I will argue that the last equation can only hold if

$$g_q(q(\theta),\theta) - g_q(q(\hat{\theta}),\hat{\theta})\frac{c_{q\theta}(q(\theta),\theta)}{c_{q\theta}(q(\hat{\theta}),\hat{\theta})} = 0 \tag{C3'}$$

at the optimum which is condition (C3). The argument is that otherwise there exists an implementable perturbation h such that $G'(0) > 0$ which contradicts optimality. First, note that by the continuity of q the left hand side of (C3') is continuous in θ. Hence, if the left hand side of (C3') was not zero for some type, say for for the sake of argument it is positive, then there is an interval (θ_1', θ_2') such that the left hand side of (C3') is positive for all $\theta \in (\theta_1', \theta_2')$ and their corresponding $\hat{\theta}$.

Let h be zero for all $\theta \notin (\theta_1', \theta_2')$. Furthermore, let h be continuous, strictly positive on (θ_1', θ_2') and differentiable with bounded derivative on (θ_1', θ_2').[39] Clearly, $G'(0) > 0$ in this case.

If the left hand side of (C3') is negative for some type, then again there has to be an interval (θ_1', θ_2') such that the left hand side of (C3') is negative for all $\theta \in (\theta_1', \theta_2')$. In this case, choose h to be zero for all $\theta \notin (\theta_1', \theta_2')$. Furthermore, let h be continuous, strictly negative on (θ_1', θ_2') and differentiable with bounded derivative on (θ_1', θ_2'). Again, $G'(0)$ is clearly positive.

It remains to show that such an h is implementable. As q^* is strictly increasing on (θ_1', θ_2') and h has a bounded derivative, q is strictly increasing on (θ_1', θ_2') for ε small enough.

As q^* is implementable and strictly increasing, $\hat{q}(\theta, q^*(\theta))$ is strictly decreasing on (θ_1', θ_2') according to lemma 5. The boundedness of h_θ implies that $\hat{q}$ is also strictly decreasing (θ_1', θ_2') for $\varepsilon > 0$ small enough: For types where q^* is differentiable $\hat{q}_\theta(\theta, q(\theta)) = \frac{\int_{\hat{q}(\theta,q(\theta))}^{q(\theta)} c_{q\theta\theta}(y,\theta)\, dy + q_\theta(\theta)c_{q\theta}(q(\theta),\theta)}{c_{q\theta}(\hat{q}(\theta,q(\theta)),\theta)}$ which converges to the derivative of $\hat{q}(\theta, q^*(\theta))$ as $\varepsilon \to 0$. Hence, $\frac{d\,\hat{q}(\theta,q(\theta))}{d\theta} < 0$ at types where q^* is differentiable for $\varepsilon > 0$ small enough. As q^* is monotone, it is differentiable almost everywhere. Hence, strict monotonicity of $\hat{q}$ (for

[39] For example, $h(\theta) = -\left(\theta - \frac{\theta_1'+\theta_2'}{2}\right)^2 + \left(\frac{\theta_1'-\theta_2'}{2}\right)^2$ for $\theta \in (\theta_1', \theta_2')$ and $h(\theta) = 0$ for all other types.

$\varepsilon > 0$ small enough) follows from continuity of $\hat{q}$ and the limit result for $d\,\hat{q}/d\theta$. The strict monotonicity of $\hat{q}$ together with $c_{qq\theta} < 0$ implies that $\phi(\theta, q(\theta))$ is strictly decreasing in θ (for $\varepsilon > 0$ small enough). Hence, q is strictly increasing on $(\hat{\theta}_1, \hat{\theta}_2)$ for $\varepsilon > 0$ small enough.[40] That is, the monotonicity constraint is satisfied by q for $\varepsilon > 0$ small enough.

It remains to check non-local incentive constraints under q for $\varepsilon > 0$ small enough. Lemma 5 implies that $\Phi(\theta, \hat{\theta}) \geq 0$ for $\theta > \theta_2$ and $\hat{\theta} \in (\hat{\theta}_1, \theta_1)$: For these types, (NLIC) was satisfied with strict inequality under q^* and therefore the same will be true for q if $\varepsilon > 0$ is small enough because $\Phi(\theta, \hat{\theta})$ is continuous in ε.[41] The same argument holds for types $\theta \in (\hat{\theta}_1, \theta_1)$ and $\hat{\theta} < \hat{\theta}_2$. For θ and $\hat{\theta}$ such that $\hat{\theta}_1 < \hat{\theta} < \theta < \theta_1$, the perturbation does not affect $\Phi(\theta, \hat{\theta})$. Hence, (NLIC) is satisfied for these types.

Now take $\theta > \theta_2$ and $\hat{\theta} < \hat{\theta}_2$. Here it is important to note that $h(\theta_1) = h(\theta_2) = 0$. This implies that $\Phi(\theta_2, \hat{\theta}_2) = 0$ both under the original and the perturbed decision. As $\Phi(\theta, \hat{\theta}) = \Phi(\theta_2, \hat{\theta}_2) + \int_{\hat{\theta}}^{\theta} \int_{q(\hat{\theta})}^{min(q(\hat{\theta}_2), q(x))} -c_{q\theta}(y, x)\, dy\, dx + \int_{\theta_2}^{\theta} \int_{q(\hat{\theta}_2)}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx = \Phi^*(\theta_2, \hat{\theta}_2) + \int_{\hat{\theta}}^{\theta} \int_{q(\hat{\theta})}^{min(q(\hat{\theta}_2), q(x))} -c_{q\theta}(y, x)\, dy\, dx + \int_{\theta_2}^{\theta} \int_{q(\hat{\theta}_2)}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx = \Phi^*(\theta, \hat{\theta})$, (NLIC) holds between θ and $\hat{\theta}$.

Next take $\theta \in [\theta_1, \theta_2]$. I will distinguish four cases. First, take $\hat{\theta}_1 < \hat{\theta} < \theta_1$. Then $\Phi(\theta, \hat{\theta}) = \Phi(\theta_1, \hat{\theta}) + \int_{\theta_1}^{\theta} \int_{q(\hat{\theta})}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx \geq \Phi(\theta_1, \hat{\theta}) = \Phi^*(\theta_1, \hat{\theta}) \geq 0$ where the first inequality holds because $\int_{q(\hat{\theta})}^{q(x)} -c_{q\theta}(y, x)\, dy \geq \int_{q(\phi(x, q(x))}^{q(x)} -c_{q\theta}(y, \theta_1)\, dy = 0$ for all $x \in [\theta_1, \theta]$ by the monotonicity of q. Second, take $\hat{\theta} \in [\hat{\theta}_2, \hat{\theta}_1]$ and let $\theta' \in [\theta_1, \theta_2]$ be the type such that $\phi(\theta', q(\theta')) = \hat{\theta}$. Then $\Phi(\theta, \hat{\theta}) = \Phi(\theta', \hat{\theta}) + \int_{\theta'}^{\theta} \int_{q(\hat{\theta})}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx \geq \Phi(\theta', \hat{\theta}) = 0$ where the inequality holds because $\hat{q}$ is decreasing in θ and $\int_{q(\phi(x, q(x))}^{q(x)} c_{q\theta}(y, \theta_1)\, dy = 0$ for all $x \in [\theta_1, \theta_2]$. Third, take $\hat{\theta} < \hat{\theta}_2$. Then $\Phi(\theta, \hat{\theta}) = \Phi(\theta_2, \hat{\theta}) + \int_{\theta}^{\theta_2} \int_{q(\hat{\theta})}^{q(x)} c_{q\theta}(y, x)\, dy\, dx + \geq \Phi(\theta_2, \hat{\theta}) = \Phi(\theta_2, \hat{\theta}_2) + \int_{\hat{\theta}}^{\theta_2} \int_{q(\hat{\theta})}^{min(q(x), q(\hat{\theta}_2))} -c_{q\theta}(y, x)\, dy\, dx = \Phi^*(\theta_2, \hat{\theta}) \geq 0$ where I utilize $\Phi(\theta_2, \hat{\theta}_2) = \Phi^*(\theta_2, \hat{\theta}_2) = 0$ (recall that $h(\theta_2) = 0$). Fourth, $\hat{\theta} \in [\theta_1, \theta)$. One can choose θ_1 and θ_2 close enough (or equivalently let h be non-zero only on a small enough subinterval of (θ_1, θ_2)) such that $c_{q\theta}(q(\theta_1), \theta_2) < 0$ which implies that (NLIC) is slack for $\theta, \hat{\theta} \in [\theta_1, \theta_2]$.

Last, take $\hat{\theta} \in [\hat{\theta}_2, \hat{\theta}_1]$ and let θ' be the type such that $\phi(\theta', q(\theta')) = \hat{\theta}$. First, $\theta \in (\hat{\theta}_1, \theta_1)$. Note that $\int_x^{\theta} c_{q\theta}(q(x), z)\, dz < 0$ for all $x \in [\hat{\theta}, \hat{\theta}_1]$ because $\theta < \theta''(x)$ where $\theta''(x)$

[40] To reduce the length of the expressions, I use the notation $\hat{\theta}_i = \phi(\theta_i, q(\theta_i))$.

[41] To be precise, $\Phi^*(\theta, \hat{\theta})$ (where Φ^* is Φ under q^*) has to be bounded away from 0 for these types: Otherwise, that is if there was a sequence of $(\theta_n, \hat{\theta}_n)$ such that $\Phi^*(\theta_n, \hat{\theta}_n) \to 0$, for the limit types of (a subsequence of) this sequence (NLIC) would hold with equality as Φ^* is continuous. This would violate lemma 5.

is the type in $[\theta_1, \theta_2]$ such that $x = \phi(\theta'', q(\theta''))$. This implies that $\Phi(\theta, \hat{\theta}) \geq \Phi(\theta, \hat{\theta}_1) = \Phi^*(\theta, \hat{\theta}_1) \geq 0$. Second, $\theta \in [\theta_1, \theta_2]$ was already dealt with above. Third, $\theta > \theta_2$. Note that $\int_x^\theta c_{q\theta}(q(x), z)\, dz > 0$ for all $x \in (\hat{\theta}_2, \hat{\theta})$ as $\theta > \theta_2$. This implies $\Phi(\theta, \hat{\theta}) \geq \Phi(\theta, \hat{\theta}_2) = \Phi(\theta_2, \hat{\theta}_2) + \int_{\theta_2}^\theta \int_{q(\hat{\theta}_2)}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx = \Phi^*(\theta_2, \hat{\theta}_2) + \int_{\theta_2}^\theta \int_{q(\hat{\theta}_2)}^{q(x)} -c_{q\theta}(y, x)\, dy\, dx = \Phi^*(\theta, \hat{\theta}_2) \geq 0$. Fourth, $\theta \in [\hat{\theta}_2, \hat{\theta})$. One can choose θ_1 and θ_2 close enough (or equivalently let h be non-zero only on a small enough subinterval of (θ_1, θ_2)) such that $c_{q\theta}(q(\hat{\theta}_2), \hat{\theta}_1) < 0$ which implies that (NLIC) is slack for $\theta, \hat{\theta} \in [\hat{\theta}_2, \hat{\theta}_1]$.

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